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V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning

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A major challenge for modern AI is to learn to understand the world and learn to act largely by observation (LeCun, 2022). This paper explores a self-supervised approach that combines internet-scale video data with a small amount of interaction data (robot trajectories), to develop models capable of understanding, predicting, and planning in the physical world. We first pre-train an action-free joint-embedding-predictive architecture, V-JEPA 2, on a video and image dataset comprising over 1 million hours of internet video. V-JEPA 2 achieves strong performance on motion understanding (77.3 top-1 accuracy on Something-Something v2) and state-of-the-art performance on human action anticipation (39.7 recall-at-5 on Epic-Kitchens-100) surpassing previous task-specific models. Additionally, after aligning V-JEPA 2 with a large language model, we demonstrate state-of-the-art performance on multiple video question-answering tasks at the 8 billion parameter scale (e.g., 84.0 on PerceptionTest, 76.9 on TempCompass). Finally, we show how self-supervised learning can be applied to robotic planning tasks by post-training a latent action-conditioned world model, V-JEPA 2-AC, using less than 62 hours of unlabeled robot videos from the Droid dataset. We deploy V-JEPA 2-AC zero-shot on Franka arms in two different labs and enable picking and placing of objects using planning with image goals. Notably, this is achieved without collecting any data from the robots in these environments, and without any task-specific training or reward. This work demonstrates how self-supervised learning from web-scale data and a small amount of robot interaction data can yield a world model capable of planning in the physical world.

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Blogpost: https://ai.meta.com/blog/v-jepa-2-world-model-benchmarks

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1 Introduction

Humans have the ability to adapt and generalize when taking on new tasks and operating in unfamiliar environments. Several cognitive learning theories suggest that humans learn an internal model of the world by integrating low-level sensory inputs to represent and predict future states (Craik, 1967; Rao and Ballard, 1999), and they further posit that this world model shapes our perception at any given moment, playing a crucial role in informing our understanding of reality (Friston, 2010; Clark, 2013; Nortmann et al., 2015). Moreover, our ability to predict the effects of our actions on future states of the world is also essential for goal-oriented planning (Sutton and Barto, 1981, 1998; Ha and Schmidhuber, 2018; Wolpert and Ghahramani, 2000). Building artificial agents that learn a world model from sensory data, such as video, could enable them to *understand* the physical world, *predict* future states, and effectively — like humans — *plan* in new situations, resulting in systems capable of tackling tasks that have not been encountered before.

Previous works have explored the development of predictive world models from interaction data consisting

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V-JEPA 2：自监督视频模型实现理解、预测与规划

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现代人工智能面临的一项重大挑战，是主要通过观察学会理解世界并采取行动（LeCun, 2022）。本文探索一种自监督方法，将互联网规模的视频数据与少量交互数据（机器人轨迹）结合，构建能够理解、预测并在物理世界中规划的模型。我们首先在包含超过 100 万小时互联网视频的视频与图像数据集上，预训练不以动作为条件的联合嵌入预测架构 V-JEPA 2。V-JEPA 2 在运动理解上表现出色（Something-Something v2 的 top-1 准确率为 77.3），在人类动作预判上达到最先进水平（Epic-Kitchens-100 的 recall@5 为 39.7），超越此前的任务专用模型。此外，将 V-JEPA 2 与大语言模型对齐后，我们在 80 亿参数规模下，于多个视频问答任务中达到最先进水平（例如 PerceptionTest 为 84.0，TempCompass 为 76.9）。最后，我们利用 Droid 数据集中不足 62 小时的无标注机器人视频，对潜在动作条件世界模型 V-JEPA 2-AC 进行后训练，展示自监督学习如何用于机器人规划任务。我们在两个不同实验室的 Franka 机械臂上零样本部署 V-JEPA 2-AC，通过以图像为目标的规划实现物体抓取和放置。值得关注的是，这无需从这些环境中的机器人采集任何数据，也无需任务专用训练或奖励。本研究展示了如何从互联网规模数据和少量机器人交互数据中进行自监督学习，得到能在物理世界中规划的世界模型。

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1 引言

人类在承担新任务、进入陌生环境时，具有适应和泛化能力。多种认知学习理论认为，人类通过整合低层感官输入，学习一个表征并预测未来状态的内部世界模型（Craik, 1967; Rao and Ballard, 1999）。这些理论进一步认为，世界模型塑造了我们在任何时刻的感知，对我们理解现实发挥关键作用（Friston, 2010; Clark, 2013; Nortmann et al., 2015）。此外，预测自身动作如何影响世界未来状态的能力，也是目标导向规划的基础（Sutton and Barto, 1981, 1998; Ha and Schmidhuber, 2018; Wolpert and Ghahramani, 2000）。构建从视频等感官数据中学习世界模型的人工智能体，可能使其能够理解物理世界、预测未来状态，并像人类一样在新情境中有效规划，从而处理此前从未遇到的任务。

此前研究从交互数据发展预测世界模型，这些数据包括

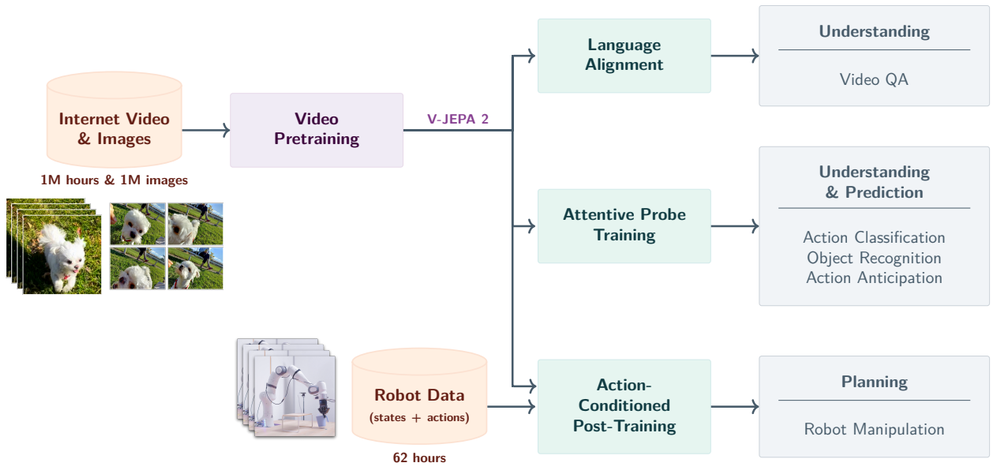


Figure 1 V-JEPA 2 Overview. Leveraging 1M hours of internet-scale video and 1M images, we pretrain the V-JEPA 2 video model using a visual mask denoising objective (Bardes et al., 2024; Assran et al., 2023), and leverage this model for downstream tasks such as action classification, object recognition, action anticipation, and Video Question Answering by aligning the model with an LLM backbone. After pretraining, we can also freeze the video encoder and train a new action-conditioned predictor with a small amount of robot interaction data on top of the learned representations, and leverage this action-conditioned model, V-JEPA 2-AC, for downstream robot manipulation tasks using planning within a model predictive control loop.

of state-action sequences, often also relying on explicit reward feedback from the environment to infer goals (Sutton and Barto, 1981; Fragkiadaki et al., 2015; Ha and Schmidhuber, 2018; Hafner et al., 2019b; Hansen et al., 2022). However, the limited availability of real-world interaction data constrains the scalability of these methods. To address this limitation, more recent works have leveraged both internet-scale video and interaction data towards training action-conditioned video generation models for robot control, but only demonstrate limited results in robot execution using model-based control (Hu et al., 2023; Yang et al., 2024b; Bruce et al., 2024; Agarwal et al., 2025). In particular, this line of research often emphasizes the evaluation of the faithfulness of the predictions and visual quality instead of planning capabilities, perhaps due to the computational cost of planning by generating video.

In this work, we build upon the self-supervised hypothesis as a means to learn world models that capture background knowledge of the world largely from observation. Specifically, we leverage the joint-embedding predictive architecture (JEPA) (LeCun, 2022), which learns by making predictions in a learned representation space. In contrast to approaches that focus on learning entirely from interaction data, self-supervised learning enables us to make use of internet-scale video — depicting sequences of states without direct observations of the actions — to learn to both represent video observations and learn a predictive model for world dynamics in this learned representation space. Furthermore, in contrast to approaches based on video generation, the JEPA approach focuses on learning representations for predictable aspects of a scene (e.g., the trajectory of an object in motion) while ignoring unpredictable details that generative objectives emphasize, since they make pixel-level predictions (e.g., the precise location of each blade of grass in a field, or each leaf on a tree). By scaling JEPA pretraining, we demonstrate that it yields video representations with state-of-the-art understanding and prediction capabilities, and that such representations can be leveraged as a basis for action-conditioned predictive models and enable zero-shot planning.

Our approach, V-JEPA 2, utilizes a stage-wise training procedure, beginning with action-free pre-training on internet-scale video, followed by post-training with a small amount of interaction data (see Figure 1). In the first stage, we use a mask-denoising feature prediction objective (Assran et al., 2023; Bardes et al., 2024), where the model predicts masked segments of a video in a learned representation space. We train

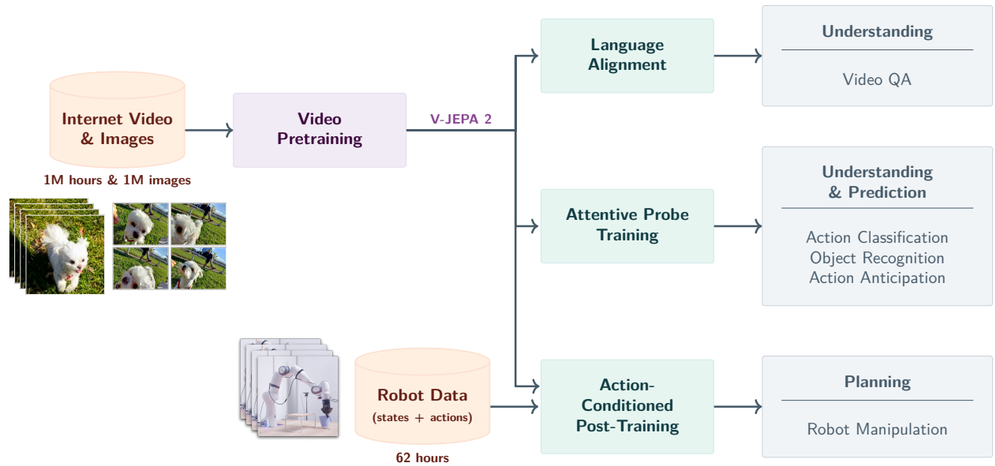


图 1 V-JEPA 2 概览。利用 100 万小时互联网规模视频与 100 万张图像，以视觉掩码去噪目标（Bardes et al., 2024; Assran et al., 2023）预训练视频模型，并用于动作分类、物体识别、动作预判，以及与 LLM 骨干对齐后的视频问答。预训练后，还可冻结视频编码器，在所学表征上使用少量机器人交互数据训练新的动作条件预测器。所得 V-JEPA 2-AC 在模型预测控制循环中进行规划，用于下游机器人操作。

状态–动作序列，往往还依赖环境的显式奖励反馈来推断目标（Sutton and Barto, 1981; Fragkiadaki et al., 2015; Ha and Schmidhuber, 2018; Hafner et al., 2019b; Hansen et al., 2022）。然而，真实世界交互数据有限，制约了这些方法的可扩展性。为解决这一限制，近期研究结合互联网规模视频与交互数据，训练用于机器人控制的动作条件视频生成模型，但在采用基于模型的控制执行机器人任务时，取得的结果仍然有限（Hu et al., 2023; Yang et al., 2024b; Bruce et al., 2024; Agarwal et al., 2025）。尤其是，这一研究方向往往侧重评估预测的忠实性和视觉质量，而非规划能力，可能是因为通过生成视频进行规划的计算成本较高。

本研究以自监督假说为基础，希望主要通过观察学习捕获世界背景知识的世界模型。具体而言，我们采用联合嵌入预测架构（JEPA）（LeCun, 2022），通过在学习到的表征空间中作预测来学习。与完全依赖交互数据的方法相比，自监督学习使我们能使用互联网规模视频——它们展示状态序列，却无法直接观测动作——同时学习视频观测的表征，并在该表征空间中学习世界动态的预测模型。此外，与视频生成方法相比，JEPA 关注场景中可预测方面的表征（例如运动物体的轨迹），忽略生成目标所强调的不可预测细节，因为生成方法进行像素级预测（例如田野中每根草或树上每片叶子的确切位置）。扩大 JEPA 预训练规模后，我们展示其视频表征具有最先进的理解与预测能力，并可作为动作条件预测模型的基础，实现零样本规划。

V-JEPA 2 采用分阶段训练：先在互联网规模视频上进行无动作条件预训练，再用少量交互数据后训练（图 1）。第一阶段使用掩码去噪特征预测目标（Assran et al., 2023; Bardes et al., 2024），在学习到的表征空间预测视频被遮挡部分。我们训练

the V-JEPA 2 encoder with up to 1 billion parameters and with more than 1 million hours of video. Our experiments confirm that scaling self-supervised video pretraining enhances the encoder's ability to achieve visual understanding, including broad motion and appearance recognition capabilities, through probe-based evaluations and by aligning the encoder with a language model for video question-answering (Krojer et al., 2024; Pătrăucean et al., 2023; Liu et al., 2024c; Cai et al., 2024; Shangguan et al., 2024).

Following pretraining on internet-scale video, we train an action-conditioned world model, V-JEPA 2-AC, on a small set of interaction data using the representations learned in the first stage. Our action-conditioned world model is a 300M-parameter transformer network employing a block-causal attention mechanism, which autoregressively predicts the representation of the next video frame conditioned on an action and previous states. With as little as 62 hours of unlabeled interaction data from the Droid dataset (Khazatsky et al., 2024), we demonstrate the feasibility of training a latent world model that, given sub-goals, can be leveraged to plan actions on a Franka robot arm and perform prehensile manipulation tasks from a monocular RGB camera zero-shot in a new environment.

To summarize, we show that joint-embedding predictive architectures learning from videos can be used to build a world model that enables *understanding* the physical world, *predicting* future states, and effectively *planning* in new situations; this is achieved by leveraging internet-scale video and a small amount of interaction data. Specifically:

- *Understanding — Probe-based Classification:* Scaling self-supervised video pretraining results in video representations applicable to many tasks. V-JEPA 2 excels at encoding fine-grained motion information, achieving strong performance on tasks requiring motion understanding, such as Something-Something v2, with 77.3 top-1 accuracy using an attentive probe.
- *Understanding — Video Question-Answering:* V-JEPA 2 encoder can be used to train a multi-modal large language model, to tackle video-question answering tasks. We observe state-of-the-art performance on 8B language model class on multiple benchmarks that require physical world understanding and temporal reasoning, such as MVP (44.5 paired accuracy), PerceptionTest (84.0 test set accuracy), TempCompass (76.9 multi-choice accuracy), TemporalBench (36.7 multi-binary short-QA accuracy) and TOMATO (40.3 accuracy). In particular, we show that a video encoder pre-trained *without* language supervision can be aligned with a language model and achieve state-of-the-art performance, contrary to conventional wisdom (Yuan et al., 2025; Wang et al., 2024b).
- *Prediction:* Large-scale self-supervised video pretraining enhances prediction capabilities. V-JEPA 2 achieves state-of-the-art performance on the Epic-Kitchens-100 human-action anticipation task using an attentive probe, with 39.7 recall-at-5, which is a 44% relative improvement over the previous best model.
- *Planning:* We demonstrate that V-JEPA 2-AC, obtained by post-training V-JEPA 2 with only 62 hours of unlabeled robot manipulation data from the popular Droid dataset, can be deployed in new environments to solve prehensile manipulation tasks using planning with given subgoals. Without training on any additional data from robots in our labs, and without any task-specific training or reward, the model successfully handles prehensile manipulation tasks, such as Grasp and Pick-and-Place with novel objects and in new environments.

The remainder of this paper is organized as follows. Section 2 describes the V-JEPA 2 pretraining procedure, including the key ingredients enabling scaling beyond the original V-JEPA recipe of Bardes et al. (2024). Section 3 then introduces our approach to training a task-agnostic action-conditioned world model, V-JEPA 2-AC, leveraging the pretrained V-JEPA 2 model. Section 4 demonstrates using V-JEPA 2-AC for robot control via model-based planning. Because V-JEPA 2-AC models world dynamics in a learned representation space, its capabilities fundamentally depend on the information captured in the V-JEPA 2 representation space, and so we further explore the performance of V-JEPA 2 for video understanding in Section 5 and prediction tasks in Section 6. Finally, in Section 7 we show that V-JEPA 2 can be aligned with a language model for video question answering. Section 8 discusses related work, and we conclude in Section 9.

V-JEPA 2 编码器扩展至 10 亿参数，并使用超过 100 万小时视频。实验确认，扩大视频自监督预训练可增强编码器的视觉理解能力，包括广泛的运动与外观识别；通过探测器评估，以及与语言模型对齐后的视频问答验证这些能力（Krojer et al., 2024; Pătrăucean et al., 2023; Liu et al., 2024c; Cai et al., 2024; Shangguan et al., 2024）。

互联网规模视频预训练后，利用第一阶段表征，在少量交互数据上训练动作条件世界模型 V-JEPA 2-AC。该模型为 3 亿参数 Transformer，采用块因果注意力，以动作与此前状态为条件，自回归预测下一帧表征。仅使用 Droid 不足 62 小时的无标注交互数据（Khazatsky et al., 2024），即可训练潜在世界模型，使其根据子目标在 Franka 机械臂上规划动作，仅凭单目 RGB 相机，在新环境中零样本执行抓持操作。

总之，从视频学习的联合嵌入预测架构，可利用互联网规模视频与少量交互数据，构建支持物理世界理解、未来状态预测及新情境有效规划的世界模型。具体而言：

- 理解——基于探测器的分类：扩大视频自监督预训练，得到适用于多任务的表征。V-JEPA 2 擅长编码细粒度运动信息，在 SSv2 等运动理解任务上表现出色，注意力探测器 top-1 达到 77.3。
- 理解——视频问答：V-JEPA 2 编码器可用于训练多模态大语言模型，处理视频问答。在需要物理世界理解和时间推理的多项基准上，80 亿参数语言模型级别达到最先进性能，包括 MVP（成对准确率 44.5）、PerceptionTest（测试准确率 84.0）、TempCompass（多选准确率 76.9）、TemporalBench（短问答多重二元准确率 36.7）和 TOMATO（准确率 40.3）。尤其表明，未经语言监督预训练的视频编码器也能与语言模型对齐并达到最先进性能，与传统认识不同（Yuan et al., 2025; Wang et al., 2024b）。
- 预测：大规模视频自监督预训练增强预测能力。V-JEPA 2 使用注意力探测器，在 Epic-Kitchens-100 人类动作预判上达到最先进水平，recall@5 为 39.7，相对此前最佳模型提高 44%。
- 规划：仅用 Droid 中 62 小时无标注机器人操作数据，对 V-JEPA 2 后训练得到 V-JEPA 2-AC，即可在新环境中根据给定子目标规划抓持操作。无需使用实验室机器人额外数据训练，无需任务专用训练或奖励，模型即可针对新物体和新环境，成功执行抓取、抓取放置等任务。

本文其余部分安排如下。第 2 节介绍 V-JEPA 2 的预训练过程，包括使其能超越 Bardes et al.（2024）原始 V-JEPA 方案扩展规模的关键要素。第 3 节介绍利用预训练 V-JEPA 2 模型，训练与任务无关的动作条件世界模型 V-JEPA 2-AC 的方法。第 4 节展示如何通过基于模型的规划，将 V-JEPA 2-AC 用于机器人控制。由于 V-JEPA 2-AC 在学习到的表征空间中建模世界动态，其能力根本上取决于 V-JEPA 2 表征空间所捕获的信息，因此第 5 节进一步研究 V-JEPA 2 的视频理解性能，第 6 节研究预测任务。最后，第 7 节展示将 V-JEPA 2 与语言模型对齐，以进行视频问答。第 8 节讨论相关工作，第 9 节总结。

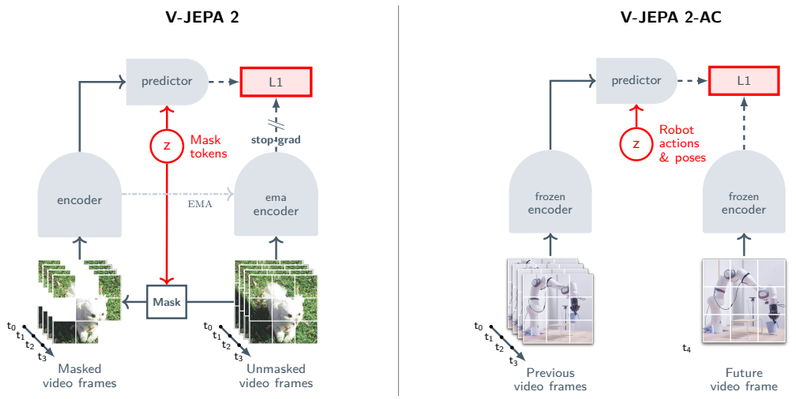


Figure 2 Multistage training. (Left) We first pretrain the V-JEPA 2 video encoder on internet-scale image and video data using a visual mask denoising objective (Bardes et al., 2024; Assran et al., 2023). A video clip is patchified into a sequence of tokens and a mask is applied by dropping a subset of the tokens. The encoder then processes the masked video sequence and outputs an embedding vector for each input token. Next, the outputs of the encoder are concatenated with a set of learnable mask tokens that specify the position of the masked patches, and subsequently processed by the predictor. The outputs of the predictor are then regressed to the prediction targets using an L1 loss. The prediction targets are computed by an ema-encoder, the weights of which are defined as an exponential moving average of the encoder weights. **(Right)** After pretraining, we freeze the video encoder and learn a new action-conditioned predictor, V-JEPA 2-AC, on top of the learned representation. We leverage an autoregressive feature prediction objective that involves predicting the representations of future video frames conditioned on past video frames, actions, and end-effector states. Our action-conditioned predictor uses a block-causal attention pattern such that each patch feature at a given time step can attend to the patch features, actions, and end-effector states from current and previous time steps.

2 V-JEPA 2: Scaling Self-Supervised Video Pretraining

We pretrain V-JEPA 2 on a visual dataset that includes over 1 million hours of video. The self-supervised training task is based on mask denoising in representation space and builds upon the V-JEPA framework (Bardes et al., 2024). In this paper, we extend the V-JEPA framework by exploring larger-scale models, increasing the size of the pretraining data, and introducing a spatial and temporal progressive resolution training strategy that enables us to efficiently pretrain models beyond short 16-frame video clips.

2.1 Methodology

Mask-Denoising in Representation Space. The V-JEPA objective aims to predict the learned representation of a video y from a view x of that video that has been masked, i.e., from which patches have been randomly dropped (Figure 2, left). The task meta-architecture consists of an encoder, $E_\theta(\cdot)$, which extracts video representations, and a predictor, $P_\phi(\cdot)$, which predicts the representation of masked video parts. The encoder and predictor are trained simultaneously using the objective,

$$\text{minimize}_{\theta,\phi,\Delta_y} \quad \|P_\phi(\Delta_y, E_\theta(x)) - \text{sg}(E_{\overline{\theta}}(y))\|_1, \tag{1}$$

where Δ_y is a learnable mask token that indicates the locations of the dropped patches. The loss uses a stop-gradient operation, $\text{sg}(\cdot)$, and an exponential moving average, $\overline{\theta}$, of the weights θ of the encoder network to prevent representation collapse. The loss is applied only to the predictions of the masked patches.

Architecture. The encoder, $E_\theta(\cdot)$, and predictor, $P_\phi(\cdot)$, are each parameterized as a vision transformer (Dosovitskiy et al., 2020) (or ViT). To encode relative position information in the vision transformer, we leverage RoPE (Rotary Position Embedding) instead of the absolute sincos position embedding used in Bardes et al.

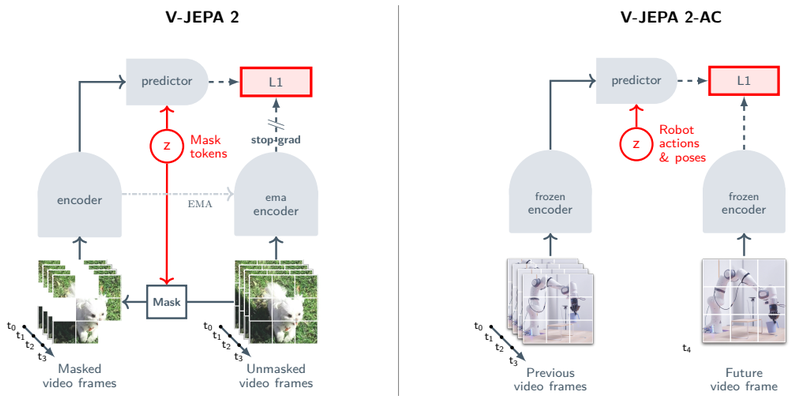


图 2 多阶段训练。（左）首先采用视觉掩码去噪目标（Bardes et al., 2024; Assran et al., 2023），在互联网规模的图像与视频数据上预训练 V-JEPA 2 视频编码器。将视频片段划分为图块词元序列，丢弃部分词元以施加掩码。编码器处理带掩码的视频序列，并为每个输入词元输出一个嵌入向量。随后，将编码器输出与一组指定被遮挡图块位置的可学习掩码词元拼接，再由预测器处理。预测器输出通过 L1 损失回归到预测目标。目标由 EMA 编码器计算，其权重定义为编码器权重的指数移动平均。（右）预训练后，冻结视频编码器，在所学表征上训练新的动作条件预测器 V-JEPA 2-AC。采用自回归特征预测目标，以过去的视频帧、动作和末端执行器状态为条件，预测未来视频帧的表征。动作条件预测器采用块因果注意力模式，使给定时间步的每个图块特征都能关注当前和此前时间步的图块特征、动作及末端执行器状态。

2 V-JEPA 2：扩大视频自监督预训练

V-JEPA 2 在包含超过 100 万小时视频的视觉数据集上预训练。自监督任务基于表征空间中的掩码去噪，建立在 V-JEPA 框架（Bardes et al., 2024）上。本文通过探索更大模型、增加预训练数据，以及引入时空渐进分辨率训练策略，扩展 V-JEPA，使预训练能高效超越 16 帧短片段。

2.1 方法

表征空间中的掩码去噪。V-JEPA 目标根据视频的遮挡视图 x，预测视频 y 的所学表征；遮挡指随机丢弃图块（图 2 左）。整体架构包括提取视频表征的编码器 $E_\theta(\cdot)$，以及预测遮挡部分表征的预测器 $P_\phi(\cdot)$。两者同时通过以下目标训练：

$$\text{minimize}_{\theta,\phi,\Delta_y} \quad \|P_\phi(\Delta_y, E_\theta(x)) - \text{sg}(E_{\overline{\theta}}(y))\|_1, \tag{1}$$

其中 Δ_y 是指示被丢弃图块位置的可学习掩码词元。损失使用停止梯度操作 $\text{sg}(\cdot)$ 和编码器权重 θ 的指数移动平均 θ，防止表征坍塌。损失仅作用于被遮挡图块的预测。

架构。编码器 $E_\theta(\cdot)$ 和预测器 $P_\phi(\cdot)$ 均参数化为视觉 Transformer（ViT）（Dosovitskiy et al., 2020）。为编码相对位置，采用旋转位置嵌入 RoPE，替代 Bardes et al. 所用的绝对正弦余弦位置嵌入，

(2024). We use a 3D extension of traditional 1D-RoPE (Su et al., 2024) by partitioning the feature dimension into three approximately equal segments (for the temporal, height, and width axes) and applying the 1D rotations separately to the segment for each axis. We found that using 3D-RoPE instead of absolute sincos position embeddings (Vaswani et al., 2017) helps stabilize training for the largest models. To process a video with our transformer encoder, we first patchify it as a sequence of tubelets of size $2 \times 16 \times 16$ ($T \times H \times W$) and employ the same multiblock masking strategy as in Bardes et al. (2024).

Key Scaling Ingredients. In this section we introduce and study four additional key ingredients which enable scaling the V-JEPA pre-training principle to obtain our V-JEPA 2 model.

1. *Data scaling:* We increase the dataset size from 2 million to 22 million videos by leveraging and curating additional data sources.
2. *Model scaling:* We scale the encoder architecture from 300 million to over 1 billion parameters, going from a ViT-L to a ViT-g (Zhai et al., 2022).
3. *Longer training:* Adopting a warmup-constant-decay learning rate schedule simplifies hyperparameter tuning and enables us to extend training from 90 thousand up to 252 thousand iterations, effectively leveraging the additional data.
4. *Higher resolution:* We leverage the warmup-constant-decay schedule to efficiently scale to higher resolution video and longer video clips by training on shorter, lower-resolution clips during the warmup and constant phases, and then increasing resolution and/or clip-length during the final decay phase.

The remainder of this section describes each of these ingredients in further detail and also quantifies the impact of each ingredient using the evaluation protocol described next.

Evaluation Protocol. Our goal with model pretraining is to infuse general visual understanding into our encoder. We therefore evaluate our model and data design choices by assessing the quality of the model's learned representation on a set of six motion and appearance classification tasks: Something-Something v2 (Goyal et al., 2017), Diving-48 (Li et al., 2018), Jester (Materzynska et al., 2019), Kinetics (Kay et al., 2017), COIN (Tang et al., 2019), and ImageNet (Deng et al., 2009). We use a frozen evaluation protocol: we freeze the encoder weights and train a task-specific 4-layers attentive probe on its representation to output a predicted class. In this section, we focus mainly on the average accuracy across the six understanding tasks. Refer to Section 5 for additional details about the tasks, evaluation protocol, and results.

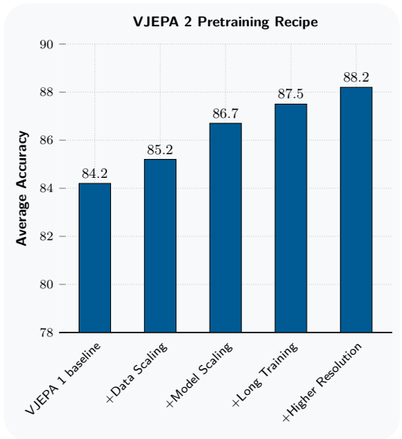


Figure 3 Scaling Ingredients. The effects of scaling interventions on average accuracy across 6 image and video classification tasks (SSv2, Diving-48, Jester, Kinetics, COIN, ImageNet) using a ViT-L/16 model as baseline.

2.2 Scaling Self-Supervised Video Learning

We first present a summary of the key findings of our scaling analysis, where we investigate the impact of the four key ingredients on downstream task average performance. Figure 3 illustrates the effects of these scaling interventions on average accuracy across 6 classification tasks, using a ViT-L/16 model pretrained on 2 million videos with the V-JEPA objective as our baseline. Increasing the dataset from 2 million to 22 million videos (VM22M) yields a 1.0-point improvement. Scaling the model from 300 million to 1 billion parameters (ViT-g/16) provides an additional 1.5-point gain. Extending training from 90K to 252K iterations contributes another 0.8-point improvement. Finally, enhancing both spatial resolution ($256 \rightarrow 384$) and temporal duration ($16 \rightarrow 64$ frames), during both pretraining and evaluation, boosts performance to 88.2%, representing a

（2024）。我们将传统一维 RoPE（Su et al., 2024）扩展到三维：把特征维度划为近似等大的三段，分别对应时间、高度和宽度，在各段独立施加一维旋转。对最大模型，三维 RoPE 相比绝对正弦余弦位置嵌入（Vaswani et al., 2017）有助于稳定训练。使用 Transformer 编码器处理视频时，先划分成大小为 $2 \times 16 \times 16$ ($T \times H \times W$) 的时空小块序列，并采用与 Bardes et al.（2024）相同的多块遮挡策略。

关键规模扩展要素。本节介绍并研究四项新增关键要素，使 V-JEPA 预训练原则能扩展为 V-JEPA 2。

1. 数据规模：利用并筛选额外数据源，将视频从 200 万增加至 2200 万。2. 模型规模：编码器从 3 亿扩展至超过 10 亿参数，从 ViT-L 变为 ViT-g（Zhai et al., 2022）。3. 更长训练：采用预热–恒定–衰减学习率，简化调参，将训练从 9 万延至 25.2 万次迭代，有效利用更多数据。4. 更高分辨率：利用该调度高效扩展视频分辨率与长度，预热和恒定阶段训练较短、较低分辨率片段，最终衰减阶段增加分辨率和／或片段长度。

本节其余部分详细介绍这些要素，并使用下述评估方案量化各项影响。

评估方案。模型预训练的目标，是使编码器具备通用视觉理解能力。因此，我们通过六项运动与外观分类任务，评估所学表征的质量，从而评价模型和数据的设计选择：Something-Something v2（Goyal et al., 2017）、Diving-48（Li et al., 2018）、Jester（Materzynska et al., 2019）、Kinetics（Kay et al., 2017）、COIN（Tang et al., 2019）和 ImageNet（Deng et al., 2009）。我们采用冻结评估方案：冻结编码器权重，在其表征上训练任务专用的四层注意力探测器，输出预测类别。本节主要关注六项理解任务的平均准确率。任务、评估方案和结果的更多细节见第 5 节。

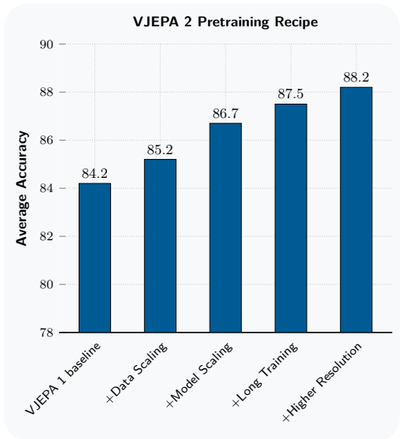


图 3 规模扩展要素。以 ViT-L/16 为基线，展示各扩展措施对六项图像与视频分类任务（SSv2、Diving-48、Jester、Kinetics、COIN、ImageNet）平均准确率的影响。

2.2 扩大视频自监督学习

首先汇总规模分析的关键发现，考察四项要素对平均下游性能的影响。图 3 展示它们对六项分类任务平均准确率的影响，基线为在 200 万视频上以 V-JEPA 目标预训练的 ViT-L/16。数据从 200 万增至 2200 万（VM22M），提升 1.0 个百分点。模型从 3 亿增至 10 亿参数，

（ViT-g/16）额外提升 1.5 个百分点。训练从 90K 延至 252K 次迭代再提升 0.8。最后，预训练和评估均增加空间分辨率（$256 \rightarrow 384$）和时间长度（$16 \rightarrow 64$ 帧），使性能达 88.2%，相当于

Table 1 VideoMix22M (VM22M) Pretraining Dataset. To build our observation pretraining dataset, we combined four different video sources and one image dataset. We use a source-specific sampling probability during training and apply retrieval-based curation on YT1B to reduce noisy content (e.g., cartoon- or clipart-style).

Source	Samples	Type	Total Hours	Apply Curation	Weight
SSv2 (Goyal et al., 2017)	168K	EgoVideo	168	No	0.056
Kinetics (Carreira et al., 2019)	733K	ExoVideo	614	No	0.188
Howto100M (Miech et al., 2019)	1.1M	ExoVideo	134K	No	0.318
YT-Temporal-1B (Zellers et al., 2022)	19M	ExoVideo	1.6M	Yes	0.188
ImageNet (Deng et al., 2009)	1M	Images	n/a	No	0.250

cumulative 4.0-point improvement over the ViT-L/16 baseline. Each individual change provides a positive impact, confirming the potential of scaling in video self-supervised learning (SSL).

2.3 Pretraining Dataset

Next, we describe the sources of videos and images that make up our pretraining dataset, and our approach to curating the dataset.

Scaling Dataset Size. We construct a large-scale video dataset by combining publicly available data sources. Using publicly-available sources in this work enables other researchers to reproduce these results. The overall dataset includes ego-centric videos from the Something-Something v2 dataset (SSv2) introduced in Goyal et al. (2017), exo-centric action videos from the Kinetics 400, 600, and 700 datasets (Kay et al., 2017; Carreira et al., 2018, 2019), YouTube tutorial videos from HowTo100M (Miech et al., 2019), and general YouTube videos from YT-Temporal-1B (Zellers et al., 2022), which we refer to as YT1B. We also include images from the ImageNet dataset (Deng et al., 2009) to increase the visual coverage of the pretraining data. To enable joint image and video pretraining, we duplicate an image temporally and treat it as a 16-frame video where all frames are identical. During training, we sample from each data source with a weighting coefficient that we determined empirically via manual tuning. The resulting dataset, which we refer to as VideoMix22M (or VM22M), consists of 22 million samples. Table 1 lists these data sources and their weights.

Figure 4 (Left) compares the performance of a ViT-L/16 pretrained on VM22M with a similar model trained on the smaller (2 million) VideoMix2M dataset from Bardes et al. (2024). Training on VM22M leads to a +1 point improvement on average performance on visual understanding tasks, compared to VM2M. Performance improvement is more prominent on appearance-based tasks such as Kinetics-400, COIN, and ImageNet, showing the importance of increasing visual coverage for those tasks.

Data Curation. YT1B is a large video dataset, consisting of 1.4 million video-hours, with no curation and minimal filtering compared to smaller video datasets (like Kinetics and Something-Something v2). Because uncurated and unbalanced data can hinder model performance (Assran et al., 2022; Oquab et al., 2023), we filter YT1B by adapting an existing retrieval-based curation pipeline to handle videos. Specifically, we extract scenes from YT1B videos, compute an embedding vector for each scene, and then use a cluster-based retrieval process (Oquab et al., 2023) to select video scenes according to a target distribution, which is composed of the Kinetics, Something-Something v2, COIN and EpicKitchen training datasets. We describe the details of the dataset construction procedure in Appendix A.2. Similar to Oquab et al. (2023), we ensure that none of the videos from the target validation sets are contained in the initial, uncurated data pool.

In Figure 4 (Right), we compare the average performance on visual understanding evaluations between a ViT-L model pretrained on uncurated YT-1B data and a comparable model trained on our Curated-YT-1B dataset. Training with the curated dataset yields a +1.4 point average performance improvement over the uncurated baseline. Notably, the Curated-YT-1B-trained model achieves competitive performance relative to the full VM22M dataset at the ViT-L scale. However, larger-scale models benefit more from VM22M training (see Appendix A.2), suggesting that combining Curated-YT-1B with other data sources enhances scalability.

表 1 VideoMix22M（VM22M）预训练数据集。整合四个视频源和一个图像数据集，训练时使用源专用采样概率。对 YT1B 采用检索式筛选，减少卡通、剪贴画风格等含噪内容。

来源	样本数	Type	总小时数	应用筛选	权重
SSv2 (Goyal et al., 2017)	168K	第一视角视频	168	No	0.056
Kinetics (Carreira et al., 2019)	733K	第三视角视频	614	No	0.188
Howto100M (Miech et al., 2019)	1.1M	第三视角视频	134K	No	0.318
YT-Temporal-1B (Zellers et al., 2022)	19M	第三视角视频	1.6M	Yes	0.188
ImageNet (Deng et al., 2009)	1M	图像	n/a	No	0.250

相较 ViT-L/16 基线累计提升 4.0 个百分点。每项改动均有正面影响，验证视频自监督学习规模扩展的潜力。

2.3 预训练数据集

接下来介绍　　组成预训练数据集的视频与图像来源，以及我们的方法
筛选数据集　　et.

扩大数据集规模。通过整合公开数据源，构建大规模视频数据集。本研究采用公开数据，使其他研究者能够复现结果。整体数据集包括 Goyal et al.（2017）提出的 Something-Something v2（SSv2）第一视角视频、Kinetics 400、600、700 的第三视角动作视频（Kay et al., 2017; Carreira et al., 2018, 2019）、HowTo100M 的 YouTube 教程视频（Miech et al., 2019），以及 YT-Temporal-1B 的一般 YouTube 视频（Zellers et al., 2022），后者简称 YT1B。还加入 ImageNet 图像（Deng et al., 2009），扩大预训练数据的视觉覆盖。为联合预训练图像和视频，将图像沿时间维复制，视为所有帧相同的 16 帧视频。训练时，按经人工调参确定的权重系数从各数据源采样。所得数据集称为 VideoMix22M（VM22M），包含 2200 万个样本。各数据源及其权重见表 1。

图 4 左图比较 VM22M 预训练 ViT-L/16 与 Bardes et al.（2024）较小的 VideoMix2M（200 万）数据训练模型。VM22M 使视觉理解平均性能提高 +1 个百分点；提升在 Kinetics-400、COIN 和 ImageNet 等外观任务更明显，说明扩大视觉覆盖对这些任务的重要性。

数据筛选。YT1B 包含 140 万小时视频，相比 Kinetics、Something-Something v2 等较小数据集，没有经过筛选，过滤也很少。未筛选且不平衡的数据可能影响性能（Assran et al., 2022; Oquab et al., 2023），因此将已有的检索式筛选流程适配到视频，以过滤 YT1B。具体而言，从 YT1B 视频提取场景，为每个场景计算嵌入向量，再采用基于聚类的检索过程（Oquab et al., 2023），按由 Kinetics、Something-Something v2、COIN、EpicKitchen 训练集构成的目标分布选择场景。数据集构建详见附录 A.2。与 Oquab et al.（2023）类似，确保目标验证集视频均不包含在最初未筛选的数据池中。

图 4 右图比较未筛选 YT-1B 与 Curated-YT-1B 预训练 ViT-L 的视觉理解平均性能。筛选数据相比未筛选基线平均提升 +1.4 个百分点。值得关注的是，在 ViT-L 规模，筛选 YT-1B 的性能已可与完整 VM22M 竞争。但更大模型更受益于 VM22M（见附录 A.2），说明结合筛选 YT-1B 与其他数据源有助于扩展规模。

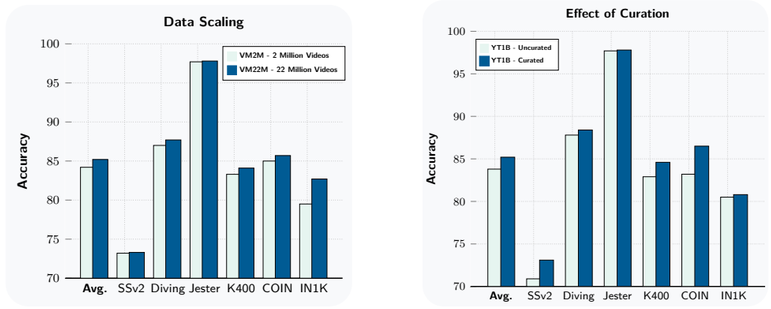


Figure 4 Data Scaling & Curation. We train and compare models on different data-mixes. Models are ViT-L/16 trained for 90K iterations using a cosine learning schedule following Bardes et al. (2024). **(Left)** We compare the performance of a ViT-L/16 model pretrained on the VM2M dataset and our VM22M dataset. Training on the VM22M dataset leads to a +1 point improvement in average performance. Performance improvement is more pronounced on appearance-based tasks such as Kinetics-400, COIN, and ImageNet **(Right)** We compare the performance of a ViT-L/16 model pretrained on YT1B and a model pretrained on our Curated-YT1B dataset, which leverages our cluster-based curation. Training on the curated dataset leads to a +1.4 point improvement on average performances, showing the effectiveness of data-curation.

2.4 Pretraining Recipe

Scaling Model Size. To explore the scaling behavior of our model, we trained a family of encoder models with parameter counts ranging from 300 million (ViT-L) to 1 billion (ViT-g) parameters. All encoder architecture details are provided in Table 12 in the appendix. Note that each encoder uses the same predictor architecture, similar to a ViT-small. We report the average performance of these encoders on visual understanding tasks in Figure 5 (Left). Scaling the model size from 300 million (ViT-L) to 1 billion (ViT-g) parameters yields a +1.5 points average performance improvement. Both motion and appearance understanding tasks benefit from scaling, with SSv2 improving by +1.6 points and Kinetics by +1.5 points (cf.Table 4). These results confirm that self-supervised video pretraining effectively leverages larger model capacities, up to the 1B-parameter ViT-g.

Training Schedule. V-JEPA 2 model training employs a warmup-constant learning rate schedule followed by a cooldown phase (Zhai et al., 2022; Hägele et al., 2024). Similarly to Hägele et al. (2024), we found that this schedule performs comparably to a half-cosine schedule (Loshchilov and Hutter, 2016); it also makes exploring long training runs more cost-effective, since multiple cooldown runs can be started from different checkpoints of the constant phase. We simplified the recipe from Bardes et al. (2024) by maintaining fixed teacher EMA and weight decay coefficients instead of using ramp-up schedule, as these variations showed minimal impact on downstream understanding tasks. Figure 3 shows that extending the training schedule from 90K to 252K iterations yields a +0.8 average performance improvement with ViT-g models, validating the benefits of extended training durations. This schedule also facilitates progressive training by incrementally increasing video resolution during the cooldown phase.

Efficient Progressive-Resolution Training. While most previous video encoders focus on short clips of 16 frames (roughly seconds) (Bardes et al., 2024; Wang et al., 2024b, 2023), we explore training with longer clips of up to 64 frames (16 seconds) at higher spatial resolutions. However, training time increases dramatically with longer durations and higher resolutions — training our ViT-g model on $64 \times 384 \times 384$ inputs would require roughly 60 GPU-years (see Figure 5, Middle). To reduce this, we adopt a progressive resolution strategy (Touvron et al., 2019; Oquab et al., 2023) that boosts training efficiency while maintaining downstream performance. Our training process begins with a warmup phase where we train on 16-frame, 256×256-resolution videos with linear learning rate warmup over 12K iterations, followed by a main training phase with a constant learning rate for 228K iterations. Then, during the cooldown phase, we increase video duration and resolution while linearly decaying the learning rate over 12K iterations. Hence the additional

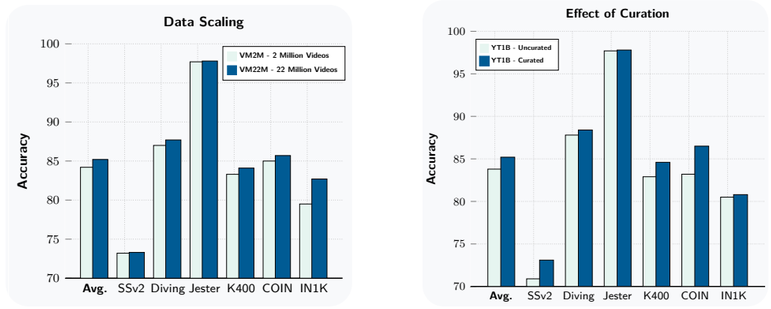


图 4 数据规模扩展与筛选。比较不同混合数据训练的模型，均为 ViT-L/16，遵循 Bardes et al.（2024），以余弦学习率训练 90K 次迭代。（左）比较 VM2M 与 VM22M 预训练：VM22M 使平均性能提高 +1 个百分点，提升在 Kinetics-400、COIN 和 ImageNet 等外观任务上更明显。（右）比较 YT1B 与采用聚类筛选的 Curated-YT1B：筛选数据使平均性能提高 +1.4 个百分点，说明数据筛选有效。

2.4 预训练方案

扩大模型规模。为研究规模扩展行为，训练从 3 亿参数（ViT-L）到 10 亿参数（ViT-g）的编码器系列，架构详见附录表 12。各编码器使用相同、类似 ViT-small 的预测器。图 5 左图报告视觉理解平均性能。参数从 3 亿增至 10 亿，平均性能提升 +1.5 个百分点。运动与外观理解均受益，SSv2 提升 +1.6，Kinetics 提升 +1.5（参见表 4）。这说明视频自监督预训练能有效利用更大模型容量，至少直至 10 亿参数 ViT-g。

训练调度。V-JEPA 2 采用预热–恒定学习率调度，随后进入收尾衰减阶段（Zhai et al., 2022; Hägele et al., 2024）。与 Hägele et al.（2024）类似，我们发现其性能与半余弦调度相当（Loshchilov and Hutter, 2016），且探索长训练更具成本效益，因为可从恒定阶段的不同检查点启动多个收尾训练。相较 Bardes et al.（2024），固定教师 EMA 和权重衰减系数，替代逐渐增加的调度，从而简化方案；这些变化对下游理解影响很小。图 3 表明，ViT-g 训练从 90K 延长至 252K 次迭代，平均性能提升 +0.8，验证延长训练的益处。该调度也便于在收尾阶段逐步增加视频分辨率。

高效的渐进分辨率训练。此前多数视频编码器关注 16 帧的短片段（约数秒）（Bardes et al., 2024; Wang et al., 2024b, 2023），我们探索以更高空间分辨率训练最长 64 帧（16 秒）的片段。然而，更长视频和更高分辨率会大幅增加训练时间：以 $64 \times 384 \times 384$ 输入训练 ViT-g 约需 60 GPU 年（见图 5 中图）。为降低成本，采用渐进分辨率策略（Touvron et al., 2019; Oquab et al., 2023），提高效率并保持下游性能。首先在 16 帧、256×256 分辨率视频上预热，前 12K 次迭代线性增加学习率；随后以恒定学习率主要训练 228K 次迭代。最后在收尾阶段增加视频长度和分辨率，并在 12K 次迭代中线性衰减学习率。因此，额外的

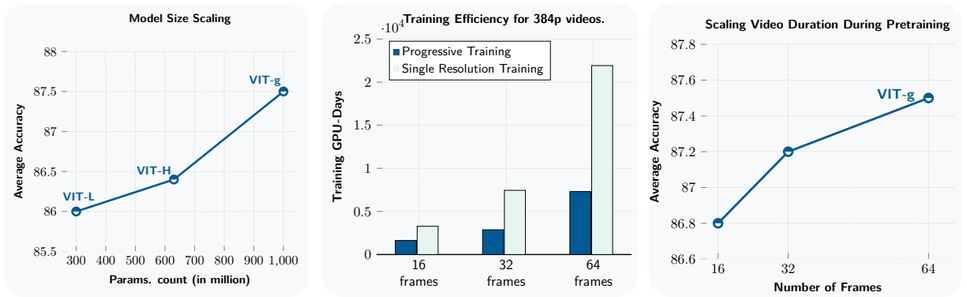


Figure 5 Model Scaling. We explore the impact of scaling model size and input video resolution. All models are trained on the VideoMix22M pretraining dataset. **(Left)** Average performance across six understanding tasks as a function of model scale. Models are trained with a constant learning rate until performance plateaus on downstream tasks. We then cool down the model using 64 frames at 256×256 resolution and report post-cooldown performance. Scaling the model size from 300M to 1B parameters yields a +1.7 point average improvement. **(Middle)** Training times (GPU-days) for ViT-g on A100 GPUs when training videos at 384×384 resolution with different numbers of frames per clip. We compare progressive resolution training (252K iterations at 16 frames / 256×256 resolution, followed by 12K cooldown iterations at 384×384 resolution) to the projected time for full-resolution training. Progressive training provides up to 8× speedup, significantly reducing the pretraining compute requirement. **(Right)** Effect of inscreasing video duration at cooldown on downstream performance for ViT-g. Even when only using 16-frame clips during inference/evaluation, increasing video duration during the cooldown phase of training improves average task performance by +0.7 points.

computational overhead associated with training on longer-duration, higher-resolution videos is only incurred during the final cooldown phase. This approach enables efficient high-resolution training: as shown in Figure 5 (Middle), we achieve an 8.4× reduction in GPU time for a model that can ingest 64-frame, 384×384 resolution inputs, compared to directly training such a model from scratch at full resolution throughout all phases of training. Furthermore, we still observe the benefits of a model that can process longer-duration and higher-resolution inputs as discussed next.

Scaling temporal and spatial video resolution. Figure 5 examines how input video resolution affects downstream task performance. When increasing clip duration from 16 to 64 frames during pretraining while maintaining a fixed 16-frame evaluation duration, we observe a +0.7 percentage point average performance improvement (Figure 5, Right). Additionally, we see that increasing the video duration and resolution during evaluation leads to a significant improvement across the tasks (refer to Table 4 and Appendix A.4.2). These results demonstrate that video self-supervised pretraining benefits from increased temporal resolution during both training and evaluation. Although we experimented with scaling to even longer video clips (128 and 256 frames), we did not observe any further improvement beyond 64 frames on this set of understanding tasks.

3 V-JEPA 2-AC: Learning an Action-Conditioned World Model

After pre-training, the V-JEPA 2 model can make predictions about missing part in videos. However, these predictions do not directly take into account the causal effect of actions that an agent might take. In the next stage of training, described in this section, we focus on making the model useful for planning by leveraging a small amount of interaction data. To that end, we learn a frame-causal action-conditioned predictor on top of the frozen V-JEPA 2 video encoder (Figure 2, right). We train our model on data from the Droid dataset (Khazatsky et al., 2024) consisting of data from experiments with a table-top Franka Panda robot arm collected through teleoperation. We refer to the resulting action-conditioned model as V-JEPA 2-AC, and in Section 4 we show that V-JEPA 2-AC can be used within a model-predictive control planning loop to plan actions in new environments.

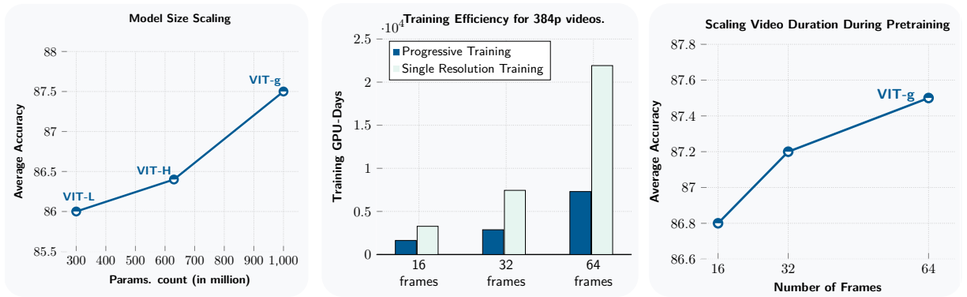


图 5 模型规模扩展。研究扩大模型和输入视频分辨率的影响，所有模型均在 VideoMix22M 上预训练。（左）六项理解任务的平均性能随模型规模变化。以恒定学习率训练至下游性能达到平台，再以 64 帧、 256×256 分辨率进行收尾训练，并报告其后性能。参数从 3 亿增加至 10 亿，平均提升 +1.7 个百分点。（中）ViT-g 在 A100 GPU 上，以分辨率 384×384 、不同片段帧数训练所需的 GPU 天数。比较渐进分辨率训练（以 16 帧／ 256×256 分辨率训练 252K 次迭代，再以 384×384 分辨率收尾训练 12K 次）与全分辨率训练的预计耗时。渐进训练最高加速 8×，显著降低预训练计算需求。（右）收尾阶段增加视频长度对 ViT-g 下游性能的影响。即使推理与评估只用 16 帧片段，延长收尾训练视频仍使平均任务性能提升 +0.7 个百分点。

训练更长、更高分辨率视频的额外计算，仅在最后收尾阶段发生。图 5 中图说明，与所有阶段从头以全分辨率训练相比，可接收 64 帧、分辨率 384×384 输入的模型，其 GPU 耗时减少 8.4×。同时仍能获得处理更长、更高分辨率输入的益处，下文讨论。

扩大视频时间与空间分辨率。图 5 考察输入分辨率对下游性能的影响。预训练片段从 16 延至 64 帧、评估仍固定 16 帧时，平均提升 +0.7 个百分点（图 5 右）。评估时增加视频长度和分辨率也显著改善各任务（见表 4 和附录 A.4.2），说明视频自监督预训练在训练与评估阶段均受益于更高时间分辨率。虽然测试了 128 和 256 帧的更长片段，但在这些理解任务中，超过 64 帧未进一步改善。

3 V-JEPA 2-AC：学习动作条件世界模型

预训练后，V-JEPA 2 能预测视频缺失部分，但这些预测未直接考虑智能体动作的因果影响。本节介绍下一训练阶段：利用少量交互数据，使模型能用于规划。在冻结 V-JEPA 2 视频编码器之上，学习帧因果的动作条件预测器（图 2 右）。采用 Droid 数据集（Khazatsky et al., 2024），其数据来自通过遥操作采集的桌面 Franka Panda 机械臂实验。所得动作条件模型称为 V-JEPA 2-AC。第 4 节展示它如何在模型预测控制规划循环中，为新环境规划动作。

3.1 Action-Conditioned World Model Training

Our goal is to take the V-JEPA 2 model after pre-training and obtain a latent world model that can be used for control of an embodied agentic system via closed-loop model-predictive control. To achieve this, we train V-JEPA 2-AC, an autoregressive model that predicts representations of future video observations conditioned on control actions and proprioceptive observations.

In this section we describe a concrete instantiation of this framework for a tabletop arm with a fixed exocentric camera, and where control actions correspond to end-effector commands. The model is trained using approximately 62 hours of unlabeled video from the raw Droid dataset, which consists of short videos, typically 3–4 seconds long, of a 7-DoF Franka Emika Panda arm equipped with a two-finger gripper. Here, *unlabeled* video refers to the fact that we do not use additional meta-data indicating any reward, what type of task was being performed in each demonstration, or whether the demonstration was successful or not in completing the task being attempted. Rather, we only use the raw video and end-effector state signals from the dataset (each video in the dataset is accompanied by meta-data indicating the end-effector state in each frame — three dimensions for position, three for orientation, and one for the gripper state).

Model inputs. In each iteration of training we randomly sample a mini-batch of 4 second video clips from the Droid dataset, and, for simplicity, discard any videos shorter than 4 seconds, leaving us with a smaller subset of the dataset comprising under 62 hours of video. The video clips are sampled with resolution 256×256 and a frame-rate of 4 frames-per-second (fps), yielding 16 frame clips denoted by $(x_k)_{k\in[16]}$, where each x_k represents a single video frame. The robot's end-effector state in each observation is denoted by the sequence $(s_k)_{k\in[16]}$, where s_k is a real-valued 7D vector defined relative to the base of the robot. The first three dimensions of s_k encode the cartesian position of the end-effector, the next three dimensions encode its orientation in the form of extrinsic Euler angles, and the last dimension encodes the gripper state. We construct a sequence of actions $(a_k)_{k\in[15]}$ by computing the change in end-effector state between adjacent frames. Specifically, each action a_k is a real-valued 7-dimensional vector representing the change in end-effector state between frames k and $k+1$. We apply random-resize-crop augmentations to the sampled video clips with the aspect-ratio sampled in the range (0.75, 1.35).

Loss function. We use V-JEPA 2 encoder $E(\cdot)$ as an image encoder and encode each frame independently in a given clip to obtain a sequence of feature maps $(z_k)_{k\in[16]}$, where $z_k \coloneqq E(x_k) \in \mathbb{R}^{H\times W\times D}$ with $H \times W$ denoting the spatial resolution of the feature map, and D the embedding dimension. In practice, our feature maps are encoded using the ViT-g encoder and have the shape $16 \times 16 \times 1408$. Note that the encoder is kept frozen during this post-training phase. The sequence of feature maps, end-effector states, and actions are temporally interleaved as $(a_k, s_k, z_k)_{k\in[15]}$ and processed with the transformer predictor network $P_\phi(\cdot)$ to obtain a sequence of next state representation predictions $(\hat{z}_{k+1})_{k\in[15]}$. The scalar-valued teacher-forcing loss function is finally computed as

$$\mathcal{L}_{\text{teacher-forcing}}(\phi) \coloneqq \frac{1}{T}\sum_{k=1}^{T}\|\hat{z}_{k+1} - z_{k+1}\|_1 = \frac{1}{T}\sum_{k=1}^{T}\left\|P_\phi\left((a_t, s_t, E(x_t))_{t\le k}\right) - E(x_{k+1})\right\|_1, \tag{2}$$

with $T = 15$. We also compute a two-step rollout loss to improve the model's ability to perform autoregressive rollouts at inference time. For simplicity of exposition and with slight overloading of notation, let $P_\phi(\hat{a}_{1:T}; s_k, z_k) \in \mathbb{R}^{H\times W\times D}$ denote the final predicted state representation obtained by autoregressively running V-JEPA 2-AC with an action sequence $(\hat{a}_i)_{i\in[T]}$, starting from $(s_k,\, z_k)$. We can now denote the rollout loss as:

$$\mathcal{L}_{\text{rollout}}(\phi) \coloneqq \|P_\phi(a_{1:T}, s_1, z_1) - z_{T+1}\|_1. \tag{3}$$

In practice we use $T = 2$ for computing the rollout loss, such that we only differentiate the predictor through one recurrent step.

The overall training objective is thus given by

$$L(\phi) \coloneqq \mathcal{L}_{\text{teacher-forcing}}(\phi) + \mathcal{L}_{\text{rollout}}(\phi), \tag{4}$$

and is minimized with respect to the predictor weights ϕ. For illustrative purposes, the training procedure is depicted in Figure 6 with $T = 4$ for both the teacher forcing and rollout loss.

3.1 动作条件世界模型训练

目标是在预训练 V-JEPA 2 后得到潜在世界模型，用于通过闭环模型预测控制操控具身智能体。为此训练 V-JEPA 2-AC 自回归模型，以控制动作和本体感觉观测为条件，预测未来视频观测表征。

本节给出这一框架用于桌面机械臂的具体实例：相机固定为第三视角，控制动作对应末端执行器指令。使用 Droid 原始数据集约 62 小时的无标注视频训练；其中短视频通常长 3–4 秒，机器人为配备双指夹爪的七自由度 Franka Emika Panda 机械臂。无标注指不使用奖励、演示任务类型或任务是否成功等额外元数据，而仅使用原始视频和末端执行器状态信号。每段视频附有各帧末端状态：位置三维、朝向三维、夹爪状态一维。

模型输入。在每次训练迭代中，从 Droid 数据集随机采样一个小批量的 4 秒视频片段。为简化处理，丢弃所有不足 4 秒的视频，得到总时长不足 62 小时的较小数据子集。以分辨率 256×256、每秒 4 帧（fps）采样视频，得到 16 帧片段，记为 $(x_k)_{k\in[16]}$，每个 x_k 表示单帧视频。各观测中的机器人末端执行器状态记为序列 $(s_k)_{k\in[16]}$，其中 s_k 是相对于机器人基座定义的七维实值向量。s_k 的前三维编码末端执行器的笛卡尔位置，接下来三维以外禀欧拉角编码朝向，最后一维编码夹爪状态。通过计算相邻帧之间末端执行器状态的变化，构造动作序列 $(a_k)_{k\in[15]}$。具体而言，每个动作 a_k 是七维实值向量，表示帧 k 与 $k+1$ 之间末端执行器状态的变化。对采样的视频片段应用随机缩放裁剪增强，宽高比从 (0.75, 1.35) 中采样。

损失函数。将 V-JEPA 2 编码器 $E(\cdot)$ 用作图像编码器，独立编码视频片段的每一帧，得到特征图序列 $(z_k)_{k\in[16]}$，其中 $z_k \coloneqq E(x_k) \in \mathbb{R}^{H\times W\times D}$，$H \times W$ 表示特征图空间分辨率，D 为嵌入维度。实际使用 ViT-g 编码器，特征图形状为 $16 \times 16 \times 1408$。在这一后训练阶段，编码器始终冻结。将特征图、末端执行器状态和动作沿时间交错排列为 $(a_k, s_k, z_k)_{k\in[15]}$，通过 Transformer 预测器 $P_\phi(\cdot)$ 处理，得到下一状态表征的预测序列 $(\hat{z}_{k+1})_{k\in[15]}$。最终计算标量教师强制损失：

$$\mathcal{L}_{\text{teacher-forcing}}(\phi) \coloneqq \frac{1}{T}\sum_{k=1}^{T}\|\hat{z}_{k+1} - z_{k+1}\|_1 = \frac{1}{T}\sum_{k=1}^{T}\left\|P_\phi\left((a_t, s_t, E(x_t))_{t\le k}\right) - E(x_{k+1})\right\|_1, \tag{2}$$

其中 $T = 15$。另计算两步展开损失，改善推理时自回归展开能力。为简化说明，稍微复用记号：令 $P_\phi(\hat{a}_{1:T};\, s_k, z_k) \in \mathbb{R}^{H\times W\times D}$ 表示从 $(s_k,\, z_k)$ 出发，以动作序列 $(\hat{a}_i)_{i\in[T]}$ 自回归运行 V-JEPA 2-AC 后得到的最终预测状态表征。展开损失可记为：

$$\mathcal{L}_{\text{rollout}}(\phi) \coloneqq \|P_\phi(a_{1:T}, s_1, z_1) - z_{T+1}\|_1. \tag{3}$$

实际中计算展开损失时使用 $T = 2$，因此只让梯度经过预测器的一次递归步骤。

因此，总训练目标为

$$L(\phi) \coloneqq \mathcal{L}_{\text{teacher-forcing}}(\phi) + \mathcal{L}_{\text{rollout}}(\phi), \tag{4}$$

并针对预测器权重 ϕ 最小化。为说明，图 6 展示教师强制与展开损失均采用 $T = 4$ 的训练过程。



Figure 6 V-JEPA 2-AC training. V-JEPA 2-AC is trained in an autoregressive fashion, utilizing a teacher forcing loss and a rollout loss. (**Left**) In the teacher forcing loss, the predictor takes the encoding of the current frame representation as input and learns to predict the representation of the next timestep. (**Right**) The rollout loss involves feeding the predictor's output back as input, allowing the model to be trained to predict several timesteps ahead. By optimizing the sum of these two losses, V-JEPA 2-AC enhances its ability to accurately forecast the future by reducing error accumulation during rollouts.

Architecture. The predictor network $P_\phi(\cdot)$ is a ∼300M parameter transformer network with 24-layers, 16 heads, 1024 hidden dimension, and GELU activations. The action, end-effector state, and flattened feature maps input to the predictor are processed with separate learnable affine transformations to map them to the hidden dimension of the predictor. Similarly, the outputs of the last attention block of the predictor go through a learnable affine transformation to map them back to the embedding dimension of the encoder. We use our 3D-RoPE implementation to represent the spatiotemporal position of each video patch in the flattened feature map, while only applying the temporal rotary positional embeddings to the action and pose tokens. We use a block-causal attention pattern in the predictor so that each patch feature at a given time step can attend to the action, end-effector state, and other patch features from the same timestep, as well as those from previous time steps.

3.2 Inferring Actions by Planning

Energy minimization. Given an image of the goal state, we leverage V-JEPA 2-AC for downstream tasks by planning. Specifically, at each time step, we plan an action sequence for a fixed time horizon by minimizing a goal-conditioned energy function. We then execute the first action, observe the new state, and repeat the process. Let s_k denote the current end-effector state, and x_k and x_g denote the current observed frame and goal image, respectively, which are separately encoded with the video encoder to obtain the feature maps z_k and z_g. Given a planning horizon, T, we optimize a sequence of robot actions, $(a_i^\star)_{i\in[T]}$, by minimizing a goal-conditioned energy function,

$$\mathcal{E}(\hat{a}_{1:T};\ z_k, s_k, z_g) \coloneqq \|P(\hat{a}_{1:T}; s_k, z_k) - z_g\|_1, \tag{5}$$

such that $(a_i^\star)_{i\in[T]} \coloneqq \operatorname{argmin}_{\hat{a}_{1:T}} \mathcal{E}(\hat{a}_{1:T};\ z_k, s_k, z_g)$. As illustrated in Figure 7, the model infers an action sequence $(a_i^\star)_{i\in[T]}$ by selecting a trajectory that minimizes the L1 distance between the world model's imagined state representation T steps into the future and its goal representation. In practice, we minimize (5) in each



图 6 V-JEPA 2-AC 训练。采用自回归训练，同时使用教师强制损失与展开损失。（左）教师强制损失中，预测器输入当前帧的编码表征，学习预测下一时间步表征。（右）展开损失将预测器输出反馈为输入，训练其预测多个未来时间步。优化两种损失之和，减少展开误差累积，改善未来预测准确性。

架构。预测器网络 $P_\phi(\cdot)$ 是具有 ∼300 百万参数的 Transformer，包含 24 层、16 个注意力头、1024 隐藏维度和 GELU 激活。输入动作、末端执行器状态和展平的特征图，分别通过独立的可学习仿射变换映射至预测器隐藏维度。最后一个注意力块的输出也通过可学习仿射变换，映射回编码器嵌入维度。采用我们的三维 RoPE 实现表示展平特征图中每个视频图块的时空位置；动作与姿态词元仅应用时间旋转位置嵌入。预测器采用块因果注意力，使各时间步的图块特征能关注同一时间步的动作、末端状态和其他图块特征，以及此前时间步的信息。

3.2 通过规划推断动作

能量最小化。给定目标状态图像，通过规划将 V-JEPA 2-AC 用于下游任务。每个时间步固定时域，最小化目标条件能量函数，规划动作序列；执行首个动作、观察新状态，再重复。令 s_k 为当前末端状态，x_k 和 x_g 分别为当前观测帧与目标图像，经视频编码器独立编码得到特征图 z_k 和 z_g。给定规划时域 T，通过最小化目标条件能量函数，优化机器人动作序列 $(a_i^\star)_{i\in[T]}$:

$$\mathcal{E}(\hat{a}_{1:T};\ z_k, s_k, z_g) \coloneqq \|P(\hat{a}_{1:T}; s_k, z_k) - z_g\|_1, \tag{5}$$

使 $(a_i^\star)_{i\in[T]} \coloneqq \operatorname{argmin}_{\hat{a}_{1:T}} \mathcal{E}(\hat{a}_{1:T};\ z_k, s_k, z_g)$。如图 7，模型选择使想象的未来 T 步状态与目标表征之间 L1 距离最小的轨迹，推断动作序列 $(a_i^\star)_{i\in[T]}$。实际中，在每次

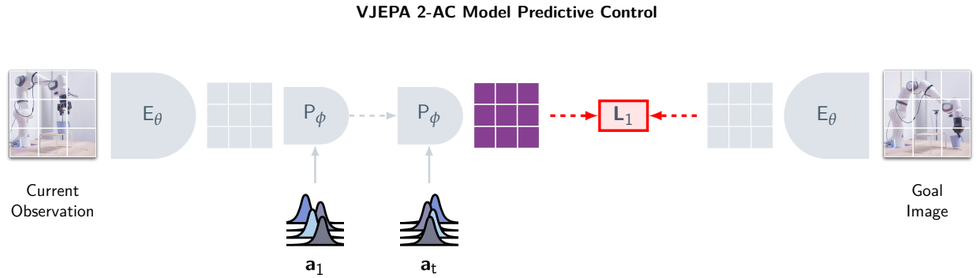


Figure 7 Planning. We plan an action sequence for a fixed time horizon T by minimizing the L1 distance between the world model's imagined state representation T steps into the future and its goal representation. The L1 loss is optimized with respect to the actions $(a_k)_{k\in[T]}$ using the cross-entropy method (Rubinstein, 1997). Specifically, in each planning step, we sample the action coordinates at each point in the planning horizon from a sequence of Gaussian distributions initialized with zero mean and unit variance. The population statistics of the top-k actions trajectories are used to update the mean and variance of the Gaussian distributions. This process is repeated for several iterations before finally returning the mean of the sequence of Gaussians as the selected action trajectory.

planning step using the Cross-Entropy Method (Rubinstein, 1997), and only execute the first action on the robot before re-planning, as in receding horizon control.

4 Planning: Zero-shot Robot Control

In this section we demonstrate how V-JEPA 2-AC can be used to implement basic robot skills like reaching, grasping, and pick-and-place via model-predictive control. We focus on tasks with visual goal specification and show that V-JEPA 2-AC generalizes zero-shot to new environments.

4.1 Experimental Setup

Baselines. We compare the performance of V-JEPA 2-AC with two baselines, one vision-language-action model trained with behavior cloning, and one video generation-based world model.

The first baseline is based on the Octo video-language-action model that allows for goal-image conditioning (Octo Model Team et al., 2024). We start from the open-source weights of the *octo-base-1.5* version of the model, which is pretrained on the *Open-X Embodiment* dataset containing over 1M trajectories.[1] We fine-tune the Octo model with behaviour cloning on the entire Droid dataset using hindsight relabeling (Andrychowicz et al., 2017; Ghosh et al., 2019) with image goals and end-effector states. In particular, we sample random segments of trajectories from the Droid dataset during training, and uniformly sample goal images up to 20 timesteps forward in the trajectory. We use the official open-source code for fine-tuning, including all standard Droid optimization hyperparameters, and leverage single side image view inputs at 256×256 resolution, a context of two previous frames, and a horizon of 4 future actions.

The second baseline we compare with is based on the Cosmos video generation model (Agarwal et al., 2025). We start with the open-source weights for the action-free Cosmos model (latent diffusion-7B with continuous tokenizer), which was trained on 20M hours of video, and we fine-tune the model on Droid using the officially-released action-conditioned fine-tuning code.[2] To improve performance when training on Droid, we (i) lowered the learning rate to match that used in the video-conditioned Cosmos recipe, (ii) removed the dropout in the video conditioning to improve the training dynamics, and (iii) increased the noise level by a factor of e^2, as we observed that the model trained with a lower noise factor struggled to leverage the information in the conditioning frame. Although the Cosmos technical report (Agarwal et al., 2025)

[1] In comparison, we train V-JEPA 2-AC on 23k trajectories from Droid, including successes and failures.
[2] https://github.com/nvidia-cosmos/cosmos-predict1

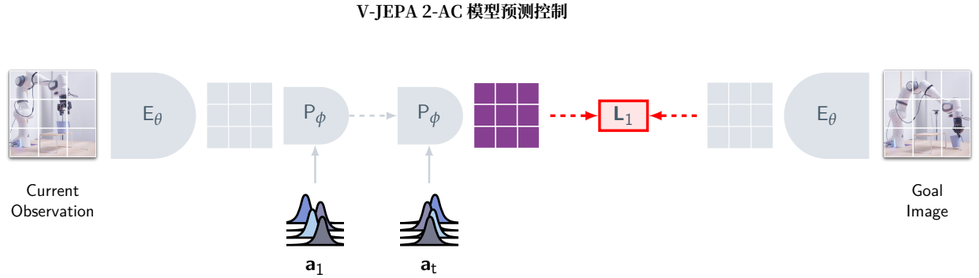


图 7 规划。固定规划时域 T，最小化世界模型想象的未来 T 步状态表征与目标表征之间的 L1 距离。使用交叉熵方法（Rubinstein, 1997），针对动作 $(a_k)_{k\in[T]}$ 优化 L1 损失。每次规划，从均值初始化为零、方差为一的高斯分布序列，采样规划时域内各时间点的动作坐标。用最优 top-k 动作轨迹的总体统计量更新高斯均值和方差。重复若干次，最终返回高斯序列的均值，作为选定动作轨迹。

规划步骤中采用交叉熵方法（Rubinstein, 1997），在机器人上只执行第一个动作，再重规划，与滚动时域控制一致。

4 规划：零样本机器人控制

本节展示用 V-JEPA 2-AC 进行模型预测控制，实现到达、抓取、抓取放置等基本技能。任务采用视觉目标，并展示模型零样本泛化到新环境。

4.1 实验设置

基线。将 V-JEPA 2-AC 与两个基线比较：行为克隆训练的视觉–语言–动作模型，以及基于视频生成的世界模型。

第一个基线基于支持目标图像条件的 Octo 视频–语言–动作模型（Octo Model Team et al., 2024）。从 octo-base-1.5 的开源权重开始，该模型在包含超过 100 万条轨迹的 Open-X Embodiment 数据集上预训练[1]。使用整个 Droid 数据集进行行为克隆微调，采用事后重标记（Andrychowicz et al., 2017; Ghosh et al., 2019），以图像目标和末端执行器状态为条件。训练时从 Droid 随机采样轨迹片段，并在轨迹未来最多 20 个时间步内均匀采样目标图像。采用官方微调代码及全部标准 Droid 优化超参数，使用分辨率 256×256 的单侧图像视角、此前两帧作为上下文，以及未来 4 个动作的时域。

第二个基线是 Cosmos 视频生成模型（Agarwal et al., 2025）。从不以动作为条件的 Cosmos 开源权重开始，即使用连续词元化器的潜在扩散 7B 模型，在 2000 万小时视频上训练；再用官方动作条件微调代码在 Droid 上微调[2]。为改善 Droid 训练，（i）降低学习率，使其与视频条件 Cosmos 方案一致；（ii）移除视频条件中的 dropout，改善训练动态；（iii）将噪声水平提高 e^2 倍，因为较低噪声系数的模型难以利用条件帧信息。尽管 Cosmos 技术报告（Agarwal et al., 2025）

[1] 相比之下，V-JEPA 2-AC 在 Droid 的 2.3 万条轨迹上训练，包含成功与失败。[2] https://github.com/nvidia-cosmos/cosmos-predict1

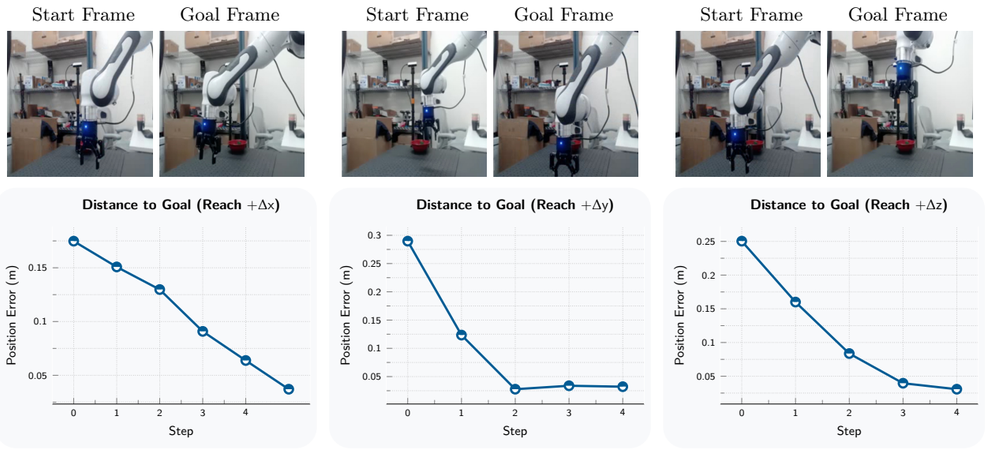


Figure 8 Single-Goal Reaching. Single-goal reaching involves moving the end-effector to a desired location in space based on a single goal image. This task measures for a basic understanding of actions as well as a 3D spatial understanding of the scene, including depth, from the monocular RGB camera. In each step, we use V-JEPA 2-AC to plan a sequence of actions by minimizing the L1 distance between the model's imagined future state representations and its representation of the goal frame. The first action is then executed before re-planning in the next time step. During planning, we only sample individual actions in the L1-Ball of radius 0.075 centered at the origin. Thus, the maximum achievable decrease in cartesian distance to the goal in a single step is 0.13 (∼13 cm).

mentions using world models for planning or model-predictive control as a future application, to the best of our knowledge this is the first reported attempt using Cosmos models for robot control.

Robot deployment. All models are deployed zero-shot on Franka Emika Panda arms with RobotiQ grippers, located in two different labs, neither of which appears in the Droid dataset. Visual input is provided through an uncalibrated low-resolution monocular RGB camera. The robots use the same exact model weights and inference code, and similar low-level controllers based on operational space control. We use blocking control for both the V-JEPA 2-AC world model and Cosmos world model (i.e., the system waits for the last commanded action to be completed before sending a new action to the controller) and experiment with both blocking and non-blocking control for Octo, and report the best performance across the two options. When planning with V-JEPA 2-AC and Cosmos, we constrain each sampled action to the L1-Ball of radius 0.075 centered at the origin, which corresponds to a maximum end-effector displacement of approximately 13 cm for each individual action, since large actions are relatively out-of-distribution for the models.

4.2 Results

Single-goal reaching. First, we evaluate on the task of single-goal reaching, which involves moving the end-effector to a desired location in space based on a single goal image. This task measures for a basic understanding of actions as well as a 3D spatial understanding of the scene (including depth) from the monocular RGB camera.

Figure 8 shows the Euclidean distance between the end-effector and its goal position during robot execution for three different single-goal reaching tasks. In all cases, the model is able to move the end-effector within less than 4 cm of its goal position, and select actions that lead to a monotonic decrease in the error. This can be seen as a form of visual servoing (Hill, 1979), wherein visual feedback from a camera is used to control a robot's motion. However, unlike classical approaches in visual servoing, V-JEPA 2-AC achieves this by training on unlabeled, real-world video data.

In Figure 9, we visualize the V-JEPA 2-AC energy landscape from equation (5) for the Δy reaching task as a

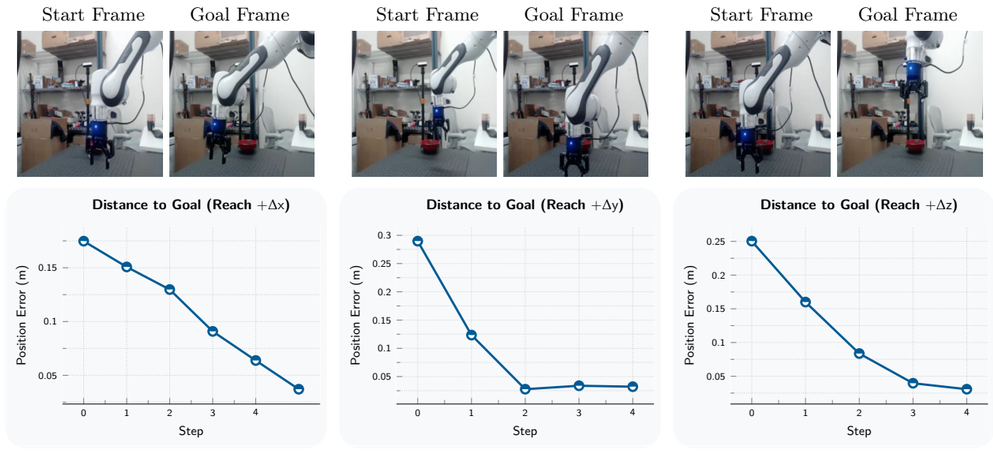


图 8 单目标到达。根据一张目标图像，将末端执行器移至期望空间位置，检验模型对动作的基本理解，以及从单目 RGB 相机理解三维场景和深度的能力。每步用 V-JEPA 2-AC 规划动作序列，最小化模型想象的未来状态表征与目标帧表征之间的 L1 距离。执行首个动作，再于下一时间步重规划。规划仅从以原点为中心、半径 0.075 的 L1 球内采样动作，因此单步最多使至目标的笛卡尔距离缩短 0.13（∼13 cm）。

将世界模型规划或模型预测控制列为未来应用；据我们所知，这是首次报告使用 Cosmos 控制机器人的尝试。

机器人部署。所有模型均零样本部署到两个不同实验室的 Franka Emika Panda 机械臂上，配备 RobotiQ 夹爪；两个实验室均未出现在 Droid 数据集中。视觉输入来自未标定、低分辨率、单目 RGB 相机。机器人使用完全相同的模型权重和推理代码，以及相似的基于操作空间控制的低层控制器。V-JEPA 2-AC 和 Cosmos 均采用阻塞控制，即等待上一条动作完成后再向控制器发送新动作；Octo 则测试阻塞与非阻塞两种控制，并报告最佳结果。使用 V-JEPA 2-AC 与 Cosmos 规划时，将各采样动作限制在以原点为中心、半径 0.075 的 L1 球内，对应单个动作最大末端位移约 13 cm，因为较大的动作相对超出模型训练分布。

4.2 结果

单目标到达。首先评估根据一张目标图像，将末端执行器移至期望空间位置的任务，衡量动作基本理解及从单目 RGB 相机理解三维场景（包括深度）的能力。

图 8 展示三项单目标到达任务执行过程中，末端执行器与目标的欧氏距离。所有情况下，模型都能将末端移至目标不足 4 cm 内，并选择使误差单调下降的动作。可视为视觉伺服（Hill, 1979），即用相机视觉反馈控制机器人运动。但与经典方法不同，V-JEPA 2-AC 通过无标注真实视频训练实现这一能力。

图 9 可视化式（5）中 V-JEPA 2-AC 在 Δy 到达任务的能量景观，作为

function of a single cartesian-control action, sweeping Δx and Δy while holding $\Delta z = 0$ fixed. The energy function achieves its minimum near the ground-truth action, providing further evidence that the model has learned to reasonably infer the effect of actions without requiring precision sensing. It is also interesting to observe that the energy landscape induced by V-JEPA 2-AC is relatively smooth and locally convex, which should facilitate planning.

Prehensile manipulation. Next, we evaluate all models on more challenging prehensile object manipulation tasks, namely *grasp*, *reach with object*, and *pick-and-place*. Success rates are reported in Table 2 and Table 3, and averaged across 10 trials with various permutations to the task across trials (e.g., object location, starting pose, etc.). For the *grasp* and *reach with object* tasks the model is shown a single goal image. For the *pick-and-place* tasks we present two sub-goal images to the model in addition to the final goal. The first goal image shows the object being grasped, the second goal image shows the object in the vicinity of the goal position. The model first optimizes actions with respect to the first sub-goal for 4 time-steps before automatically switching to the second sub-goal for the next 10 time-steps, and finally the third goal for the last 4 time-steps. Examples of robot execution for the pick-and-place task are shown in Figure 10. Start and goal frames for all individual tasks in Lab 1 are shown in Appendix B.2. The *grasp* task requires precise control from visual feedback to correctly grip the object. The *reach with object* task requires the model to navigate while holding an object, which necessitates a basic understanding of intuitive physics to avoid dropping the object. Finally, the *pick-and-place* task tests for the ability to compose these atomic skills.

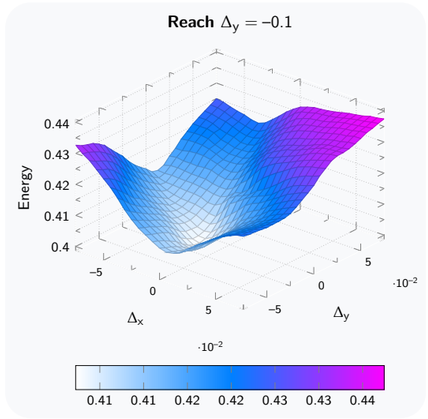


Figure 9 V-JEPA 2-AC Energy Landscape. Energy landscape for single-goal reaching task with respect to end-effector cartesian-control action (sweeping Δx and Δy while holding $\Delta z = 0$ fixed); ground truth action relating goal image to start frame is located at $(\Delta x, \Delta y) = (0, -0.1)$. We see that the energy function achieves its minimum around $(\Delta x, \Delta y) \approx (0, -0.05)$, indicating that the model has learned to reasonably infer the effect of actions without requiring precision sensing.

While all models achieve a high success-rate on *reach*, differences in performance are more apparent on tasks involving object interaction. We observe that the success-rate for all models depends on the type of object being manipulated. For instance, we find that the cup is mostly easily grasped by placing one finger inside the object and gripping around the rim, however if the control actions produced by the model are not accurate enough, the robot will miss the rim of the cup and fail to grasp the object. When manipulating the box, there are many more feasible grasping configurations, however, the model requires more precise gripper control to ensure that the fingers are open wide enough to grasp the object. We see that, for all models, the variation in success-rate with respect to the object type is due to the combination of sub-optimal actions and the unique challenges associated with manipulating each respective object. Nonetheless, we see that the V-JEPA 2-AC model achieves the highest success-rate across all tasks, highlighting the feasibility of latent planning for robot manipulation.

In Table 3, we compare planning performance when using V-JEPA 2-AC versus the Cosmos action-conditioned video generation model based on latent diffusion. In both cases we leverage the cross-entropy method (Rubinstein, 1997) for optimizing the sequence of actions using a single NVIDIA RTX 4090 GPU, and construct the energy function by encoding the goal frame in the latent space of the model, as in equation (5). With 80 samples, 10 refinement steps, and a planning horizon of 1, it takes 4 minutes to compute a single action in each planning step with Cosmos. While we achieve a high success rate of 80% on the *reach* tasks when using Cosmos, performance on object interaction tasks is weaker. Note that under a planning time of 4 minutes per action, a full pick & place trajectory requires over one hour of robot execution. By contrast, with 10× more samples in each refinement step, the V-JEPA 2-AC world model requires only 16 seconds per action and leads to higher performance across all considered robot skills. We can potentially reduce the planning time for both models in future work by leveraging additional computing resources for planning, reducing the number

随单个笛卡尔控制动作变化的函数，固定 $\Delta z = 0$，遍历 Δx 和 Δy。能量在真实动作附近最小，进一步说明模型无需精确感知也能合理推断动作效果。有趣的是，V-JEPA 2-AC 能量景观相对光滑、局部凸，应有助于规划。

抓持操作。接下来，在更具挑战性的抓持物体操作任务上评估所有模型，包括抓取、持物到达和抓取放置。成功率见表 2 和表 3，取 10 次试验的平均值，各次试验改变任务配置（例如物体位置、起始姿态等）。抓取和持物到达任务中，向模型展示一张目标图像；抓取放置任务中，除最终目标外，还展示两个子目标图像。第一个目标展示物体被抓住，第二个展示物体处于目标位置附近。模型先针对第一个子目标优化动作 4 个时间步，再自动切换到第二个子目标执行 10 个时间步，最后以第三个目标执行 4 个时间步。图 10 展示抓取放置任务的机器人执行示例。实验室 1 中所有单项任务的起始帧与目标帧见附录 B.2。抓取要求根据视觉反馈精确控制，以正确夹住物体。持物到达要求模型抓住物体并移动，需要基本的直觉物理理解以避免物体掉落。最后，抓取放置检验组合这些基础技能的能力。

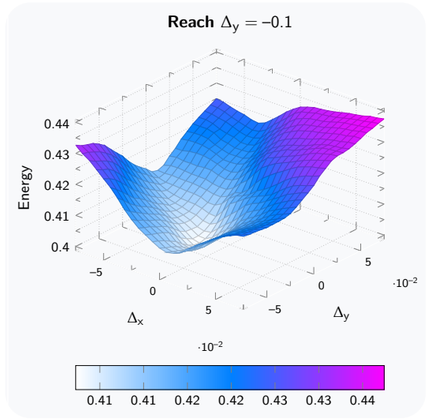


图 9 V-JEPA 2-AC 能量景观。单目标到达任务的能量随末端笛卡尔控制动作变化：固定 $\Delta z = 0$，遍历 Δx 和 Δy。将起始帧变为目标图像的真实动作位于 $(\Delta x, \Delta y) = (0, -0.1)$。能量在 $(\Delta x, \Delta y) \approx (0, -0.05)$ 附近最小，说明模型无需精确感知也能合理推断动作效果。

所有模型在到达任务上成功率较高，但物体交互任务更能体现性能差异。各模型成功率均依赖物体类型。例如，杯子最容易通过以下方式抓取，

将一根手指放入物体内部并夹住杯沿；但如果模型生成的控制动作不够准确，机器人会错过杯沿而无法抓住。操作盒子时，可行抓取配置更多，但需要更精确的夹爪控制，确保手指张开得足够宽。所有模型的成功率随物体类型变化，均来自次优动作与各物体独特操作难点的共同作用。尽管如此，V-JEPA 2-AC 在所有任务中成功率最高，体现了潜在空间规划用于机器人操作的可行性。

表 3 比较使用 V-JEPA 2-AC 与基于潜在扩散的 Cosmos 动作条件视频生成模型的规划性能。两者均在单张 NVIDIA RTX 4090 GPU 上使用交叉熵方法（Rubinstein, 1997）优化动作序列，并按式（5）将目标帧编码到模型潜在空间来构造能量函数。Cosmos 在使用 80 个样本、10 次优化迭代和规划时域 1 时，每次规划计算一个动作需 4 分钟。虽然 Cosmos 在到达任务上达到 80% 的较高成功率，但物体交互任务表现较弱。每个动作规划 4 分钟，意味着完整抓取放置轨迹需超过一小时机器人执行时间。相比之下，V-JEPA 2-AC 每次优化使用多 10× 倍的样本，但每个动作仅需 16 秒，且在全部所考察技能上性能更高。未来可通过增加规划计算资源、减少数量，降低两种模型的规划时间，

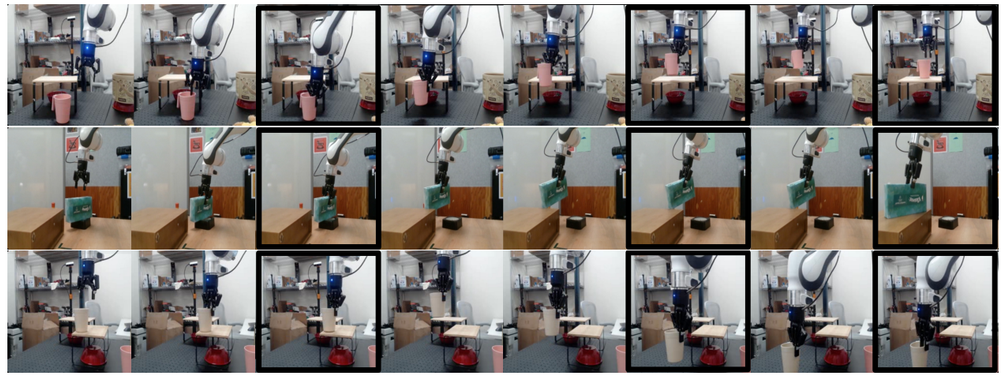


Figure 10 Pick-&-Place. Closed-loop robot execution of V-JEPA 2-AC for multi-goal pick-&-place tasks. Highlighted frames indicate when the model achieves a sub-goal and switches to the next goal. The first goal image shows the object being grasped, the second goal image shows the object in the vicinity of the desired location, and the third goal image shows the object placed in the desired position. The model first optimizes actions with respect to the first sub-goal for 4 time-steps before automatically switching to the second sub-goal for the next 10 time-steps, and finally the third goal for the last 4 time-steps. Robot actions are inferred through goal-conditioned planning. The V-JEPA 2-AC model is able to perform zero-shot pick-&-place tasks on two Franka arms in different labs, with various object configurations and cluttered environments.

Table 2 Zero-Shot Robot Manipulation. All models are deployed zero-shot on two Franka arms with RobotiQ grippers located in different labs. Given image-goals for each considered task, all models run closed loop to infer a sequence of actions to achieve the goal. Success rates are reported out of 10 trials with various permutations to the task across trials (e.g., object location, starting pose, etc.).

			Grasp		Reach w/ Obj.		Pick-&-Place	
Method		**Reach**	Cup	Box	Cup	Box	Cup	Box
Octo (Octo Model Team et al., 2024)	Lab 1	100%	20%	0%	20%	70%	20%	10%
	Lab 2	100%	10%	0%	10%	70%	10%	10%
	Avg	*100%*	*15%*	*0%*	*15%*	*70%*	*15%*	*10%*
V-JEPA 2-AC (ours)	Lab 1	100%	70%	30%	90%	80%	80%	80%
	Lab 2	100%	60%	20%	60%	70%	80%	50%
	Avg	*100%*	*65%*	*25%*	*75%*	*75%*	*80%*	*65%*

of samples and refinement steps used at each time step, training a feed-froward policy in the world-models' imagination to initialize the planning problem, or potentially leveraging gradient-based planning in the case of V-JEPA 2-AC.

4.3 Limitations

Sensitivity to camera positioning. Since the V-JEPA 2-AC model is trained to predict representations of the next video frame given an end-effector Cartesian control action, without any explicit camera calibration, it must therefore implicitly infer the action coordinate axis from the monocular RGB camera input. However, in many cases, the robot base is not visible in the camera frame, and thus the problem of inferring the action coordinate axis is not well defined, leading to errors in the world model. In practice, we manually tried different camera positions before settling on one that worked well across all of our experiments. We conduct a quantitative analysis of the V-JEPA 2-AC world model's sensitivity to camera position in Appendix B.4.

Long horizon planning. Long horizon planning with world models is limited by a number of factors. First, autoregressive prediction suffers from error accumulation: the accuracy of the representation-space

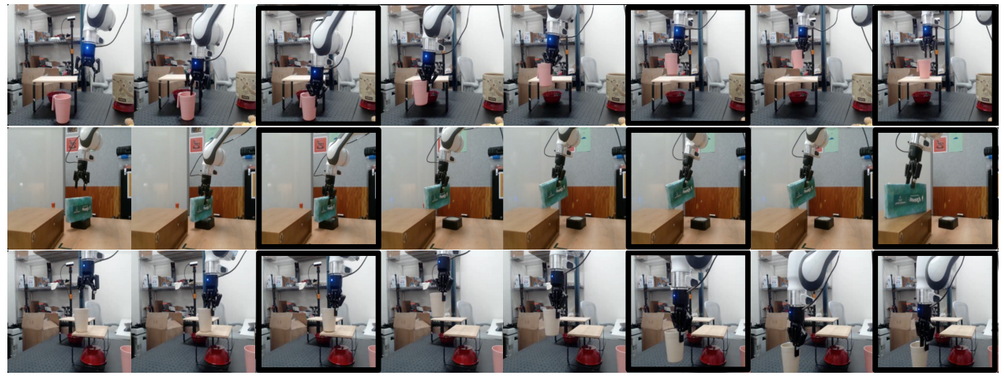


图 10 抓取放置。V-JEPA 2-AC 闭环执行多目标抓取放置任务。高亮帧表示模型达到子目标并切换下一目标。第一目标图像展示物体被抓住，第二展示物体接近期望位置，第三展示物体已放在期望位置。模型针对第一子目标优化 4 个时间步，自动切换第二目标执行 10 步，最后以第三目标执行 4 步。机器人动作通过目标条件规划推断。V-JEPA 2-AC 可在不同实验室的两台 Franka 机械臂上，针对不同物体配置及杂乱环境，零样本执行抓取放置。

表 2 零样本机器人操作。全部模型零样本部署到不同实验室的两台 Franka 机械臂，配备 RobotiQ 夹爪。给定图像目标，闭环推断实现目标的动作序列。报告 10 次试验成功率，各次改变物体位置、起始姿态等任务配置。

			抓取		持物到达		抓取放置	
方法		**到达**	Cup	Box	Cup	Box	Cup	Box
Octo (Octo Model Team et al., 2024)	实验室 1	100%	20%	0%	20%	70%	20%	10%
	实验室 2	100%	10%	0%	10%	70%	10%	10%
	Avg	*100%*	*15%*	*0%*	*15%*	*70%*	*15%*	*10%*
V-JEPA 2-AC（我们的方法）	实验室 1	100%	70%	30%	90%	80%	80%	80%
	实验室 2	100%	60%	20%	60%	70%	80%	50%
	Avg	*100%*	*65%*	*25%*	*75%*	*75%*	*80%*	*65%*

每步使用的样本数和优化迭代数；在世界模型想象中训练前馈策略，以初始化规划；或对 V-JEPA 2-AC 使用基于梯度的规划。

4.3 局限

对相机位置的敏感性。V-JEPA 2-AC 根据末端执行器笛卡尔控制动作预测下一帧表征，不进行显式相机标定，因此必须从单目 RGB 输入隐式推断动作坐标轴。但许多情况下画面看不到机器人基座，导致坐标轴推断问题定义不充分，世界模型因此出错。实际中，我们人工尝试多个相机位置，最终选择在全部实验中表现良好的位置。附录 B.4 定量分析该模型对相机位置的敏感性。

长时域规划。世界模型长时域规划受多种因素限制。首先，自回归预测存在误差累积：表征空间中的

Table 3 Planning Performance. Comparing closed-loop robot manipulation using MPC with V-JEPA 2-AC world model and Cosmos world model. In both cases we leverage the cross-entropy method (Rubinstein, 1997) for optimizing the sequence of actions using a single NVIDIA RTX 4090 GPU. For each robot skill, we evaluate each model across 10 tasks and average the results. With 80 samples, 10 refinement steps, and a planning horizon of 1, it takes 4 minutes to compute a single action in each planning step with Cosmos, which is an action-conditioned video generation model based on latent diffusion. Note that under a planning time of 4 minutes per action, a full pick & place trajectory takes over one hour. By contrast, with 10× more samples in each refinement step, the V-JEPA 2-AC world model requires only 16 seconds per action and leads to higher performance across all considered robot skills.

Lab 2	Planning Details				Grasp			Pick-&-Place	
Method	#Samples	Iter.	Horizon	Time	**Reach**	Cup	Box	Cup	Box
Cosmos (Agarwal et al., 2025)	80	10	1	4 min.	80%	0%	20%	0%	0%
V-JEPA 2-AC (ours)	800	10	1	16 sec.	100%	60%	20%	80%	50%

predictions decreases with longer autoregressive rollouts, thereby making it more difficult to reliably plan over long horizons. Second, long-horizon planning increases the size of the search space: the number of possible action trajectories increases exponentially given a linear increase in the planning horizon, thereby making it computationally challenging to plan over long horizons. On the other hand, long-horizon planning is necessary for solving non-greedy prediction tasks, e.g., pick-and-place without image sub-goals. Future work exploring world models for long-horizon planning will enable the solution of many more complex and interesting tasks.

Image goals. Following many previous works in goal-conditioned robot manipulation (Finn and Levine, 2017; Lynch et al., 2020; Chebotar et al., 2021; Jang et al., 2022; Liu et al., 2022; Gupta et al., 2022), our current formulation of the optimization target assumes that we have access to visual goals. However, when deploying robots in-the-wild, it may be more natural to express goals in other forms, such as with language. Future work that aligns latent action-conditioned world models with language models will step towards more general task specification via natural language.

5 Understanding: Probe-based Classification

The capabilities of a representation-space world model, such as V-JEPA 2-AC discussed above, are inherently limited by the state information encoded in the learned representation space. In this section and subsequent sections, we probe the representations learned by V-JEPA 2 and compare the V-JEPA 2 encoder to other vision encoders on visual classification.

Visual classification tasks can focus either on *appearance understanding* or *motion understanding*. While appearance understanding tasks can generally be solved using information visible in a single frame of an input video clip (even when the classification labels describe actions), motion understanding tasks require several frames to correctly classify a video (Goyal et al., 2017). To ensure a balanced evaluation of both motion and appearance, we have selected three motion understanding tasks, namely Something-Something v2 (SSv2), Diving-48, and Jester, which require the model to understand human gestures and movements. For appearance understanding, we have chosen Kinetics400 (K400), COIN, and ImageNet (IN1K), which involve recognizing actions, scenes, and objects. Empirically, we show that V-JEPA 2 outperforms state-of-the-art visual encoders on motion understanding tasks, while being competitive on appearance understanding tasks.

Attentive Probe. We train an 4-layers attentive probe on top of the frozen encoder output using the training data from each task. Our attentive probe is composed of four transformer blocks, the last of which replaces standard self-attention with a cross-attention layer using a learnable query token. Following standard practice, several clips with a fixed number of frames are sampled from a video during inference. The classification logits are then averaged across clips. We keep the resolution similar to the one used for V-JEPA 2 pretraining. We ablate the number of layers of our attentive probe in Appendix C.2, and also provide full details on the number of clips, clip size, and other hyperparameters used in the downstream tasks.

表 3 规划性能。比较以 V-JEPA 2-AC 和 Cosmos 世界模型进行 MPC 闭环机器人操作。两者均在单张 NVIDIA RTX 4090 上，通过交叉熵方法（Rubinstein, 1997）优化动作序列。每项技能各评估 10 个任务并取平均。Cosmos 是基于潜在扩散的动作条件视频生成模型，使用 80 个样本、10 次优化迭代、规划时域 1 时，每步计算单个动作需 4 分钟；因此完整抓取放置耗时超过一小时。相比之下，V-JEPA 2-AC 每次优化使用多 10× 倍的样本，但单动作仅需 16 秒，在所有技能上性能更高。

实验室 2	规划细节				抓取			抓取放置	
方法	样本数	迭代次数	规划时域	Time	**到达**	Cup	Box	Cup	Box
Cosmos (Agarwal et al., 2025)	80	10	1	4 分钟	80%	0%	20%	0%	0%
V-JEPA 2-AC（我们的方法）	800	10	1	16 秒	100%	60%	20%	80%	50%

预测准确性随自回归展开长度下降，使长时域规划更难可靠进行。其次，长时域扩大搜索空间：规划时域线性增长时，可能动作轨迹数指数增长，带来计算挑战。另一方面，长时域是解决非贪心任务的必要条件，例如不提供图像子目标的抓取放置。未来研究支持长时域规划的世界模型，将能解决更多复杂而有趣的任务。

图像目标。遵循许多目标条件机器人操作研究（Finn and Levine, 2017; Lynch et al., 2020; Chebotar et al., 2021; Jang et al., 2022; Liu et al., 2022; Gupta et al., 2022），当前优化目标假设可获得视觉目标。但实际环境部署时，用语言等其他形式表达目标可能更自然。未来将潜在动作条件世界模型与语言模型对齐，可迈向通过自然语言指定更一般任务。

5 理解：基于探测器的分类

表征空间世界模型，例如上述 V-JEPA 2-AC，其能力本质上受限于表征所编码的状态信息。本节及后续章节探测 V-JEPA 2 表征，并在视觉分类中与其他编码器比较。

视觉分类可关注外观或运动理解。外观理解一般可凭输入视频单帧中可见的信息完成，即使标签描述的是动作；运动理解则需要多帧才能正确分类（Goyal et al., 2017）。为平衡评估两类能力，选择三项要求理解人体手势与运动的任务：SSv2、Diving-48、Jester。外观理解采用 Kinetics400、COIN、ImageNet，涉及动作、场景和物体识别。实验说明 V-JEPA 2 在运动理解上优于最先进视觉编码器，在外观理解上也具有竞争力。

注意力探测器。使用各任务训练数据，在冻结编码器输出上训练四层注意力探测器，由四个 Transformer 块构成，最后一块以带可学习查询词元的交叉注意力替代标准自注意力。遵循常见做法，推理时从视频采样多个固定帧数片段，对各片段的分类 logits 取平均。分辨率与 V-JEPA 2 预训练接近。附录 C.2 消融探测器层数，并给出下游任务片段数、片段长度及其他超参数的完整细节。

Table 4 Action and Object Classification. We report the classification performance of V-JEPA 2 models pretrained on 64 frames at resolution 256×256 for all models, except V-JEPA 2 ViT-g_{384} which was pretrained at resolution 384×384, on action and object classification, and compare their performance with state-of-art image and video encoders. All models follow the same evaluation protocol except for V-JEPA 2 ViT-g_{384}. We use 256×256 resolution with $16 \times 2 \times 3$ inputs for SSv2 (16 frames clip, 2 temporal crops, 3 spatial crops), $16 \times 8 \times 3$ for K400, $32 \times 8 \times 3$ for COIN and $32 \times 4 \times 3$ for Diving-48 and Jester. V-JEPA 2 ViT-g_{384} uses a higher resolution of 384×384 for all six tasks, and additionally uses $64 \times 2 \times 3$ inputs for SSv2 and 32x8x3 inputs for COIN. Our V-JEPA 2 ViT-g significantly outperforms other vision encoders on motion understanding tasks and is competitive on appearance tasks. It achieves the best average performance of 87.5 across all image and videos encoders. V-JEPA 2 ViT-g_{384} further improves results consistently across tasks, reaching 88.2 average performance. $*$: $\mathbf{PE_{core}G}$ achieves an accuracy 89.8% on ImageNet using an attentive probe and input resoluton of 448px (Bolya et al., 2025). We use an input resolution of 256px and a different probe architecture in our case.

			Motion Understanding			*Appearance Understanding*		
Method	**Param.**	**Avg.**	**SSv2**	**Diving-48**	**Jester**	**K400**	**COIN**	**IN1K**
Results Reported in the Literature								
VideoMAEv2 (Wang et al., 2023)	1B	–	56.1	–	–	82.8	–	71.4
InternVideo2-1B (Wang et al., 2024b)	1B	–	67.3	–	–	87.9	–	–
InternVideo2-6B (Wang et al., 2024b)	6B	–	67.7	–	–	88.8	–	–
VideoPrism (Zhao et al., 2024)	1B	–	68.5	71.3	–	87.6	–	–
Image Encoders Evaluated Using the Same Protocol								
DINOv2 (Darcet et al., 2024)	1.1B	81.1	50.7	82.5	93.4	83.6	90.7	86.1
$\mathbf{PE_{core}G}$ (Bolya et al., 2025)	1.9B	82.3	55.4	76.9	90.0	88.5	**95.3**	87.6*
SigLIP2 (Tschannen et al., 2025)	1.2B	81.1	49.9	75.3	91.0	87.3	95.1	**88.0**
Video Encoders Evaluated Using the Same Protocol								
V-JEPA ViT-H (Bardes et al., 2024)	600M	85.2	74.3	87.9	97.7	84.5	87.1	80.0
InternVideo2$_{s2}$-1B (Wang et al., 2024b)	1B	87.0	69.7	86.4	97.0	**89.4**	93.8	85.8
V-JEPA 2 ViT-L	300M	86.0	73.7	89.0	97.6	85.1	86.8	83.5
V-JEPA 2 ViT-H	600M	86.4	74.0	89.8	97.7	85.3	87.9	83.8
V-JEPA 2 ViT-g	1B	87.5	75.3	90.1	97.7	86.6	90.7	84.6
V-JEPA 2 ViT-g_{384}	1B	**88.2**	**77.3**	**90.2**	**97.8**	87.3	91.1	85.1

Evaluation protocol. We compare the performance of V-JEPA 2 on motion and appearance tasks with several other visual encoders: DINOv2 with registers (Darcet et al., 2024) is the current state-of-the-art model for self-supervised learning with images, while SigLIP2 (Tschannen et al., 2025) and the Perception Encoder $PE_{core}G$ (Bolya et al., 2025) are two state-of-the-art models for image-text contrastive pretraining. We also consider two video encoders: the self-supervised V-JEPA (Bardes et al., 2024), and InternVideo2$_{s2}$-1B (Wang et al., 2024b) which relies primarily on vision-text contrastive pretraining.

We use the same evaluation protocol for every baseline and for V-JEPA 2, learning an attentive probe on top of the frozen encoder, similar to Bardes et al. (2024). We adapt image-based models to video following the procedure used in Oquab et al. (2023), concatenating the features of each input frame. For InternVideo2$_{s2}$-1B, we use its image positional embedding for the ImageNet task, and for video tasks we interpolate its positional embedding from 4 frames to 8, producing a token count similar to V-JEPA 2. Despite using a common evaluation protocol, the baseline encoders are trained on different data (e.g., DINOv2 on LVD-142M, $PE_{core}G$ on MetaCLIP) and are thus not directly comparable. We can therefore only compare different approaches at a system level; i.e., with a consistent evaluation protocol despite differences in training protocol and data. We also include existing results from the literature using a similar frozen protocol, but with potentially different attentive head architecture. In particular, we share reported results of VideoMAEv2 (Wang et al., 2023), InternVideo-1B and 6B (Wang et al., 2024b), and VideoPrism (Zhang et al., 2024c) on the classification tasks we consider, when available. We provide complete evaluation and hyperparameters in Appendix C.1.

Results. Table 4 reports the classification performance of V-JEPA 2, the other encoders we evaluated, and other notable results reported in the literature. V-JEPA 2 ViT-g (at 256 resolution) significantly outperforms other vision encoders on motion understanding tasks. It achieves a top-1 accuracy of 75.3 on SSv2 compared to

表 4 动作与物体分类。报告 V-JEPA 2 在动作和物体分类上的性能，并与最先进的图像和视频编码器比较。除 V-JEPA 2 ViT-g_{384} 以分辨率 384×384 预训练外，所有模型均以 64 帧、分辨率 256×256 预训练。除 V-JEPA 2 ViT-g_{384} 外，所有模型遵循相同评估方案。使用分辨率 256×256：SSv2 输入为 $16 \times 2 \times 3$（16 帧片段、2 个时间裁剪、3 个空间裁剪），K400 为 $16 \times 8 \times 3$，COIN 为 $32 \times 8 \times 3$，Diving-48 和 Jester 为 $32 \times 4 \times 3$。V-JEPA 2 ViT-g_{384} 在全部六项任务中使用更高分辨率 384×384，并且在 SSv2 中使用 $64 \times 2 \times 3$ 输入，在 COIN 中使用 32×8×3 输入。V-JEPA 2 ViT-g 在运动理解任务上显著优于其他视觉编码器，在外观任务上也具有竞争力。在所有图像和视频编码器中，它取得最高平均性能 87.5。V-JEPA 2 ViT-g_{384} 在各任务上进一步改善，平均性能达到 88.2。 $*$：$PE_{core}G$ 使用注意力探测器和 448px 输入分辨率，在 ImageNet 上达到 89.8% 准确率（Bolya et al., 2025）。我们采用 256px 输入分辨率和不同的探测器架构。

			运动理解			外观理解		
方法	**参数量**	**Avg.**	**SSv2**	**Diving-48**	**Jester**	**K400**	**COIN**	**IN1K**
文献报告的结果								
VideoMAEv2 (Wang et al., 2023)	1B	–	56.1	–	–	82.8	–	71.4
InternVideo2-1B (Wang et al., 2024b)	1B	–	67.3	–	–	87.9	–	–
InternVideo2-6B (Wang et al., 2024b)	6B	–	67.7	–	–	88.8	–	–
VideoPrism (Zhao et al., 2024)	1B	–	68.5	71.3	–	87.6	–	–
采用相同方案评估的图像编码器								
DINOv2 (Darcet et al., 2024)	1.1B	81.1	50.7	82.5	93.4	83.6	90.7	86.1
$PE_{core}G$ (Bolya et al., 2025)	1.9B	82.3	55.4	76.9	90.0	88.5	**95.3**	87.6*
SigLIP2 (Tschannen et al., 2025)	1.2B	81.1	49.9	75.3	91.0	87.3	95.1	**88.0**
采用相同方案评估的视频编码器								
V-JEPA ViT-H (Bardes et al., 2024)	600M	85.2	74.3	87.9	97.7	84.5	87.1	80.0
InternVideo2$_{s2}$-1B (Wang et al., 2024b)	1B	87.0	69.7	86.4	97.0	**89.4**	93.8	85.8
V-JEPA 2 ViT-L	300M	86.0	73.7	89.0	97.6	85.1	86.8	83.5
V-JEPA 2 ViT-H	600M	86.4	74.0	89.8	97.7	85.3	87.9	83.8
V-JEPA 2 ViT-g	1B	87.5	75.3	90.1	97.7	86.6	90.7	84.6
V-JEPA 2 ViT-g_{384}	1B	**88.2**	**77.3**	**90.2**	**97.8**	87.3	91.1	85.1

评估方案。将 V-JEPA 2 的运动和外观性能与其他视觉编码器比较：带寄存器的 DINOv2（Darcet et al., 2024）是当前先进的图像自监督模型，SigLIP2（Tschannen et al., 2025）和 Perception Encoder $PE_{core}G$（Bolya et al., 2025）是先进的图像–文本对比预训练模型。另考虑两个视频编码器：自监督 V-JEPA（Bardes et al., 2024）与主要依赖视觉–文本对比预训练的 InternVideo2$_{s2}$-1B（Wang et al., 2024b）。

对所有基线及 V-JEPA 2 采用相同评估方案，与 Bardes et al.（2024）类似，在冻结编码器之上学习注意力探测器。按 Oquab et al.（2023）的方法，通过拼接各输入帧的特征，将图像模型适配到视频。对于 InternVideo2$_{s2}$-1B，在 ImageNet 任务中使用其图像位置嵌入；视频任务中将位置嵌入从 4 帧插值到 8 帧，使词元数与 V-JEPA 2 接近。尽管评估方案统一，基线编码器的训练数据不同（例如 DINOv2 使用 LVD-142M，$PE_{core}G$ 使用 MetaCLIP），因此无法直接比较。我们只能在系统层面比较各方法，即在训练方案和数据不同的情况下使用一致的评估方案。还纳入文献中采用类似冻结方案的结果，但其注意力头架构可能不同。特别是，在结果可用时，报告 VideoMAEv2（Wang et al., 2023）、InternVideo-1B 和 6B（Wang et al., 2024b）、VideoPrism（Zhang et al., 2024c）在所考察分类任务上的已有结果。完整评估设置与超参数见附录 C.1。

结果。表 4 报告 V-JEPA 2、所评估其他编码器及文献中代表性分类结果。256 分辨率的 ViT-g 在运动理解上显著优于其他编码器，SSv2 top-1 为 75.3，相比

Table 5 Prediction: Human Action Anticipation. Comparison with the state-of-the-art on the EK100 Action Anticipation benchmark. We report mean-class recall-at-5 for verb, noun and action on the validation set of EK100. V-JEPA 2 performance scales linearly with model size and outperforms previous state-of-the-art across all model sizes.

Method	Param.	*Action Anticipation*		
		Verb	Noun	**Action**
InAViT (Roy et al., 2024)	160M	51.9	52.0	25.8
Video-LLaMA (Zhang et al., 2023)	7B	52.9	52.0	26.0
PlausiVL (Mittal et al., 2024)	8B	55.6	54.2	27.6
Frozen Backbone				
V-JEPA 2 ViT-L	300M	57.8	53.8	32.7
V-JEPA 2 ViT-H	600M	59.2	54.6	36.5
V-JEPA 2 ViT-g	1B	61.2	55.7	38.0
V-JEPA 2 ViT-g$_{384}$	1B	**63.6**	**57.1**	**39.7**

69.7 for InternVideo and 55.4 for PE$_{Core}$G. V-JEPA 2 is also competitive on appearance tasks, reaching 84.6 on ImageNet (a +4.6 point improvement over V-JEPA). Overall, V-JEPA 2 obtains the best average performance across all six tasks, compared to other video and image encoders. The higher-resolution, longer-duration V-JEPA 2 ViT-g$_{384}$ shows further improvement across all tasks, reaching 88.2 average performance.

6 Prediction: Probe-based Action Anticipation

Action anticipation consists in predicting the future action given a contextual video clip leading up to some time before the action. Using the Epic-Kitchens-100 (EK100) benchmark (Damen et al., 2022), we demonstrate that V-JEPA 2 action anticipation performance increases consistently with model size. Furthermore, despite only using an attentive probe trained on top of V-JEPA 2 representations, we show that V-JEPA 2 significantly outperforms prior state-of-the-art approaches that were specifically designed for this task.

Task. The EK100 dataset is comprised of 100 hours of cooking activities recorded from an egocentric perspective across 45 kitchen environments. Each video in EK100 is annotated with action segments, which include a start timestamp, an end timestamp, and an action label. There are 3,568 unique action labels, each consisting of a verb and a noun category, with a total of 97 verb categories and 300 noun categories. The EK100 action anticipation task involves predicting noun, verb, and action (i.e., predicting verb and noun jointly) from a video clip, referred to as context, that occurs before the start timestamp of an action segment. The interval between the end of the context and the beginning of the action segment is the anticipation time, which is set to 1 second by default. Given that different future actions may be possible from a given context, mean-class recall-at-5 is used as the metric to measure performance (Damen et al., 2022).

Anticipation Probe. An attentive probe is trained on top of the frozen V-JEPA 2 encoder and predictor to anticipate future actions. Specifically, we sample a video clip that ends 1 second before an action starts. This video context is fed to the V-JEPA 2 encoder. The predictor takes the encoder representation, along with the mask tokens corresponding to the frame 1 second into the future, and predicts the representation of the future video frame. The outputs of the predictor and encoder are concatenated along the token dimension and fed to an attentive probe with a similar architecture to those used in Section 5, with the difference being that the anticipation probe's final cross-attention layer learns three query tokens (as opposed to one), and each query output is fed to a different linear classifier to predict the action category, the verb category, and the noun category respectively. A focal loss (Lin et al., 2017) is applied to each classifier independently and then summed before backpropagating through the shared attention blocks of the probe. We provide additional details and evaluation hyperparameters in Appendix D.1.

Baselines. We compare our model with three baselines that are trained specifically for action anticipation: InAViT (Roy et al., 2024) is a a supervised approach that leverages explicit hand-object interaction modeling,

表 5 预测：人类动作预判。与 EK100 最先进方法比较，报告验证集动词、名词、动作的类别平均 recall@5。V-JEPA 2 性能随模型大小线性扩展，全部规模均优于此前最先进方法。

方法	参数量	动作预判		
		Verb	Noun	动作
InAViT (Roy et al., 2024)	160M	51.9	52.0	25.8
Video-LLaMA (Zhang et al., 2023)	7B	52.9	52.0	26.0
PlausiVL (Mittal et al., 2024)	8B	55.6	54.2	27.6
冻结骨干网络				
V-JEPA 2 ViT-L	300M	57.8	53.8	32.7
V-JEPA 2 ViT-H	600M	59.2	54.6	36.5
V-JEPA 2 ViT-g	1B	61.2	55.7	38.0
V-JEPA 2 ViT-g$_{384}$	1B	**63.6**	**57.1**	**39.7**

InternVideo 为 69.7，PE$_{Core}$G 为 55.4。外观任务也有竞争力，ImageNet 达 84.6，比 V-JEPA 提升 +4.6 个百分点。总体而言，V-JEPA 2 在六项任务平均性能上胜过其他图像和视频编码器。更高分辨率、更长视频的 ViT-g$_{384}$ 在各任务进一步改善，平均达到 88.2。

6 预测：基于探测器的动作预判

动作预判根据动作发生之前的上下文视频，预测未来动作。采用 Epic-Kitchens-100（Damen et al., 2022），展示 V-JEPA 2 的预判性能随模型规模一致提升。此外，虽仅在表征之上训练注意力探测器，模型仍显著优于此前专为该任务设计的最先进方法。

任务。EK100 包含在 45 个厨房环境中以第一视角拍摄的 100 小时烹饪活动。每段视频标注动作片段，包括起始时间戳、结束时间戳和动作标签。共有 3568 个不同动作标签，每个由动词与名词类别组成，总计 97 个动词类别、300 个名词类别。动作预判任务要求根据发生在动作片段开始之前的视频上下文，预测名词、动词与动作（即联合预测动词和名词）。上下文结束与动作开始之间的间隔为预判时间，默认 1 秒。由于同一上下文可能对应多个未来动作，使用类别平均 recall@5 作为指标（Damen et al., 2022）。

预判探测器。在冻结的 V-JEPA 2 编码器和预测器之上训练注意力探测器，预判未来动作。采样在动作开始前 1 秒结束的视频片段，送入编码器。预测器以编码器表征及对应未来 1 秒视频帧的掩码词元为输入，预测未来帧表征。沿词元维拼接预测器与编码器输出，送入架构类似第 5 节的注意力探测器；区别是预判探测器最后的交叉注意力层学习三个查询词元，而非一个，各查询输出送入不同线性分类器，分别预测动作、动词和名词类别。对每个分类器独立计算 focal loss（Lin et al., 2017），相加后通过共享注意力块反向传播。更多细节与评估超参数见附录 D.1。

基线。与三种专用动作预判方法比较：InAViT（Roy et al., 2024）是显式建模手–物体交互的监督方法，

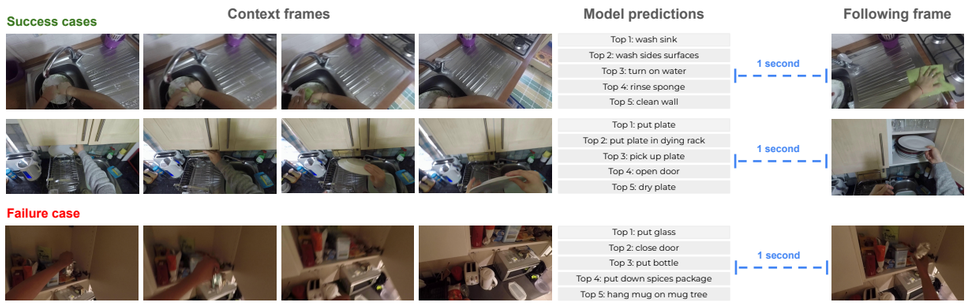


Figure 11 Visualization of EK100 prediction. (Left): four selected frames from the context frames. (Middle): model predictions, ordered by likelihood. (Right): following frame after the 1 second anticipation time. We show two examples where the model is successful and one example where the model fails.

and Video-LLaMA (Zhang et al., 2023) and PlausiVL (Mittal et al., 2024) are both approaches that leverage a large language model, with up to 7 billion parameters.

Results. Table 5 summarizes the results on the EK100 action anticipation benchmark. We compare V-JEPA 2 ViT-L, ViT-H and ViT-g encoders, increasing parameter count from 300 millions to 1 billion. All three leverage 32 frames with 8 frames per second at resolution 256×256 as video context. We also report results of ViT-g_{384} which uses a resolution of 384×384. V-JEPA 2 demonstrates a linear scaling behavior with respect to model size, in terms of action prediction recall-at-5. V-JEPA 2 ViT-L with 300 million parameters achieves 32.7 recall-at-5. Increasing the size of the model to 1 billion parameters leads to a +5.3 point improvement with an action recall-at-5 of 38.0. Furthermore, V-JEPA 2 benefits from using a context with higher resolution, and V-JEPA 2 ViT-g_{384} at resolution 384×384 improves recall-at-5 by an additional +1.7 points over the other models using 256×256 resolution.

V-JEPA 2 outperforms the previous state-of-the-art model PlausiVL by a significant margin, even with its 300 million parameters compared to the 8 billion parameters used in PlausiVL. In particular, V-JEPA 2 ViT-g_{384} demonstrates a +12.1 points improvement over PlausiVL on action recall-at-5, corresponding to a 44% relative improvement.

In Figure 11 we visualize V-JEPA 2 predictions on three samples from the EK100 validation set, two where the model is successful and one where the model fails. For both successful examples, V-JEPA 2 not only retrieves the correct action correctly with top 1 confidence, but also proposes coherent top 2 to 5 actions, based on the given context. For example, in the top row, the correct action is "wash sink", but "turn on water" or "clean wall" would both have been valid actions given the presence of a tap and a wall. The model also predicts "rinse sponge", which is the current action being performed, probably assuming that this action could still be going on after 1 second. For the failure case, V-JEPA 2 still proposes coherent actions such as "close door" and "put down spices package", but misses the exact nature of the object: "tea package".

Limitations. V-JEPA 2 and the EK100 benchmark have several limitations. First, V-JEPA 2 does not fully solve EK100, there are failure cases where the model either gets the verb, the noun, or both wrong. We study the distribution of these failures in Appendix D.2. Second, we focus here on predicting actions with a 1 second anticipation time. The accuracy of V-JEPA 2 degrades when predicting at longer time horizons, see Appendix D.2. Third, the EK100 benchmark is limited to kitchen environments, with a closed well-defined vocabulary, and we do not know how well V-JEPA 2 generalizes to other environments. This limits the utility and applicability of models trained on EK100. Lastly, actions in EK100 are chosen from a fixed set of categories, making it impossible to generalize to action categories not present in the training set.

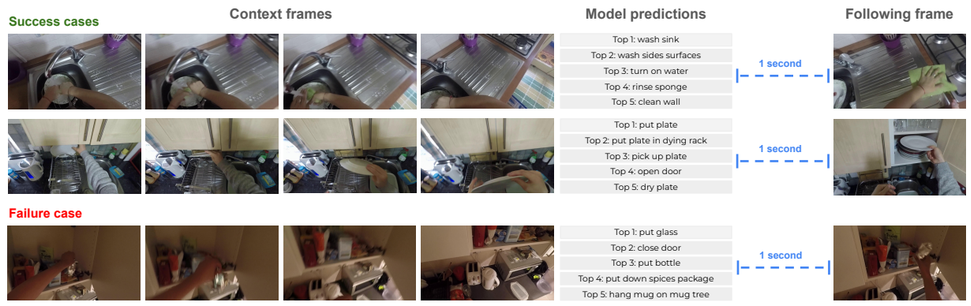


图 11 EK100 预测可视化。（左）上下文选四帧。（中）按可能性排序的模型预测。（右）预判时间 1 秒之后的帧。展示两个成功样本和一个失败样本。

Video-LLaMA（Zhang et al., 2023）和 PlausiVL（Mittal et al., 2024）均利用最多 70 亿参数的大语言模型。

结果。表 5 汇总 EK100 动作预判结果。比较 V-JEPA 2 ViT-L、ViT-H 和 ViT-g，参数从 3 亿增至 10 亿。三者均以 32 帧、8 fps、分辨率 256×256 作为视频上下文；另报告采用分辨率 384×384 的 ViT-g_{384} 结果。以动作预测 recall@5 衡量，V-JEPA 2 随模型规模呈线性扩展行为。3 亿参数的 ViT-L 达到 32.7；增至 10 亿后提升 +5.3 个百分点，动作 recall@5 为 38.0。此外，提高上下文分辨率也有益：分辨率为 384×384 的 ViT-g_{384}，相比使用 256×256 分辨率的模型，进一步提升 +1.7 个百分点。

V-JEPA 2 显著优于此前最先进的 PlausiVL，即使其仅有 3 亿参数，而 PlausiVL 为 80 亿。特别是 ViT-g_{384} 的动作 recall@5 比 PlausiVL 高 +12.1 个百分点，相对改善为 44%。

图 11 可视化 EK100 验证集的三个 V-JEPA 2 预测样本，两个成功，一个失败。成功样本中，模型不仅将正确动作排在置信度首位，也根据上下文提出合理的第 2 至第 5 个候选。例如上排正确动作为“清洗水槽”，但鉴于水龙头与墙壁存在，“打开水”或“清洁墙壁”也合理。模型还预测当前正在执行的“冲洗海绵”，可能认为一秒后该动作仍在继续。失败样本中，模型提出“关门”“放下调料包”等合理动作，却未识别物体的具体性质：“茶包”。

局限。V-JEPA 2 与 EK100 存在若干局限。首先，模型未完全解决 EK100，会错误预测动词、名词或两者，失败分布见附录 D.2。其次，此处主要预测提前 1 秒的动作，更长时域会降低准确率，见附录 D.2。第三，EK100 仅限厨房环境，词汇表封闭且明确定义，尚不清楚模型能否泛化至其他环境，限制了在该数据集训练的模型的实用性与适用范围。最后，动作从固定类别集合中选择，无法泛化到训练集中未出现的动作类别。

7 Understanding: Video Question Answering

In this section, we explore V-JEPA 2's ability to perform open-language video question answering (VidQA). To enable language capabilities, we train a Multimodal Large Language Model (MLLM) using V-JEPA 2 as the visual encoder in the *non-tokenized early fusion* (Wadekar et al., 2024) setup popularized by the LLaVA family of models (Li et al., 2024b). In this family of MLLMs, a visual encoder is *aligned* with a large language model by projecting the output patch embeddings of the vision encoder to the input embedding space of the LLM. The MLLM is then trained either end-to-end, or with a frozen vision encoder. The majority of the encoders used in MLLMs for VidQA are typically image encoders, which are applied independently per-frame for video inputs (Qwen Team et al., 2025; Zhang et al., 2024b). Popular instances of such encoders are CLIP (Radford et al., 2021), SigLIP (Tschannen et al., 2025), and Perception Encoder (Bolya et al., 2025), which are chosen primarily due to their semantic alignment with language, obtained by pretraining with image-caption pairs. To the best of our knowledge, our work is the first to use a video encoder that is pretrained *without* any language supervision, to train an MLLM for VidQA.

MLLM performance on downstream tasks is also highly dependent on the alignment data. In these experiments we use a dataset of 88.5 million image- and video-text pairs, similar to what was used to train PerceptionLM (Cho et al., 2025). To demonstrate the effectiveness of the V-JEPA 2 encoder, first we compare V-JEPA 2 with other state-of-the-art vision encoders in a *controlled* data setup in Section 7.2, using a subset of 18 million samples. Then, in the same controlled setup, we show that scaling the vision encoder and input resolution size both consistently improve VidQA performance in Section 7.3. Finally, we scale the alignment data, using the full 88.5 million samples to test the limits of language alignment with V-JEPA 2 in Section 7.4. Our results demonstrate that in a controlled data setup, V-JEPA 2 obtains competitive performance on open-ended VidQA tasks compared to other vision encoders. Upon scaling the alignment data, V-JEPA 2 achieves state-of-the-art performance on several VidQA benchmarks.

7.1 Experiment Setup

Video Question Answering Tasks. We evaluate on PerceptionTest (Pătrăucean et al., 2023), which assesses model performance across different skills such as memory, abstraction, physics, and semantics. Additionally, we evaluate on the MVP dataset (Krojer et al., 2024) for physical world understanding, which utilizes a minimal-video pair evaluation framework to mitigate text and appearance biases. We also evaluate on TempCompass, TemporalBench and TOMATO (Liu et al., 2024c; Cai et al., 2024; Shangguan et al., 2024) to investigate temporal understanding, and memory capabilities of models. Finally, we report results on general understanding ability using MVBench (Li et al., 2024c), which has a bias towards single-frame appearance features (Krojer et al., 2024; Cores et al., 2024), and TVBench (Cores et al., 2024), which is proposed in the literature as an alternative for general and temporal understanding, mitigating those biases.

Visual Instruction Tuning. To evaluate the V-JEPA 2 representations on visual-question answering tasks, we align V-JEPA 2 with an LLM using the visual instruction tuning procedure from the LLaVA framework (Liu et al., 2024a). This process involves converting the visual encoder outputs (or visual tokens) into LLM inputs using a learnable *projector* module, which is typically an MLP. We train MLLMs through a progressive three-stage process following Liu et al. (2024b): Stage 1, where we train the projector solely on image captioning data; Stage 2, where we train the full model on large-scale image question answering, and Stage 3, where we further train the model on large-scale video captioning and question answering. Through this staged training approach, the LLM incrementally improves its understanding of visual tokens. The vision encoder can either be frozen or finetuned along with the rest of the MLLM. We explore both settings as freezing the vision encoder gives a cleaner signal about the quality of the visual features, while finetuning the vision encoder yields better overall performance. Further details of the visual instruction training are described in Appendix E.

7.2 Comparing with Image Encoders

To isolate the contribution of vision encoders to MLLM performance and compare with V-JEPA 2, we introduce a *controlled* setup: we train individual MLLMs with different state-of-the-art encoders using the

7 理解：视频问答

本节研究 V-JEPA 2 进行开放语言视频问答（VidQA）的能力。为赋予其语言能力，我们采用 LLaVA 模型家族普及的非词元化早期融合设置（Wadekar et al., 2024; Li et al., 2024b），以 V-JEPA 2 作为视觉编码器训练多模态大语言模型（MLLM）。在这类 MLLM 中，通过将视觉编码器输出的图块嵌入投影到 LLM 的输入嵌入空间，使视觉编码器与大语言模型对齐。随后对 MLLM 进行端到端训练，或冻结视觉编码器训练。用于 VidQA 的 MLLM 多数采用图像编码器，对视频输入逐帧独立处理（Qwen Team et al., 2025; Zhang et al., 2024b）。常用编码器包括 CLIP（Radford et al., 2021）、SigLIP（Tschannen et al., 2025）和 Perception Encoder（Bolya et al., 2025）；选择它们主要因为图像–描述对预训练使其与语言具有语义对齐。据我们所知，本研究首次采用未经任何语言监督预训练的视频编码器，训练用于 VidQA 的 MLLM。

MLLM 下游性能也高度依赖对齐数据。实验采用 8850 万对图像／视频–文本数据，与 PerceptionLM 训练数据类似（Cho et al., 2025）。为验证 V-JEPA 2 编码器的有效性，先在第 7.2 节使用 1800 万样本子集，在受控数据设置下与其他先进视觉编码器比较。随后第 7.3 节在同一受控设置中展示，增大视觉编码器与输入分辨率均一致改善 VidQA 性能。最后第 7.4 节使用全部 8850 万样本扩大对齐数据，测试 V-JEPA 2 的语言对齐极限。结果表明，在受控数据设置中，V-JEPA 2 的开放式 VidQA 性能可与其他视觉编码器竞争；扩大对齐数据后，在多项 VidQA 基准上达到最先进水平。

7.1 实验设置

视频问答任务。采用 PerceptionTest（Pătrăucean et al., 2023）评估记忆、抽象、物理与语义等能力；使用 MVP（Krojer et al., 2024）评估物理世界理解，其最小视频对框架用于缓解文本和外观偏差。还采用 TempCompass、TemporalBench 和 TOMATO（Liu et al., 2024c; Cai et al., 2024; Shangguan et al., 2024）研究时间理解与记忆能力。最后，以 MVBench（Li et al., 2024c）报告一般理解能力；该基准偏向单帧外观特征（Krojer et al., 2024; Cores et al., 2024）。同时采用文献提出的 TVBench（Cores et al., 2024），作为缓解上述偏差的一般与时间理解评估替代方案。

视觉指令微调。为在视觉问答任务上评估 V-JEPA 2 表征，采用 LLaVA 框架的视觉指令微调过程（Liu et al., 2024a），使 V-JEPA 2 与 LLM 对齐。通过可学习的投影器模块（通常为 MLP），将视觉编码器输出（视觉词元）转为 LLM 输入。遵循 Liu et al.（2024b），分三个渐进阶段训练 MLLM：阶段 1，仅用图像描述数据训练投影器；阶段 2，在大规模图像问答数据上训练完整模型；阶段 3，在大规模视频描述和问答数据上继续训练。此流程使 LLM 逐步增进对视觉词元的理解。视觉编码器既可冻结，也可与 MLLM 其余部分一起微调。我们研究两者，因为冻结编码器更清楚地反映视觉特征质量，而微调带来更好的总体性能。视觉指令训练详见附录 E。

7.2 与图像编码器比较

为隔离视觉编码器对 MLLM 的贡献并与 V-JEPA 2 比较，引入受控设置：使用不同先进编码器训练各 MLLM，采用

Table 6 Comparison between off-the-shelf image encoders and V-JEPA 2 in frozen encoder setting. All experiments use the same LLM backbone (Qwen2-7B-Instruct), data, and training setup with a **frozen** vision encoder. PerceptionTest accuracy is reported on the validation set post SFT.

Method	Params Enc / LLM	Avg.	PerceptionTest SFT / Acc	MVP Paired-Acc	TempCompass multi-choice	TemporalBench (MBA-short QA)	TVBench Acc	TOMATO Acc	MVBench Acc
Off-the-shelf image encoders									
DINOv2 ViT-g$_{518}$	1.1B/7B	45.7	67.1	22.4	62.3	26.8	47.6	32.0	61.8
SigLIP2 ViT-g$_{384}$	1.1B/7B	48.1	**72.4**	26.2	66.8	25.7	48.7	33.2	64.0
PE ViT-G/14$_{448}$	1.9B/7B	49.1	72.3	26.7	67.0	27.5	51.6	34.0	64.7
V-JEPA 2 ViT-g$_{512}$	1B/7B	**52.3**	72.0	**31.1**	**69.2**	**33.3**	**55.9**	**37.0**	**67.7**

Table 7 Scaling Vision Encoder Size and Resolution. We scale the vision encoder from 300 million to 1 billion parameters and input resolution from 256 pixels to 512 pixels. All experiments use the same LLM backbone (Qwen2-7B-Instruct), data, and end-to-end training (unfrozen vision encoder) setup. PerceptionTest accuracy is reported on the validation set post SFT. Increasing V-JEPA 2 encoder scale and resolution improve average performances on VidQA tasks.

Method	Params Enc / LLM	Avg.	PerceptionTest SFT / Acc	MVP Paired-Acc	TempCompass multi-choice	TemporalBench (MBA-short QA)	TVBench Acc	TOMATO Acc	MVBench Acc
End-to-end Evaluation									
V-JEPA 2 ViT-L$_{256}$	300M/7B	51.7	74.6	32.3	70.1	30.2	50.9	36.5	67.1
V-JEPA 2 ViT-H$_{256}$	600M/7B	52.0	74.7	30.6	70.9	29.8	54.6	35.1	68.0
V-JEPA 2 ViT-g$_{256}$	1B/7B	52.3	75.5	31.9	70.7	28.3	54.2	37.3	68.3
V-JEPA 2 ViT-g$_{384}$	1B/7B	54.0	76.5	33.0	**71.7**	**33.1**	56.5	**39.0**	68.5
V-JEPA 2 ViT-g$_{512}$	1B/7B	**54.4**	**77.7**	**33.7**	71.6	32.3	**57.5**	38.5	**69.5**

same LLM backbone and training setup. In this controlled setup, we use Qwen2-7B-Instruct (Yang et al., 2024a) and freeze the vision encoder. We use 18 million image and video-text aligned samples. We first compare V-JEPA 2, pretrained at resolution 512×512 with DINOv2 (Oquab et al., 2023), SigLIP-2 (Tschannen et al., 2025), and Perception Encoder (Bolya et al., 2025).

We observe that V-JEPA 2 exhibits competitive performance in the frozen setup, outperforming DINOv2, SigLIP, and Perception Encoder (PE) in all of the tested benchmarks (Table 6) except PerceptionTest where V-JEPA 2 slightly underperforms SigLIP and PE. The improvement is especially noticeable on MVP, TemporalBench, and TVBench — benchmarks that are primarily focused on temporal understanding. Additionally, since we only change the vision encoder, we provide evidence that a video encoder trained without language supervision can outperform encoders trained with language supervision, in contrast to conventional wisdom (Tong et al., 2024; Li et al., 2024b; Liu et al., 2024d; Yuan et al., 2025). The results also indicate that using a video encoder instead of an image encoder for VidQA improves spatiotemporal understanding, highlighting the need to develop better video encoders.

7.3 Scaling Vision Encoder Size and Input Resolution

Prior work (Fan et al., 2025) suggests that scaling the vision encoder and input resolution significantly improves VQA performance for self-supervised image encoders. Thus, we scale V-JEPA 2 from 300M to 1B parameters and the input resolution from 256 to 512 pixels, and show the results in Table 7. When increasing vision encoder capacity from 300M to 1B parameters for a fixed input resolution of 256 pixels, we observe improvements of 0.9 points on PerceptionTest, 3.3 points on TVBench, and 1.2 points on MVBench. Additionally, increasing the input resolution to 512 pixels yields further improvements across all downstream tasks, such as an improvement of 2.2 points on PerceptionTest, 4.0 points on TemporalBench, and 3.3 points on TVBench. These results suggest that further scaling the vision encoder and input resolution is a promising

表 6 冻结编码器设置下，现成图像编码器与 V-JEPA 2 比较。所有实验采用相同 Qwen2-7B-Instruct 骨干、数据和训练设置，视觉编码器冻结。PerceptionTest 报告 SFT 后的验证准确率。

方法	参数量 编码器/LLM	Avg.	PerceptionTest SFT / Acc	MVP Paired-Acc	TempCompass multi-choice	TemporalBench (MBA-short QA)	TVBench Acc	TOMATO Acc	MVBench Acc
现成图像编码器									
DINOv2 ViT-g$_{518}$	1.1B/7B	45.7	67.1	22.4	62.3	26.8	47.6	32.0	61.8
SigLIP2 ViT-g$_{384}$	1.1B/7B	48.1	**72.4**	26.2	66.8	25.7	48.7	33.2	64.0
PE ViT-G/14$_{448}$	1.9B/7B	49.1	72.3	26.7	67.0	27.5	51.6	34.0	64.7
V-JEPA 2 ViT-g$_{512}$	1B/7B	**52.3**	72.0	**31.1**	**69.2**	**33.3**	**55.9**	**37.0**	**67.7**

表 7 扩大视觉编码器与分辨率。编码器从 3 亿增至 10 亿参数，输入从 256 增至 512 像素。全部实验使用相同 Qwen2-7B-Instruct 骨干、数据及端到端训练设置（视觉编码器未冻结）。PerceptionTest 报告 SFT 后验证准确率。扩大 V-JEPA 2 编码器和分辨率改善 VidQA 平均性能。

方法	参数量 编码器/LLM	Avg.	PerceptionTest SFT / Acc	MVP Paired-Acc	TempCompass multi-choice	TemporalBench (MBA-short QA)	TVBench Acc	TOMATO Acc	MVBench Acc
端到端评估									
V-JEPA 2 ViT-L$_{256}$	300M/7B	51.7	74.6	32.3	70.1	30.2	50.9	36.5	67.1
V-JEPA 2 ViT-H$_{256}$	600M/7B	52.0	74.7	30.6	70.9	29.8	54.6	35.1	68.0
V-JEPA 2 ViT-g$_{256}$	1B/7B	52.3	75.5	31.9	70.7	28.3	54.2	37.3	68.3
V-JEPA 2 ViT-g$_{384}$	1B/7B	54.0	76.5	33.0	**71.7**	**33.1**	56.5	**39.0**	68.5
V-JEPA 2 ViT-g$_{512}$	1B/7B	**54.4**	**77.7**	**33.7**	71.6	32.3	**57.5**	38.5	**69.5**

相同的 LLM 骨干和训练设置。受控设置使用 Qwen2-7B-Instruct（Yang et al., 2024a），冻结视觉编码器，使用 1800 万对图像／视频–文本对齐样本。首先将以分辨率 512×512 预训练的 V-JEPA 2，与 DINOv2（Oquab et al., 2023）、SigLIP-2（Tschannen et al., 2025）、Perception Encoder（Bolya et al., 2025）比较。

冻结设置中，V-JEPA 2 表现具有竞争力；除 PerceptionTest 略低于 SigLIP 和 PE 外，在全部测试基准上优于 DINOv2、SigLIP、Perception Encoder（表 6）。改善在主要关注时间理解的 MVP、TemporalBench 和 TVBench 上尤其明显。此外，由于仅改变视觉编码器，这些结果说明：未经语言监督训练的视频编码器能够胜过带语言监督的编码器，与传统认识不同（Tong et al., 2024; Li et al., 2024b; Liu et al., 2024d; Yuan et al., 2025）。结果也表明，VidQA 使用视频而非图像编码器能改善时空理解，强调开发更好视频编码器的必要性。

7.3 扩大视觉编码器与输入分辨率

先前研究（Fan et al., 2025）表明，增大视觉编码器与输入分辨率，可显著改善自监督图像编码器的 VQA 性能。因此，将 V-JEPA 2 从 3 亿扩展至 10 亿参数，输入分辨率从 256 扩展至 512 像素，结果见表 7。固定 256 像素输入、容量从 3 亿增至 10 亿时，PerceptionTest 提升 0.9 个百分点、TVBench 提升 3.3、MVBench 提升 1.2。进一步增至 512 像素，各下游任务继续改善，例如 PerceptionTest 提升 2.2、TemporalBench 提升 4.0、TVBench 提升 3.3。这说明继续扩大视觉编码器与输入分辨率是有前景的

Table 8 Comparison with state-of-the-art. We use the full 88.5M-sample alignment dataset and train using the same methodology as PLM 8B Cho et al. (2025), using a Llama 3.1 backbone. We observe significant improvements in downstream evaluations, obtaining state-of-the-art results in the 8B model class. PerceptionTest accuracy is reported on the test set with SFT for V-JEPA 2; all other results are zero-shot.

Method	Params Enc / LLM	Avg.	PerceptionTest Test Acc	MVP Paired-Acc	TempCompass multi-choice	TemporalBench (MBA-short QA)	TOMATO Acc	TVBench Acc	MVBench Acc
≤ 8B Video Language Models Results Reported in the Literature									
InternVL-2.5 (Chen et al., 2024)	300M/7B	52.1	68.9	39.9	68.3	24.3	29.4	61.6	72.6
Qwen2VL (Wang et al., 2024a)	675M/7B	47.0	66.9	29.2	67.9	20.4	31.5	46.0	67.0
Qwen2.5VL (Qwen Team et al., 2025)	1B/7B	49.7	70.5	36.7	71.7	24.5	24.6	50.5	69.6
PLM 8B (Cho et al., 2025)	1B/8B	56.7	82.7	39.7	72.7	28.3	33.2	**63.5**	**77.1**
V-JEPA 2 ViT-g$_{384}$ LLama 3.1 8B	1B/8B	**59.5**	**84.0**	**44.5**	**76.9**	**36.7**	**40.3**	60.6	73.5

direction for improving VidQA performance.

7.4 Improving the State-of-the-art by Scaling Data

After developing a better understanding of the capabilities of V-JEPA 2 for training an MLLM in the controlled setup, we study the effect of increasing alignment dataset size to improve the state-of-the-art of VidQA. Step changes on downstream task performance are often achieved by increasing the scale of the training data, as observed by Cho et al. (2025). To that end, we increase the scale of MLLM training data from 18 million to the full 88.5 million (4.7×). While increasing the model resolution helps in downstream performance, it comes with the challenge of accommodating a large number of visual tokens in the LLM input. We therefore choose V-JEPA 2 ViT-g$_{384}$, leading to 288 visual tokens per frame. We follow the same recipe as Cho et al. (2025) to train V-JEPA 2 ViT-g$_{384}$, using Llama 3.1 as the backbone. To simplify the training process, we use an MLP projector without pooling. Details on the scaling training setup are described in Appendix E.

Scaling the data uniformly improves the downstream benchmark performance, resulting in state-of-the-art results (Table 8) on multiple benchmarks — PerceptionTest, MVP, TempCompass, TemporalBench and TOMATO. Compared to the current state-of-the-art PerceptionLM 8B (Cho et al., 2025), we observe an increase of 1.3 points on accuracy for PerceptionTest test set, 4.8 points on paired accuracy for MVP, 4.2 points on accuracy for TempCompass, 8.4 points on Multi-binary accuracy for short-QA segment for TemporalBench and 7.1 points on accuracy for TOMATO. V-JEPA 2 does not outperform PerceptionLM on TVBench and MVBench, however it still significantly outperforms other related baselines (InternVL 2.5, Qwen2VL and Qwen2.5VL). These results underscore the need to scale training data for vision-language alignment and provide evidence that an encoder pretrained without language supervision, such as V-JEPA 2, can achieve state-of-the-art results with sufficient scale.

8 Related Work

World models and planning. As early as the work of Sutton and Barto (1981) and Chatila and Laumond (1985), AI researchers have sought to build agents that use internal models of the world — modeling both dynamics of the world, as well as mapping the static environment — to enable efficient planning and control. Previous work has investigated world models in simulated tasks (Fragkiadaki et al., 2015; Ha and Schmidhuber, 2018; Hafner et al., 2019b,a; Hansen et al., 2022, 2023; Hafner et al., 2023; Schrittwieser et al., 2020; Samsami et al., 2024), as well as real-world locomotion and manipulation tasks (Lee et al., 2020; Nagabandi et al., 2020; Finn et al., 2016; Ebert et al., 2017, 2018; Yen-Chen et al., 2020). World model approaches either learn predictive models directly in pixel-space (Finn et al., 2016; Ebert et al., 2017, 2018; Yen-Chen et al., 2020), in a learned representation space (Watter et al., 2015; Agrawal et al., 2016; Ha and Schmidhuber, 2018; Hafner et al., 2019b; Nair et al., 2022; Wu et al., 2023b; Tomar et al., 2024; Hu et al., 2024; Lancaster et al., 2024), or utilizing more structured representation spaces such as keypoint representations (Manuelli et al., 2020; Das

表 8 与最先进方法比较。使用完整 8850 万样本对齐数据，以 Llama 3.1 为骨干，采用与 PLM 8B（Cho et al., 2025）相同的训练方法。下游评估显著改善，在 80 亿模型级别达到最先进结果。V-JEPA 2 的 PerceptionTest 准确率在 SFT 后的测试集报告，其余结果均为零样本。

方法	参数量 编码器/LLM	Avg.	PerceptionTest Test Acc	MVP Paired-Acc	TempCompass multi-choice	TemporalBench (MBA-short QA)	TOMATO Acc	TVBench Acc	MVBench Acc
≤ 8B 文献报告的视频–语言模型结果									
InternVL-2.5 (Chen et al., 2024)	300M/7B	52.1	68.9	39.9	68.3	24.3	29.4	61.6	72.6
Qwen2VL (Wang et al., 2024a)	675M/7B	47.0	66.9	29.2	67.9	20.4	31.5	46.0	67.0
Qwen2.5VL (Qwen Team et al., 2025)	1B/7B	49.7	70.5	36.7	71.7	24.5	24.6	50.5	69.6
PLM 8B (Cho et al., 2025)	1B/8B	56.7	82.7	39.7	72.7	28.3	33.2	**63.5**	**77.1**
V-JEPA 2 ViT-g$_{384}$ LLama 3.1 8B	1B/8B	**59.5**	**84.0**	**44.5**	**76.9**	**36.7**	**40.3**	60.6	73.5

改善 VidQA 性能的方向。

7.4 通过扩大数据推进最先进水平

在受控设置中更充分理解 V-JEPA 2 训练 MLLM 的能力后，我们研究扩大对齐数据集，以推进 VidQA 的最先进水平。正如 Cho et al.（2025）所观察，增加训练数据规模往往使下游性能出现跃升。因此，将 MLLM 训练数据从 1800 万扩大到全部 8850 万（4.7×）。提高模型分辨率虽有益，却也要求 LLM 输入容纳大量视觉词元。因此选择每帧产生 288 个视觉词元的 V-JEPA 2 ViT-g$_{384}$, 。遵循 Cho et al.（2025）的方案，以 Llama 3.1 为骨干训练 ViT-g$_{384}$, 。为简化训练，采用不带池化的 MLP 投影器。规模扩展训练设置详见附录 E。

扩大数据规模一致改善下游基准性能，在 PerceptionTest、MVP、TempCompass、TemporalBench 和 TOMATO 等多项基准上达到最先进水平（表 8）。相比目前最先进的 PerceptionLM 8B（Cho et al., 2025），PerceptionTest 测试集准确率提升 1.3 个百分点，MVP 成对准确率提升 4.8，TempCompass 准确率提升 4.2，TemporalBench 短问答部分的多重二元准确率提升 8.4，TOMATO 准确率提升 7.1。V-JEPA 2 在 TVBench 与 MVBench 上未超过 PerceptionLM，但仍显著优于其他相关基线（InternVL 2.5、Qwen2VL、Qwen2.5VL）。结果强调扩大视觉–语言对齐数据的必要性，并说明 V-JEPA 2 这类未经语言监督预训练的编码器，在规模足够时也能达到最先进水平。

8 相关工作

世界模型与规划。早在 Sutton and Barto（1981）及 Chatila and Laumond（1985）的工作中，人工智能研究者就尝试构建利用内部世界模型的智能体，通过建模世界动态并绘制静态环境，实现高效规划和控制。先前研究在仿真任务（Fragkiadaki et al., 2015; Ha and Schmidhuber, 2018; Hafner et al., 2019b,a; Hansen et al., 2022, 2023; Hafner et al., 2023; Schrittwieser et al., 2020; Samsami et al., 2024），以及真实世界的移动与操作任务（Lee et al., 2020; Nagabandi et al., 2020; Finn et al., 2016; Ebert et al., 2017, 2018; Yen-Chen et al., 2020）中研究世界模型。这些方法可直接在像素空间学习预测模型（Finn et al., 2016; Ebert et al., 2017, 2018; Yen-Chen et al., 2020），在学习到的表征空间中建模（Watter et al., 2015; Agrawal et al., 2016; Ha and Schmidhuber, 2018; Hafner et al., 2019b; Nair et al., 2022; Wu et al., 2023b; Tomar et al., 2024; Hu et al., 2024; Lancaster et al., 2024），或采用更具结构的表征空间，例如关键点表征（Manuelli et al., 2020; Das

et al., 2020). Previous approaches that have demonstrated real world performance on robotics tasks have trained task-specific world models, and they rely on interaction data from the environment in which the robot is deployed. Evaluation is focused on demonstrating performance of world modeling approaches within the explored task space, instead of generalization to new environments or unseen objects. In this work we train a task-agnostic world model, and demonstrate generalization to new environments and objects.

Some recent works leverage both internet-scale video and interaction data towards training general purpose (task-agnostic) action-conditioned video generation models for autonomous robots (Bruce et al., 2024; Agarwal et al., 2025; Russell et al., 2025). However, thus far these approaches only demonstrate the ability to generate visually valid-looking plans given actions of the robot, but they have not demonstrated the ability to use those models to actually control the robot.

Other works have explored the integration of generative modeling into policy learning (Du et al., 2024; Wu et al., 2023a; Zhao et al., 2025; Zhu et al., 2025; Du et al., 2023; Zheng et al., 2025; Rajasegaran et al., 2025). Differently from this line of work, our goal is to leverage a world model through model-predictive control instead of policy learning to avoid the imitation learning phase that requires expert trajectories. Both approaches are orthogonal and could be combined in future works. Closest to our work, Zhou et al. (2024); Sobal et al. (2025) show that you can learn a world model stage-wise or end-to-end and use it to solve planning tasks zero-shot. While those previous works focus on small-scale planning evaluation, we show that similar principles can be scaled and used to solve real-world robotic tasks.

Vision-Language-Action models for Robotic Control. Recent imitation learning approaches in real-world robotic control have made significant progress towards learning policies that show increasingly good generalization capabilities. This is achieved by leveraging video-languange models that have been pre-trained on internet scale video and text data, which are then fine-tuned (or adapted) to also predict actions by using behavior cloning from expert demonstrations (Driess et al., 2023; Brohan et al., 2023; Black et al., 2024; Kim et al., 2024; Bjorck et al., 2025; Black et al., 2025). Although these approaches show promising generalization results, it is unclear whether they will be able to learn to predict behaviors that were not demonstrated in the training data since they lack an explicit predictive model of the world and do leverage inference-time computation for planning. They require high-quality large scale teleoperation data, and can only utilize successful trajectories. In contrast, we focus on leveraging any interaction data whether it comes from a successful or failed interaction with the environment.

Vision Foundation Models. Video foundation models in computer vision have shown that large-scale observation datasets comprised of images and/or videos can be leveraged to learn generalist vision encoders that perform well along a wide range of downstream tasks using self-supervised learning approaches from images (Grill et al., 2020; Assran et al., 2023; Oquab et al., 2023; Fan et al., 2025), videos (Bardes et al., 2024; Carreira et al., 2024; Wang et al., 2023; Rajasegaran et al., 2025), with weak language supervision (Wang et al., 2024b; Bolya et al., 2025), or a combination thereof (Tschannen et al., 2025; Fini et al., 2024). Previous works, however, tend to focus on understanding evaluation using probe-based evaluation or visual question answering tasks after aligning with a large-language model. While such tasks have served to drive progress, it remains an important goal of a visual system to enable an agent to interact with the physical world (Gibson, 1979). Beyond results on visual understanding tasks, we investigate how large-scale self-supervised learning from video can enable solving planning tasks in new environments in a zero-shot manner.

et al., 2020）。此前在真实机器人任务上取得结果的方法，训练任务专用世界模型，依赖部署环境的交互数据。评估关注已探索任务空间内的表现，而非泛化至新环境或未见物体。本研究训练任务无关的世界模型，并展示对新环境和物体的泛化。

近期研究结合互联网规模视频与交互数据，训练通用、与任务无关的动作条件视频生成模型，用于自主机器人（Bruce et al., 2024; Agarwal et al., 2025; Russell et al., 2025）。但至今只展示了给定机器人动作时，生成视觉上合理规划的能力，尚未展示用这些模型实际控制机器人的能力。

其他研究探索将生成建模整合进策略学习（Du et al., 2024; Wu et al., 2023a; Zhao et al., 2025; Zhu et al., 2025; Du et al., 2023; Zheng et al., 2025; Rajasegaran et al., 2025）。与此不同，我们通过模型预测控制而非策略学习来使用世界模型，避免需要专家轨迹的模仿学习阶段。两种方法相互独立，未来可结合。最接近本研究的是 Zhou et al.（2024）和 Sobal et al.（2025），它们展示可分阶段或端到端学习世界模型，并零样本解决规划任务。此前研究主要进行小规模规划评估，我们说明类似原则可以扩展规模并用于真实机器人任务。

用于机器人控制的视觉–语言–动作模型。近期真实世界机器人控制中的模仿学习，在学习具有越来越好泛化能力的策略方面取得明显进展。它们利用在互联网规模视频与文本上预训练的视频–语言模型，再用专家演示进行行为克隆，通过微调或适配使模型预测动作（Driess et al., 2023; Brohan et al., 2023; Black et al., 2024; Kim et al., 2024; Bjorck et al., 2025; Black et al., 2025）。虽然泛化结果可喜，但尚不清楚它们能否预测训练数据中未演示过的行为，因为它们缺少显式世界预测模型，并在推理时利用计算进行规划。它们需要高质量、大规模遥操作数据，且只能利用成功轨迹。相比之下，我们关注利用任意交互数据，无论与环境的交互成功还是失败。

视觉基础模型。计算机视觉中的视频基础模型表明，可利用由图像和／或视频组成的大规模观测数据集，通过图像自监督学习（Grill et al., 2020; Assran et al., 2023; Oquab et al., 2023; Fan et al., 2025）、视频自监督学习（Bardes et al., 2024; Carreira et al., 2024; Wang et al., 2023; Rajasegaran et al., 2025）、弱语言监督（Wang et al., 2024b; Bolya et al., 2025）或其组合（Tschannen et al., 2025; Fini et al., 2024），学习在广泛下游任务上表现良好的通用视觉编码器。但先前研究往往侧重理解能力评估，通过探测器进行评估，或在与大语言模型对齐后开展视觉问答。尽管这些任务推动了进展，使智能体能够与物理世界交互仍是视觉系统的重要目标（Gibson, 1979）。除视觉理解结果外，我们还研究大规模视频自监督学习如何支持在新环境中零样本解决规划任务。

9 Conclusion

This study demonstrates how joint-embedding predictive architectures, learning in a self-supervised manner from web-scale data and a small amount of robot interaction data, can yield a world model capable of understanding, predicting, and planning in the physical world. V-JEPA 2 achieves state-of-art performances on action classification requiring motion understanding and human action anticipation. V-JEPA 2 also outperforms previous vision encoders on video questions-answering tasks when aligned with a large-language model. Additionally, post-training an action-conditioned world model, V-JEPA 2-AC, using V-JEPA 2's representation, enables successful zero-shot prehensile manipulation tasks, such as Pick-and-Place, with real-world robots. These findings indicate V-JEPA 2 is a step towards developing advanced AI systems that can effectively perceive and act in their environment.

Future work. There are several important avenues for future work to address limitations of V-JEPA 2. First, in this work we have focused on tasks requiring predictions up to roughly 16 seconds into the future. This enables planning for simpler manipulation tasks, like grasp and reach-with-object, from a single goal image. However, to extend this to longer-horizon tasks such as pick-and-place or even more complex tasks, without requiring sub-goals will require further innovations in modeling. Developing approaches for hierarchical models capable of making predictions across multiple spatial and temporal scales, at different levels of abstraction, is a promising direction.

Second, as mentioned in Section 4, V-JEPA 2-AC currently relies upon tasks specified as image goals. Although this may be natural for some tasks, there are other situations where language-based goal specification may be preferable. Extending the V-JEPA 2-AC to accept language-based goals, e.g., by having a model that can embed language-based goals into the V-JEPA 2-AC representation space, is another important direction for future work. The results described in Section 7, aligning V-JEPA 2 with a language model, may serve as a starting point.

Finally, in this work we scaled V-JEPA 2 models up to a modest 1B parameters. The results in Section 2 demonstrated consistent performance improvements while scaling to this level. Previous work has investigated scaling vision encoders to as large as 20B parameters (Zhai et al., 2022; Carreira et al., 2024). Additional work is needed in this direction to develop scalable pre-training recipes that lead to sustained performance improvements with scale.

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9 结论

本研究说明，联合嵌入预测架构从互联网规模数据与少量机器人交互数据中进行自监督学习，可以得到能在物理世界中理解、预测和规划的世界模型。V-JEPA 2 在需要运动理解的动作分类及人类动作预判上达到最先进水平；与大语言模型对齐后，视频问答也优于此前视觉编码器。此外，在 V-JEPA 2 表征上对动作条件世界模型 V-JEPA 2-AC 后训练，可使真实机器人成功零样本执行抓取放置等抓持操作。这表明 V-JEPA 2 向能够有效感知环境并采取行动的先进人工智能系统迈进了一步。

未来工作。有几项重要方向可解决 V-JEPA 2 的局限。首先，本研究关注预测未来约 16 秒以内的任务，支持根据单张目标图像规划抓取、持物到达等较简单操作。但若不提供子目标，要扩展到抓取放置或更复杂的长时域任务，需要建模创新。开发能在多个时空尺度、不同抽象层次上预测的层级模型，是有前景的方向。

其次，如第 4 节所述，V-JEPA 2-AC 当前依赖图像目标指定任务。某些任务适合这种方式，另一些则更适合语言目标。将模型扩展为接受语言目标，例如把语言目标嵌入其表征空间，是另一重要方向。第 7 节将 V-JEPA 2 与语言模型对齐的结果，可作为起点。

最后，本研究只将 V-JEPA 2 扩至相对适中的 10 亿参数，第 2 节说明扩至该规模时性能一致改善。此前研究已探索最高 200 亿参数的视觉编码器（Zhai et al., 2022; Carreira et al., 2024）。需要进一步开发可扩展预训练方案，使性能随规模持续提高。

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Appendix

A V-JEPA 2 Pretraining

A.1 Pretraining Hyperparameters

As detailed in Section 2.4, our training pipeline consisted of two phases: 1) a constant learning rate phase and 2) a cooldown phase. For all models, we trained in the first phase until we observed plateauing or diminishing performance on the IN1K, COIN, and SSv2 tasks. At this point, we initiated the cooldown phase.

Table 9 Pretraining Hyperparameters. Common parameters for pretraining large computer vision models. We report these parameters for both the primary training phase and the cooldown phase.

Parameter	Primary Phase	Cooldown Phase
Number of frames	16	64
Frames per Second	4.0	4.0
Crop Size	256	[256, 384, 512]
Random Resize Aspect Ratio	[0.75 1.35]	[0.75, 1.35]
Random Resize Scale	[0.3, 1.0]	[0.3, 1.0]
Steps	Variable	12000
Warmup Steps	12000	N/A
Batch Size (global)	3072	3072
Starting Learning Rate	1e-4	5.25e-4
Final Learning Rate	5.25e-4	1e-6
Weight Decay	0.04	0.04
EMA	0.99925	0.99925
Spatial Mask Scale	[0.15, 0.7]	[0.15, 0.7]
Temporal Mask Scale	[1.0, 1.0]	[1.0, 1.0]
Mask Aspect Ratio	[0.75 1.5]	[0.75, 1.5]
Tubelet Size	2	2
Patch Size	16	16

Training in the first phase began with a learning rate warmup for 12,000 steps followed by a constant learning rate for the rest of the phase. We checked evaluations every 60,000 steps. The cooldown phase began with a learning rate at 5.25e-4, which was linearly ramped down to the final learning rate. Throughout both phases, all other hyperparameters were kept constant.

In the cooldown phase we increased the number of frames per clip while keeping the frames-per-second constant, as we saw a substantial benefit from feeding the model more frames (see Figure 5). In addition, we also increased the crop size of the model in this phase, which gave a substantial benefit to tasks like IN1K, which goes from 84.6 at a 256 crop to 85.1 at a 384 crop. Hyperparameters for both phases are summarized in Table 9.

Throughout the Appendix, we refer to a "abbreviated" training recipe that corresponds to a 90,000-step training following the procedure of Bardes et al. (2024). There are a few key differences with the abbreviated recipe. The first is the learning rate: the abbreviated recipe begins with a linear warmup followed by a cosine decay. The second are the schedules for weight decay and EMA, which are linearly ramped from a starting to a final value. The last is the total number of steps, which is restricted to 90,000. We use the abbreviated schedule for several of our ablations on data mixtures, as this allows us to interrogate the effects of data curation on a shorter compute budget.

A.2 Pretraining data

We began curation of YT1B by applying scene extraction via the PySceneDetect library,[3] which splits videos into clips at scene transitions. We discard scenes shorter than 4 seconds, retaining 316 million scenes. The

[3] https://github.com/Breakthrough/PySceneDetect

附录

A V-JEPA 2 预训练

A.1 预训练超参数

如第 2.4 节，训练包含恒定学习率和收尾两个阶段。所有模型第一阶段训练至 IN1K、COIN、SSv2 性能进入平台或下降，再启动收尾训练。

表 9 预训练超参数。报告大型视觉模型主要训练阶段和收尾阶段的常用参数。

参数	主要阶段	收尾阶段
帧数	16	64
每秒帧数	4.0	4.0
裁剪尺寸	256	[256, 384, 512]
随机缩放宽高比	[0.75 1.35]	[0.75, 1.35]
随机缩放比例	[0.3, 1.0]	[0.3, 1.0]
步数	变量	12000
预热步数	12000	N/A
全局批量大小	3072	3072
起始学习率	1e-4	5.25e-4
最终学习率	5.25e-4	1e-6
权重衰减	0.04	0.04
EMA	0.99925	0.99925
空间遮挡比例	[0.15, 0.7]	[0.15, 0.7]
时间遮挡比例	[1.0, 1.0]	[1.0, 1.0]
遮挡宽高比	[0.75 1.5]	[0.75, 1.5]
时空小块大小	2	2
图块大小	16	16

第一阶段前 12000 步预热学习率，其余采用恒定学习率，每 60000 步评估。收尾阶段学习率从 5.25e-4 开始，线性降至终值。两阶段其他超参数均保持恒定。

收尾阶段增加每个片段帧数，但帧率不变，因为更多帧有显著收益（图 5）。还增加裁剪尺寸，对 IN1K 等任务明显有益：256 裁剪时为 84.6，384 裁剪时为 85.1。两个阶段的超参数见表 9。

附录中的“缩短”训练方案指遵循 Bardes et al.（2024）的 90000 步训练。它有几项关键区别：学习率先线性预热再余弦衰减；权重衰减和 EMA 从初值线性增加至终值；总步数限制为 90000。数据混合消融使用该较短方案，以在较少计算预算内研究数据筛选的影响。

A.2 预训练数据

YT1B 筛选首先使用 PySceneDetect 库[3] 提取场景，按场景切换将视频划为片段。丢弃不足 4 秒的场景，保留 3.16 亿个。随后，

[3] https://github.com/Breakthrough/PySceneDetect

Table 10 Abbreviated Pretraining Hyperparameters. Common parameters for pretraining large computer vision models, targeting our abbreviated recipe.

Parameter	Abbreviated Recipe
Number of frames	16
Frames per Second	4.0
Crop Size	256
Random Resize Aspect Ratio	[0.75 1.35]
Random Resize Scale	[0.3, 1.0]
Steps	90000
Warmup Steps	12000
Batch Size (global)	3072
Starting Learning Rate	2e-4
Larning Rate	6.25e-4
Final Learning Rate	1e-6
Starting Weight Decay	0.04
Final Weight Decay	0.4
Starting EMA	0.999
Final EMA	1.0
Spatial Mask Scale	[0.15, 0.7]
Temporal Mask Scale	[1.0, 1.0]
Mask Aspect Ratio	[0.75 1.5]
Tubelet Size	2
Patch Size	16

DINOv2 ViT-L model is then applied on the middle frame of each clip to extract scene embeddings. YT1B embeddings are then clustered into 1.5 million clusters, using the same clustering strategy as Oquab et al. (2023). Embeddings are also extracted in the same manner for all videos in the target distribution, then assigned to the closest YT1B cluster. We only keep those clusters to which at least one target video was assigned—about 210k clusters out of the original 1.5 million. The retained clusters contain 115 million scenes.

Cluster-based retrieval matches the support, but not the weighting, of the target distribution. We use a weighted sampling scheme to rebalance the data to better match the target distribution. W sample from the clusters using a weighted sampling strategy: $w_c = \sum_{d=1}^{D} w_d \times \frac{N_{d,c}}{N_d}$ where w_c is the weighting coefficient for the cth cluster, w_d is the weighting coefficient for the dth target dataset (from Table 11, $N_{d,c}$ is the number of samples from the dth dataset in the cth cluster, N_d is the total number of samples in the dth dataset, and D is the total count of target datasets. We assigned the retrieval weights approximately based on how many scenes were retrieved by each target dataset, with some extra weighting assigned to EpicKitchen. This gave a final curated dataset with statistics more closely matching those of handcrafted datasets from the literature. We found that in isolation, using curated YT1B in place of its uncurated counterpart gave much better results on downstream understanding tasks (see Figure 4).

Table 11 Data Curation Statistics. We summarized the number of extracted scenes and hours of videos across clusters extracted from YT1B. The final line includes duplicates among retrievals of K710, SSv2, COIN, and EpicKitchen.

Retrieval Target	Cluster Count	Number of Scenes	Retrieval Weight
Uncurated YT1B	1.5M	316M	
K710	170k	100M	0.7
SSv2	41k	19M	0.125
COIN	37k	21M	0.125
EpicKitchen	4k	13k	0.05
Final Curated (includes duplicates)	210k	115M	

The overall statistics from how many clusters and scenes were retrieved with this strategy are summarized in Table 11. The overall dataset is weighted towards clusters retrieved with K710. This, combined with its retrieval weight of 0.7, gives the overall curated dataset a heavy Kinetics weighting that we saw reflected in

表 10 缩短训练 预训练超参数。大型视觉模型预训练的常用参数。
模型，面向我们的 缩短方案。

参数	缩短方案
帧数	16
每秒帧数	4.0
裁剪尺寸	256
随机缩放宽高比	[0.75 1.35]
随机缩放比例	[0.3, 1.0]
步数	90000
预热步数	12000
全局批量大小	3072
起始学习率	2e-4
学习率	6.25e-4
最终学习率	1e-6
起始权重衰减	0.04
最终权重衰减	0.4
起始 EMA	0.999
最终 EMA	1.0
空间遮挡比例	[0.15, 0.7]
时间遮挡比例	[1.0, 1.0]
遮挡宽高比	[0.75 1.5]
时空小块大小	2
图块大小	16

随后用 DINOv2 ViT-L 对各片段中间帧提取场景嵌入。采用与 Oquab et al.（2023）相同的聚类策略，将 YT1B 嵌入分为 150 万个簇。以相同方式提取目标分布全部视频的嵌入，并分配至最近的 YT1B 簇。只保留至少分配了一个目标视频的簇，约为原 150 万簇中的 21 万；保留簇包含 1.15 亿个场景。

基于聚类的检索匹配目标分布的支撑集，却不匹配其权重。我们采用加权采样重新平衡数据，使其更符合目标分布。从各簇按加权策略采样：$w_c = \sum_{d=1}^{D} w_d \times \frac{N_d, N_d^c}{c}$，其中 w_c 为第 c 个簇的权重系数，w_d 为第 d 个目标数据集的权重（来自表 11，N_d），为第 d 个数据集在第 c 个簇中的样本数，N_d 为第 d 个数据集的总样本数，D 为目标数据集总数。检索权重大致按各目标集检索出的场景数分配，并额外提高 EpicKitchen 权重。最终筛选数据集的统计性质更接近文献中人工构建的数据集。单独使用时，以筛选 YT1B 替代未筛选版，显著改善下游理解任务结果（见图 4）。

表 11 数据筛选统计。汇总从 YT1B 簇提取的场景数与视频小时数。最后一行包含 K710、SSv2、COIN、EpicKitchen 各检索结果间的重复。

检索目标	簇数	场景数	检索权重
未筛选 YT1B	1.5M	316M	
K710	170k	100M	0.7
SSv2	41k	19M	0.125
COIN	37k	21M	0.125
EpicKitchen	4k	13k	0.05
最终筛选数据（包含重复）	210k	115M	

该策略检索出的簇和场景总统计见表 11。数据集偏重 K710 检索出的簇，再结合其 0.7 检索权重，使整体筛选集有较强 Kinetics 权重；这种倾向反映在

K400 performance for our ablation experiments (see Appendix A.4.1). As shown in Table 1 in the main body, we combined this Curated YT1B with SSv2, Kinetics, HowTo100M, and ImageNet to create our final VM22M dataset.

A.3 Scaling Model Size

Details of the model architecture are shown in Table 12. All models are parameterized as vision transformers Dosovitskiy et al. (2020), using the standard 16×16 patch size. When scaling model size, we increase the encoder from a ViT-L (300M parameters) to a ViT-g (1B parameters), while the predictor size is kept fixed across all pre-training experiments.

Table 12 Model architecture details. Family of encoders and predictor architectures used during V-JEPA 2 pretraining, with some of the major parameters.

Model	Params	Width	Depth	Heads	MLP	Embedder
Encoders: $E_\theta(\cdot)$						
ViT-L	300M	1024	24	16	4096	$2 \times 16 \times 16$ strided conv
ViT-H	600M	1280	32	16	5120	$2 \times 16 \times 16$ strided conv
ViT-g	1B	1408	40	22	6144	$2 \times 16 \times 16$ strided conv
Predictor: $P_\phi(\cdot)$						
ViT-s	22M	384	12	12	1536	N.A.

A.4 Additional Results

A.4.1 Effect of Data Curation

Table 13 shows the results of data curation on a subset of downstream classification tasks. For this table, we trained models at the ViT-L and ViT-g scale using the abbreviated training recipe of the original V-JEPA (Bardes et al., 2024). When training smaller scale models (ViT-L), training a model on the curated variant of YT1B leads to across-the-board improvements over the uncurated variant. However, when moving to a mixed data setting (i.e., adding images and hand-selected videos), performance actually drops for a subset of tasks when using curated data, with performance on SSv2 at 72.8 for VM22M (Mixed+Curated YT1B) vs. 73.3 for Mixed+Uncurated YT1B. In some cases, the model trained with Curated YT1B alone is better than the one with mixed data, such as on the COIN (86.5 vs. 86.25) and K400 (84.6 vs. 83.7) evaluation tasks. This result is somewhat surprising, as despite including the K710 training data in the Mixed setting, we find that it does not improve performance over Curated YT1B for the K400 evaluation task.

Table 13 Effects of Data Curation on Video Understanding. Results are reported at both the ViT-L and ViT-g model scales. Models at both scales were pretrained using the abbreviated schedule of Bardes et al. (2024).

Training Data	IN1K	COIN	SSv2	K400
ViT-L				
Uncurated YT1B	80.6	83.2	70.9	82.9
Curated YT1B	80.8	**86.5**	73.1	**84.6**
Mixed+Uncurated YT1B	**82.9**	86.25	**73.3**	83.0
VM22M	**82.9**	86.0	72.8	83.7
ViT-g				
Uncurated YT1B	81.8	86.4	73.6	85.1
Curated YT1B	81.7	88.4	74.8	**86.5**
Mixed+Uncurated YT1B	83.7	88.5	75.5	85.9
VM22M	**83.9**	**89.2**	**75.6**	86.2

However, this behavior is not constant across scales. At the ViT-g scale, VM22M (Mixed+Curated YT1B) outperforms Mixed+Uncurated YT1B on all tasks.

消融实验的 K400 性能中（见附录 A.4.1）。如正文表 1，将筛选 YT1B 与 SSv2、Kinetics、HowTo100M、ImageNet 结合，得到最终 VM22M。

A.3 扩大模型规模

架构详见表 12。所有模型为视觉 Transformer（Dosovitskiy et al., 2020），采用标准图块大小 16×16。规模扩展时，编码器从 ViT-L（3 亿）增至 ViT-g（10 亿），所有预训练实验中预测器大小固定。

表 12 模型架构细节。V-JEPA 2 预训练所用编码器系列和预测器架构，以及主要参数。

模型	参数量	宽度	深度	注意力头数	MLP	嵌入提取器
编码器： $E_\theta(\cdot)$						
ViT-L	300M	1024	24	16	4096	$2 \times 16 \times 16$ 步幅卷积
ViT-H	600M	1280	32	16	5120	$2 \times 16 \times 16$ 步幅卷积
ViT-g	1B	1408	40	22	6144	$2 \times 16 \times 16$ 步幅卷积
预测器： $P_\phi(\cdot)$						
ViT-s	22M	384	12	12	1536	N.A.

A.4 补充结果

A.4.1 数据筛选的影响

表 13 展示数据筛选在部分下游分类任务上的结果。表中模型为 ViT-L 和 ViT-g，采用原始 V-JEPA 的缩短训练方案（Bardes et al., 2024）。对较小模型 ViT-L，使用筛选版 YT1B 训练，相比未筛选版能全面提升性能。但混合数据设置下（即加入图像和人工选择的视频），筛选数据反而使部分任务性能下降：SSv2 上 VM22M（混合+筛选 YT1B）为 72.8，混合+未筛选 YT1B 为 73.3。有些情况下，仅用筛选 YT1B 的模型优于混合数据模型，例如 COIN（86.5 对 86.25）和 K400（84.6 对 83.7）。这一结果有些出人意料：即便混合设置加入了 K710 训练数据，K400 评估性能仍未超过仅使用筛选 YT1B 的结果。

表 13 数据筛选对视频理解的影响。报告 ViT-L 与 ViT-g 两种规模，均使用 Bardes et al.（2024）的缩短调度预训练。

训练数据	IN1K	COIN	SSv2	K400
ViT-L				
未筛选 YT1B	80.6	83.2	70.9	82.9
筛选 YT1B	80.8	**86.5**	73.1	**84.6**
混合+未筛选 YT1B	**82.9**	86.25	**73.3**	83.0
VM22M	**82.9**	86.0	72.8	83.7
ViT-g				
未筛选 YT1B	81.8	86.4	73.6	85.1
筛选 YT1B	81.7	88.4	74.8	**86.5**
混合+未筛选 YT1B	83.7	88.5	75.5	85.9
VM22M	**83.9**	**89.2**	**75.6**	86.2

但不同规模下行为不同：ViT-g 上，VM22M（混合+筛选 YT1B）在所有任务均优于混合+未筛选 YT1B。

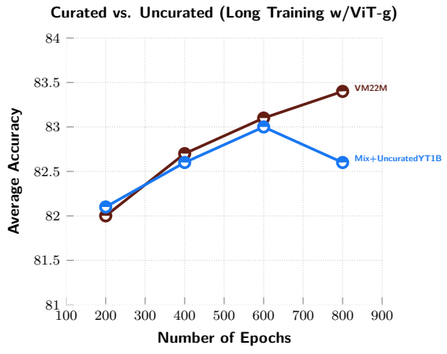


Figure 12 Effect of data curation for V-JEPA 2 pre-training. We show model performance averaged across the IN1K, COIN, SSv2, and K400 tasks as a function of pre-training "epochs" (equivalent to 300 optimization steps). Models trained with and without uncurated data achieve similar performance until epoch 600, at which point the performance of the model trained with uncurated YT1B beings degrading.

When following these tasks with the long training schedule, we continue to see differences between VM22M and Mixed+Uncurated YT1B at the ViT-g model scale, as shown in Figure 12, which compares the performance of the models while averaging across the IN1K, COIN, SSv2, and K400 image understanding tasks. Initially, the two models improve at roughly the same rate, but their performance diverges after epoch 600 where the model using uncurated data fails to continue improving.

A.4.2 Effect of Long Training Schedule and cooldown

In Table 14 we demonstrate the effects of the two-stage training process. When comparing to the ViT-g results in Table 13, we see that the abbreviated schedule is superior to the constant learning rate schedule prior to the cooldown phase. The primary benefits are achieved during the cooldown phase, which uses 64 frames for pretraining in combination with a ramped down learning rate. This leads to a large benefit of over a full point across all evaluations.

Table 14 Effects of Long Training and cooldown. Results are reported at ViT-g model with cooldown at different resolutions.

Training Stage	IN1K	COIN	SSv2	K400
Phase 1 (epoch 800, no cooldown)	83.8	89.1	75.1	85.8
Phase 2 (annealed, 256×256 resolution)	84.6	**90.7**	75.3	86.6
Phase 2 (annealed, 384×384 resolution)	**85.1**	90.2	**76.5**	**87.3**

A.4.3 Effect of Video Length at Evaluation.

Figure 13 examines how input video duration affects downstream task performance during evaluation. Using a model pretrained on 64-frame clips, we observe a +9.7 percentage point average improvement when increasing the video duration from 16 to 64 frames during evaluation. Note that this ablation uses a single clip evaluation protocol (i.e. we sample only one clip per video) instead of the standard multiclip evaluations due to memory constraints.

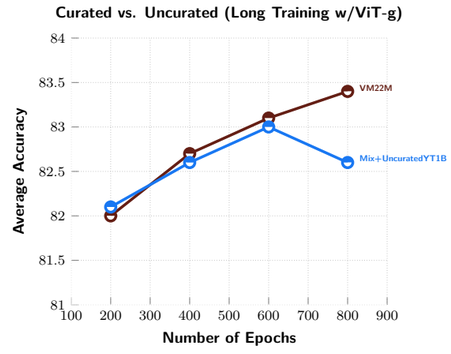


图 12 数据筛选对 V-JEPA 2 预训练的影响。绘制 IN1K、COIN、SSv2、K400 平均性能随预训练“轮次”（每轮等于 300 个优化步骤）的变化。包含和不包含未筛选数据的模型，性能直到第 600 轮相近；此后，使用未筛选 YT1B 的模型开始退化。

采用长训练调度时，ViT-g 规模下 VM22M 与混合+未筛选 YT1B 仍有差别。图 12 比较 IN1K、COIN、SSv2、K400 平均性能。起初两者提升速度相近，但在第 600 轮之后分化，未筛选数据模型无法继续改善。

A.4.2 长训练调度和收尾的影响

表 14 展示两阶段训练效果。对比表 13 的 ViT-g，收尾前缩短方案优于恒定学习率方案。主要收益来自收尾阶段：以 64 帧预训练并降低学习率，使全部评估显著提高超过一个百分点。

表 14 长训练的影响 训练与收尾。报告 ViT-g 在不同分辨率收尾训练的结果。

训练阶段	IN1K	COIN	SSv2	K400
阶段 1（第 800 轮，未收尾）	83.8	89.1	75.1	85.8
阶段 2（退火，分辨率 256×256）	84.6	**90.7**	75.3	86.6
阶段 2（退火，分辨率 384×384）	**85.1**	90.2	**76.5**	**87.3**

A.4.3 评估视频长度的影响

图 13 考察评估时视频长度的影响。使用 64 帧预训练模型，评估视频从 16 增至 64 帧，平均提升 +9.7 个百分点。因内存限制，该消融采用单片段方案，每视频只采样一个片段，而非标准多片段评估。

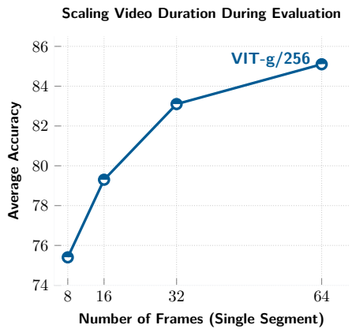


Figure 13 Effect of video duration during evaluation. Task performance further improves by running inference on longer video clips. All evaluations use ViT-g models that were annealed with 64 frames at resolution 256×256. Due to memory constraints, results are reported using a single clip evaluation protocol. Increasing the number of frames processed at inference time boosts average performance by up to +9.7 points.

B V-JEPA 2-AC Post-training

B.1 Post-Training Hyperparameters

The V-JEPA 2-AC model is trained with the AdamW (Loshchilov and Hutter, 2017) optimizer using a warmup-constant-decay learning-rate schedule, and a constant weight-decay of 0.04. We linearly warmup the learning rate from 7.5×10^{-5} to 4.25×10^{-4} over 4500 iterations, then hold it constant for 85500 iterations, and finally decay it to 0 over 4500 iterations. We use a batch size of 256 comprising 4 second video clips sampled randomly from trajectories in the Droid raw dataset at a frame rate of 4 fps. We train on the left extrinsic camera views from Droid — one could also train on videos from right camera views, however we found that training on both left and right camera views, without additionally conditioning on the camera position, degraded performance. For simplicity, we discard any videos shorter than 4 seconds, leaving us with less than 62 hours of video for training. We apply random-resize-crop augmentations to the sampled video clips with the aspect-ratio sampled in the range (0.75, 1.35).

B.2 Robot Task Definitions

Figure 14 shows examples of start and goal frames for prehensile manipulation task with a cup in Lab 1. For the *grasp* and *reach with object* tasks the model is shown a single goal image. For the *pick-and-place* tasks we present two sub-goal images to the model in addition to the final goal. The first goal image shows the object being grasped, the second goal image shows the object in the vicinity of the goal position. The model first optimizes actions with respect to the first sub-goal for 4 time-steps before automatically switching to the second sub-goal for the next 10 time-steps, and finally the third goal for the last 4 time-steps. When planning with V-JEPA 2-AC, we use 800 samples, 10 refinement steps based on the top 10 samples from the previous iteration, and a planning horizon of 1. Since all considered tasks are relatively greedy, we found a short planning horizon to be sufficient for our setup. While longer planning horizons also worked reasonably well, they require more planning time.

B.3 Visualizing World Model Predictions

To visualize the model's predictions, we train a frame decoder on the Droid dataset that maps the V-JEPA 2 representations to human-interpretable pixels. Specifically, we process 4 frame clips with the frozen V-JEPA 2 video encoder, decode each frame separately using our decoder network, and then update the weights of the decoder using a mean-squared error (L2) pixel reconstruction loss. The decoder is a feedforward network (fully deterministic regression model that does not use any sampling internally) with output dimension $256 \times 256 \times 3$, parameterized as a ViT-L. We train the decoder for 150000 optimization steps using AdamW with a fixed

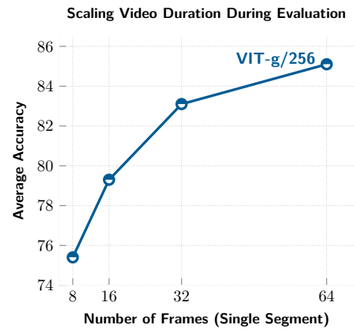


图 13 评估视频长度的影响。更长片段推理进一步改善任务性能。所有评估使用以 64 帧、分辨率 256×256 退火训练的 ViT-g。因内存限制，报告单片段结果。增加推理帧数，平均性能最高提升 +9.7 个百分点。

B V-JEPA 2-AC 后训练

B.1 后训练超参数

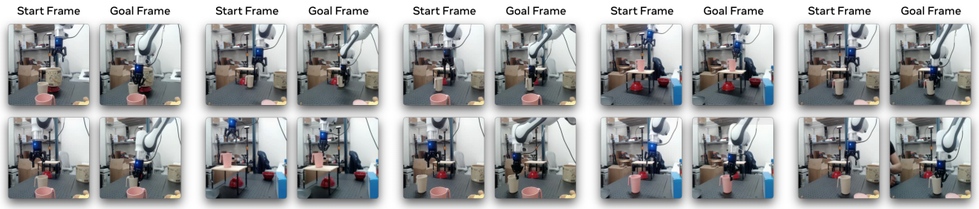
V-JEPA 2-AC 使用 AdamW 优化器（Loshchilov and Hutter, 2017），采用预热–恒定–衰减学习率调度，权重衰减恒为 0.04。前 4500 次迭代将学习率从 7.5×10^{-5} 线性升至 4.25×10^{-4}，随后保持 85500 次迭代，最后用 4500 次迭代衰减至 0。批量大小为 256，由从 Droid 原始轨迹随机采样的 4 秒视频片段组成，帧率为 4 fps。使用 Droid 左侧外部相机视角训练；也可使用右侧视角，但我们发现，在不额外以相机位置为条件时，同时使用左右视角会降低性能。为简化处理，丢弃不足 4 秒的视频，剩余训练视频不足 62 小时。对采样视频施加随机缩放裁剪增强，宽高比从 (0.75, 1.35) 中采样。

B.2 机器人任务定义

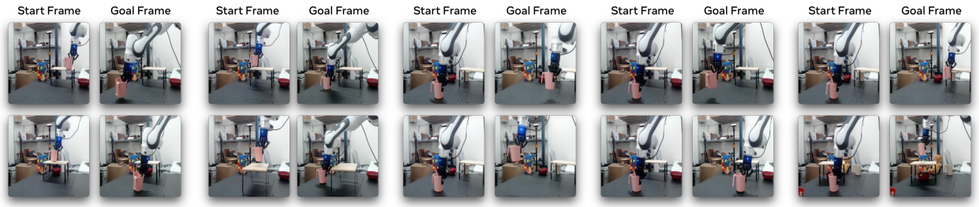
图 14 展示实验室 1 中针对杯子的抓持操作任务的起始帧与目标帧。抓取和持物到达任务仅展示一张目标图像；抓取放置除最终目标外还展示两个子目标。第一个展示物体被抓住，第二个展示物体位于目标位置附近。模型针对第一个子目标优化动作 4 个时间步，自动切换到第二个目标执行 10 步，最后以第三个目标执行 4 步。V-JEPA 2-AC 规划使用 800 个样本、10 次优化迭代，每次基于前次最优 10 个样本，规划时域为 1。由于所有任务都相对适合贪心策略，短时域已足够。更长时域也有合理表现，但需要更多规划时间。

B.3 世界模型预测可视化

为可视化预测，在 Droid 上训练帧解码器，将 V-JEPA 2 表征映射为人可理解的像素。冻结编码器处理四帧片段，解码器独立解码各帧，再用均方误差（L2）像素重建损失更新解码器。解码器为前馈网络，是内部不进行采样的完全确定性回归模型，输出维度 $256 \times 256 \times 3$，参数化为 ViT-L。使用 AdamW 训练 150000 个优化步骤，固定



(a) Grasp Cup



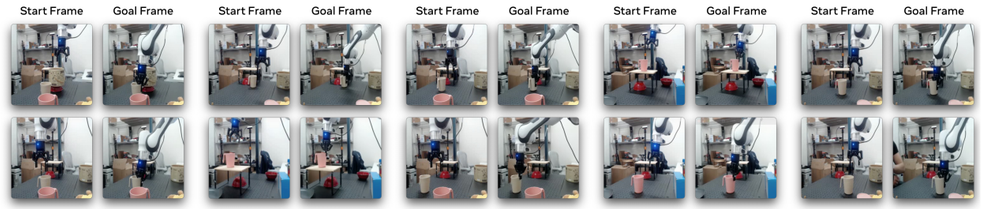
(b) Reach with Cup



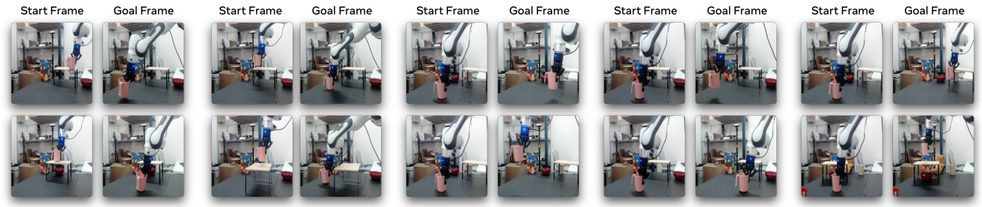
(c) Pick and Place Cup

Figure 14 Prehensile Manipulation Task Definition. Start and goal frames for prehensile manipulation tasks with a cup in Lab 1. For the *grasp* and *reach with object* tasks the model is shown a single goal image. For the *pick-and-place* tasks we present two sub-goal images to the model in addition to the final goal. The first goal image shows the object being grasped, the second goal image shows the object in the vicinity of the goal position. The model first optimizes actions with respect to the first sub-goal for 4 time-steps before automatically switching to the second sub-goal for the next 10 time-steps, and finally the third goal for the last 4 time-steps.

weight decay of 0.1, gradient clipping of 1.0, and a batch size of 1024 frames. We linearly warmup the learning rate for 2000 steps to a peak value of 5×10^{-4} and then decay it following a cosine schedule. For inference, we take the decoder trained on the V-JEPA 2 encoder and apply it off-the-shelf to the representations produced by the V-JEPA 2-AC predictor. The decision to only use use a simple feedforward architecture and



(a) Grasp Cup



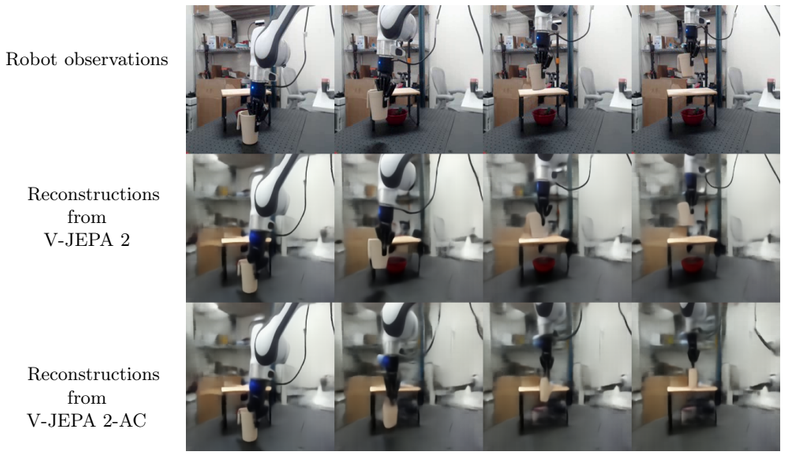
(b) Reach with Cup



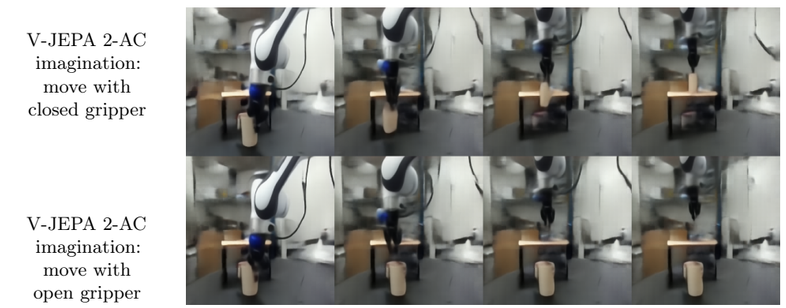
(c) Pick and Place Cup

图 14 抓持操作任务定义。展示实验室 1 中杯子任务的起始帧和目标帧。抓取与持物到达仅展示单一目标图像；抓取放置除最终目标外还展示两个子目标：第一个为物体被抓住，第二个为物体接近目标位置。模型先针对第一子目标优化 4 步，自动切换第二目标执行 10 步，最后以第三目标执行 4 步。

权重衰减为 0.1，梯度裁剪为 1.0，批量大小为 1024 帧。前 2000 步将学习率线性预热至峰值 5×10^{-4}，再按余弦调度衰减。推理时直接将基于 V-JEPA 2 编码器训练的解码器应用于 V-JEPA 2-AC 预测器表征。只用简单前馈架构的决定，以及

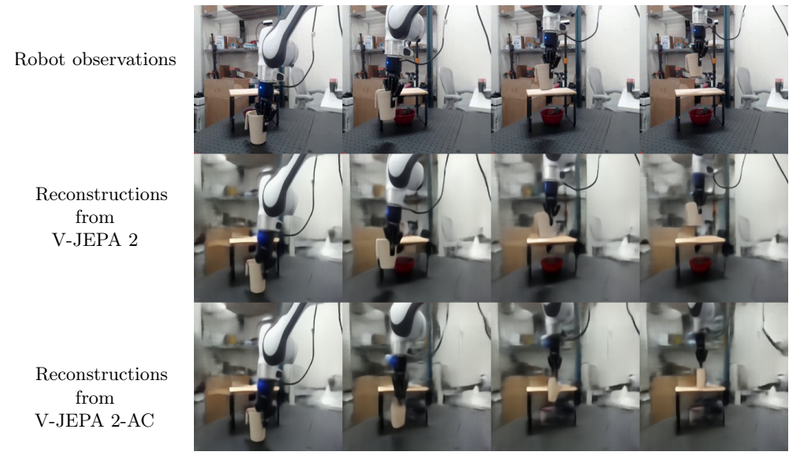


(a) Comparing accuracy of predictions to ground truth trajectory. (Top Row) Video frames of a ground-truth trajectory from a robot in our lab. **(Middle row)** Each frame is encoded by V-JEPA 2 encoder, and then decoded using the feedforward frame decoder. Reconstructions of the V-JEPA 2 representations show that the encoder captures the salient parts of the scene necessary for vision-based control; blurry background generation can be partially attributed to the low-capacity of our feedforward frame decoder. **(Bottom Row)** Autoregressive rollout produced by V-JEPA 2-AC world model using the ground-truth action sequence given first frame as context, and then decoded using feedforward frame decoder. Reconstructions of the V-JEPA 2-AC rollout show that the action-conditioned world model successfully animates the robot while keeping the background and non-interactated objects (e.g., the shelf) unaffected. However, we do observe error accumulation as the world model predicts the location of the cup to be slightly lower than that of the real trajectory in the final frame.

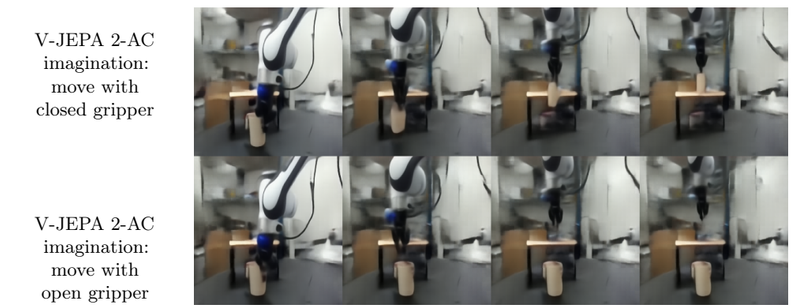


(b) Ablating predictions with open versus closed gripper. We explore how the V-JEPA 2-AC predictions change when driving the model with identical action sequences, but in one cause using a closed gripper **(top row)** and in the other with an open gripper **(bottom row)**. The world model predicts the location of the cup to be unchanged across time steps when using an open gripper action sequence, suggesting a reasonable understanding of intuitive physics (e.g., object constancy, shape constancy, and gravity).

Figure 15 Decoding representations. To visualize the model's predictions, we train a frame decoder on the Droid dataset that maps the V-JEPA 2 representations to human-interpretable pixels. The decoder is a feedforward network (fully deterministic regression model that does not use any sampling internally) trained with a mean-squared error pixel reconstruction loss. By applying the frame decoder, trained on the V-JEPA 2 encoder, to the representations produced by the V-JEPA 2-AC predictor, we can visualize world model rollouts for various action sequences.



（a）将预测准确性与真实轨迹比较。（上排）实验室机器人真实轨迹的视频帧。（中排）各帧先由 V-JEPA 2 编码器编码，再用前馈帧解码器解码。重建说明编码器捕获了视觉控制所需的显著场景部分；模糊背景部分源于前馈解码器容量较低。（下排）以第一帧为上下文，使用真实动作序列进行 V-JEPA 2-AC 自回归展开，再通过前馈帧解码器解码。重建表明动作条件世界模型成功预测机器人运动，同时不影响背景和未交互物体（如架子）。但存在误差累积：最后一帧中，世界模型预测的杯子位置略低于真实轨迹。



（b）夹爪打开与闭合的预测消融。使用相同动作序列，分别令夹爪闭合（上排）或打开（下排），考察 V-JEPA 2-AC 预测变化。打开夹爪时，模型预测杯子位置在各时间步保持不变，说明具有合理的直觉物理理解，如物体恒常性、形状恒常性和重力。

图 15 表征解码。为可视化预测，在 Droid 上训练帧解码器，将 V-JEPA 2 表征映射为人可理解的像素。解码器为不进行内部采样的完全确定性前馈回归网络，以均方像素重建误差训练。将基于 V-JEPA 2 编码器训练的解码器应用到 V-JEPA 2-AC 预测器表征，可可视化不同动作序列的世界模型展开。

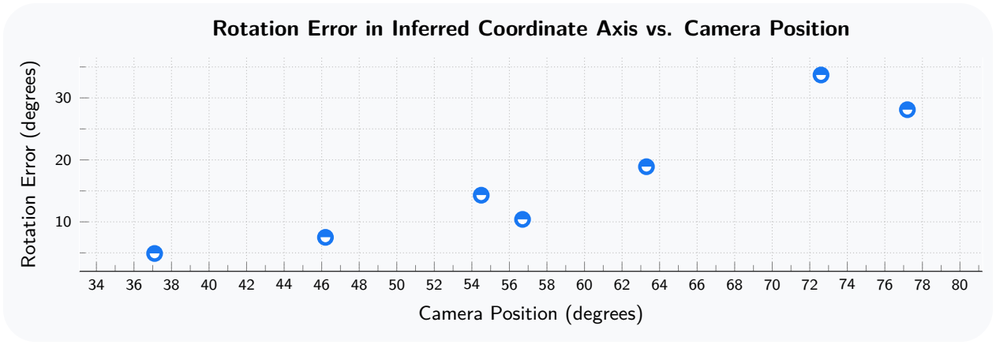


Figure 16 Sensitivity to camera position. Rotation error (in the x-y plane) of the action coordinate axis inferred by V-JEPA 2-AC as a function of camera position, with 0 degrees corresponding to a camera located at the robot base, and 90 degrees corresponding to a camera located left of the robot base. While ideally, the model's inferred coordinate axis would be invariant to camera position, here we observe that the model's inferred coordinate axis is sensitive to the camera position.

decode representations at the frame level (as opposed to video level), is to better leverage the decoder as an interpretability tool to analyze the V-JEPA 2-AC rollouts for a set of robot action sequences.

In Figure 15a, we show the video frames of a ground-truth trajectory from a robot in our lab (top row), the decoded V-JEPA 2 encoder representations of each frame (middle row), and the decoded V-JEPA 2-AC world model rollout using the ground-truth action sequence and a single starting frame as context (bottom row). Reconstructions of the V-JEPA 2 representations (middle row) show that the encoder captures the salient parts of the scene necessary for vision-based control; blurry background generation can be partially attributed to the low-capacity of our feedforward frame decoder. Reconstructions of the V-JEPA 2-AC rollout show that the action-conditioned world model successfully animates the robot while keeping the background and non-interactated objects (e.g., the shelf) unaffected. We also see that, with a closed gripper, the model correctly predicts the movement of the cup with the arm, suggesting a reasonable understanding of intuitive physics (e.g., object constancy, shape constancy, and gravity), but we do observe error accumulation as the world model predicts the location of the cup to be slightly lower than that of the real trajectory in the final frame. In Figure 15b we explore how the V-JEPA 2-AC predictions change when driving the model with identical action sequences, but in one cause using a closed gripper (top row) and in the other with an open gripper (bottom row). The world model predicts the location of the cup to be unchanged across time steps when using an open gripper action sequence.

B.4 Assessing Sensitivity to Camera Position

In practice, we manually tried different camera positions before settling on one that worked best for our experiments; then the camera is kept in the same location for all experiments, across all tasks. In this section, we conduct a quantitative analysis of the V-JEPA 2-AC world model's sensitivity to camera position. While ideally, the model's inferred coordinate axis would be invariant to camera position, here we observe that the model's inferred coordinate axis is sensitive to the camera position; this is problematic as large errors in the inferred coordinate axis can degrade success rate on downstream tasks.

We sweep several camera positions around the robot base, which we describe as a clockwise angular position around the center of the table, with 0 degrees being located at the robot base, and 90 degrees being left of the robot base. Since we train on the left exocentric camera views from the Droid dataset, we sweep camera positions between roughly 35 degrees and 85 degrees. Next, for each camera position, we collect a 201 step trajectory of random robot movements within the horizontal x-y plane. For each pair of adjacent frames in this 201 step trajectory, we compute the optimal action inferred by V-JEPA 2-AC, i.e., the action that minimizes the energy function in eq. (5) given a 1-step rollout. This allows us to construct a dataset for

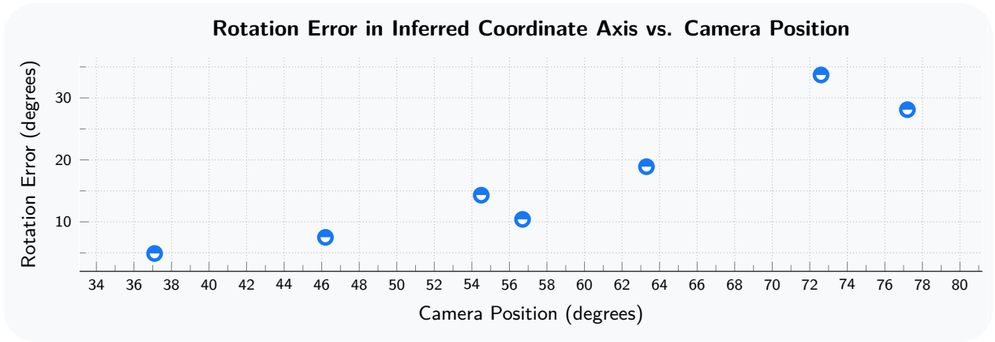


图 16 相机位置敏感性。V-JEPA 2-AC 推断动作坐标轴在 x-y 平面的旋转误差随相机位置变化。0 度为相机位于机器人基座，90 度为基座左侧。理想情况下坐标轴应不受相机位置影响，但实际观察到敏感性。

解码表征可解释性工具 在帧层面而非视频层面，是为了更好地将解码器用作分析一组机器人动作序列的 V-JEPA 2-AC 展开。

图 15a 展示实验室机器人的真实轨迹视频帧（上排）、每帧 V-JEPA 2 编码器表征的解码结果（中排），以及仅以单张起始帧为上下文、使用真实动作序列进行 V-JEPA 2-AC 世界模型展开后的解码结果（下排）。V-JEPA 2 表征的重建（中排）说明编码器捕获了视觉控制所需的场景显著部分；背景生成模糊，部分原因是前馈帧解码器容量较低。V-JEPA 2-AC 展开的重建表明，动作条件世界模型成功预测了机器人的运动，同时保持背景和未交互物体（例如架子）不受影响。我们还看到，夹爪闭合时，模型正确预测杯子随机械臂运动，说明它对直觉物理有合理理解（例如物体恒常性、形状恒常性与重力）。但仍观察到误差累积：在最后一帧，世界模型预测的杯子位置略低于真实轨迹。图 15b 研究相同动作序列下，使用闭合夹爪（上排）或打开夹爪（下排）时 V-JEPA 2-AC 预测如何变化。使用夹爪打开的动作序列时，世界模型预测杯子的位置在各时间步保持不变。

B.4 评估相机位置敏感性

实际中，人工尝试多个相机位置，最终选择最适合实验的位置；随后所有任务和实验均保持相机位置不变。本节定量分析 V-JEPA 2-AC 对相机位置的敏感性。理想情况下，模型推断的坐标轴应与相机位置无关，但这里观察到它对位置敏感；较大的坐标轴误差会降低下游任务成功率。

我们遍历机器人基座附近的多个相机位置，以桌面中心为圆心的顺时针角位置描述：0 度位于机器人基座处，90 度位于基座左侧。由于模型在 Droid 左侧外部相机视角上训练，测试相机位置约为 35 至 85 度。随后在每个位置，采集水平 x-y 平面内随机机器人运动的 201 步轨迹。对相邻帧对，计算 V-JEPA 2-AC 推断的最优动作，即一步展开时使式（5）能量最小的动作。由此构造数据集，用于

each camera position consisting of *real action* versus *inferred action* pairs. We only focus on the Δx and Δy cartesian control actions (first two dimensions of the action vector) for our analysis. Let $A \in \mathbb{R}^{200\times 2}$ denote the inferred actions and $B \in \mathbb{R}^{200\times 2}$ denote the ground truth actions. Based on this, we can solve a linear least squares problem to identify the linear transformation $W^\star \in \mathbb{R}^{2\times 2}$ that maps inferred actions A to real actions B,

$$W^\star = \underset{W\in\mathbb{R}^{2\times 2}}{\operatorname{argmin}} \|AW - B\|_2.$$

The mean absolute prediction error for all camera position is roughly 1.6cm (compared to a ground truth delta pose of roughly 5cm), suggesting that the error is systematic. In addition, we observe that for each camera position, the matrix $W^\star$ has condition number ≈ 1.5, i.e., modulo a fixed scalar coefficient, $W^\star$ is approximately a rotation matrix, and thus we can compute the rotation error in the inferred coordinate axis by using

$$W^\star \approx \overline{W}^\star = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix},$$

with $\overline{W}^\star := UV^\top$ where U and V are the left and right singular vectors of $W^\star$, respectively.

Figure 16 shows the camera position plotted against the rotation error in the V-JEPA 2-AC inferred coordinate axis. We observe that the rotation error in the inferred coordinate axis is almost a linear function of the camera position. We can most clearly see the effects of rotation errors in the inferred coordinate axis in our single-goal reaching experiments in Figure 8. While the model is always able to move the arm within 4 cm of the goal based on visual feedback from the monocular RGB camera, rotation errors in the inferred coordinate axis result in relatively suboptimal actions at each planning step, yielding a non-maximal, albeit monotonic, decrease in the distance to goal at each step.

Interestingly, since errors in the inferred coordinate axis are primarily rotation-based, one can use this approach to "calibrate" their world model by simply rotating all inferred actions by $W^\star$, and thereby introduce the desired invariance to camera position. Such an unsupervised calibration phase would involve the robot performing random actions, solving a linear least squares problem by comparing its inferred optimal actions to the actual actions it executed, and then multiplying its inferred actions by the rotation matrix before sending them to the controller during task execution. While such an approach is interesting, we emphasize that we *do no such calibration* in our experiments.

C Visual Classification

We describe in more detail the evaluation procedure used for the classification tasks described in Section 5.

C.1 Hyperparameters

Probe Architecture. We train an attentive probe on top of the frozen encoder output using the training data from each downstream task. Our attentive probe is composed of four transformer blocks, each using 16 heads in the attention layer. The first three blocks use standard self-attention; the final block uses a cross-attention layer with a learnable query token. The output of the cross-attention layer in the final block is added back to the query token as a residual connection before applying the rest of the block (LayerNorm, followed by MLP with a single GeLU activation). The transformer blocks are followed by a final linear classifier layer.

Evaluation setup parameters. All models follow the same evaluation protocol and use a resolution of 256×256, except our V-JEPA 2 ViT-g_{384}. For video evaluations, we sampled multiple clip segments from each input video. During validation, we also extracted three spatial views from each segment (instead of one view during training). The number of clip segments, frame step parameter, and global batch size vary for each eval; parameters used for each evaluation can be found in Table 15. By default, we use $16 \times 2 \times 3$ inputs for SSv2 (16 frames clip, 2 temporal crops, 3 spatial crops), $16 \times 8 \times 3$ for K400, $32 \times 8 \times 3$ for COIN, and $32 \times 4 \times 3$ for Diving-48 and Jester. V-JEPA 2 ViT-g_{384} uses a higher resolution of 384×384 for K400, COIN, Diving-48, and Jester, 512×512 for ImageNet and 384×384 with $64 \times 2 \times 3$ inputs for SSv2.

每个相机位置的数据由真实动作与推断动作对组成。分析只关注 Δx 和 Δy 笛卡尔控制动作，即动作向量前两维。令 $A \in \mathbb{R}^{200\times 2}$ 为推断动作，$B \in \mathbb{R}^{200\times 2}$ 为真实动作。由此求解线性最小二乘，识别将推断动作 A 映射为真实动作 B 的线性变换 $W^\star \in \mathbb{R}^{2\times 2}$：

$$W^\star = \underset{W\in\mathbb{R}^{2\times 2}}{\operatorname{argmin}} \|AW - B\|_2.$$

所有相机位置的平均绝对预测误差约 1.6 cm，而真实姿态变化约 5 cm，说明误差具有系统性。另外，各位置矩阵 $W^\star$ 的条件数 ≈ 1.5，即除固定标量外，$W^\star$ 近似旋转矩阵，因此可计算坐标轴旋转误差，使用

$$W^\star \approx \overline{W}^\star = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix},$$

with $\overline{W}^\star := UV^\top$，其中 U 和 V 分别为 $W^\star$ 的左、右奇异向量。

图 16 绘制相机位置与 V-JEPA 2-AC 推断坐标轴旋转误差的关系。旋转误差几乎是相机位置的线性函数。图 8 的单目标到达实验最清楚地体现其影响：模型总能根据单目 RGB 视觉反馈将机械臂移至目标 4 cm 内，但坐标轴旋转误差使每步动作相对次优。因此每步到目标距离虽单调下降，下降幅度却未达到最大。

有趣的是，坐标轴误差主要来自旋转，因此可将全部推断动作旋转 $W^\star$ 来“标定”世界模型，引入所需的相机位置不变性。无监督标定包括：机器人执行随机动作，比较推断最优动作与实际动作，求解线性最小二乘问题；任务执行时，先将推断动作乘旋转矩阵，再送入控制器。虽然该方法值得研究，但强调：我们的实验未进行这种标定。

C 视觉分类

详细介绍第 5 节分类任务的评估过程。

C.1 超参数

探测器架构。使用各下游任务训练数据，在冻结编码器输出上训练注意力探测器。它由四个 Transformer 块构成，每层 16 个注意力头。前三块采用标准自注意力，最后块采用带可学习查询词元的交叉注意力。最后层交叉注意力输出以残差连接加回查询词元，再应用块的其余部分：LayerNorm，随后是含单个 GELU 的 MLP。四个块后连接最终线性分类器。

评估设置参数。除 V-JEPA 2 ViT-g_{384} 外，所有模型采用相同评估方案和分辨率 256×256。视频评估时从每个输入视频采样多个片段；验证时每个片段提取三个空间视图，而训练时仅提取一个。各评估的片段数、帧步长和全局批量大小不同，参数见表 15。默认输入：SSv2 为 $16 \times 2 \times 3$（16 帧片段、2 个时间裁剪、3 个空间裁剪），K400 为 $16 \times 8 \times 3$，COIN 为 $32 \times 8 \times 3$，Diving-48 和 Jester 为 $32 \times 4 \times 3$。V-JEPA 2 ViT-g_{384} 在 K400、COIN、Diving-48、Jester 上使用更高分辨率 384×384，ImageNet 上使用 512×512，SSv2 上使用分辨率 384×384 和输入 $64 \times 2 \times 3$。

Table 15 Visual Classification eval params. Default parameters used for the visual classification evaluations, with non-default values for each eval (* denotes default). All attentive probes use 4 transformer blocks with 16 heads.

Parameter	Default (K400)	ImageNet	SSv2	COIN	Jester/Diving-48
Number of frames	16	16	16	32	32
Segments / Clip	8	1	2	8	4
Views / Segment	3	1	*	*	*
Frame Step	4	n/a	*	*	2
Epochs	20	*	*	*	100
Batch Size (global)	256	1024	*	128	128
Resolution	256×256	*	*	*	*
Classifier Heads	20 (4x5)	*	*	*	3 (3x1)
Classifier Learning Rates	[5e-3 3e-3 1e-3 3e-4 1e-4]	*	*	*	[1e-3 3e-4 1e-4]
Classifier Weight Decay	[.8 .4 .1 .01]	*	*	*	[.8]

Table 16 Input layers for Jester/Diving-48. For each encoder size, indices of the four encoder layers whose tokens are used as input to the linear classifier in the Jester and Diving-48 evaluations.

Encoder	# Layers	Attended Layers
ViT-L	24	17, 19, 21, 23
ViT-H	32	25, 27, 29, 31
ViT-g	40	24, 29, 34, 39

ImageNet evaluation. For ImageNet, we repeat each input image to produce a 16-frame video clip. We also use a larger global batch size (1024 instead of 256 or 128), and do not use multiple clips or views per sample.

Jester and Diving-48 evaluation. Our Jester and Diving-48 action classification evaluation tasks differ from the other understanding evaluations in several ways, primarily in that we employ a multilayer strategy. Instead of attending to the tokens from only the last layer of the encoder, we extract tokens from four encoder layers (the last layer and three intermediate layers) and attend to all of them. (Table 16 shows the layers we used for each encoder size.) We also train the probes for these two evaluations with only three classification heads (instead of 20 for the other evaluations), but train for 100 epochs (instead of 20) as these evaluations benefit from longer training. We use a global batch size of 128 for both evaluations.

Optimization. For each evaluation, we simultaneously train multiple classifier heads with different hyperparameters (learning rate and weight decay), reporting the accuracy of the best-performing classifier. For most of our evaluations (Kinetics, SSv2, COIN, and ImageNet), we train for 20 epochs and use 20 heads, each using one of five learning rate values and four weight decay values, and the learning rate decays according to a cosine schedule. We provide a summary of all hyperparameters in Table 15.

C.2 Additional Results

Probe Size. Since we use a four-layer attentive probe for these evaluations, we investigate whether using a smaller probe impacts evaluation performance. We re-run our six understanding evaluations (for two model sizes, ViT-L and ViT-g) with a smaller probe consisting of a single cross-attention block using 16 attention heads. Unlike in Section 5, we use 16 frames for all evaluations, including Diving-48 and Jester. See Table 18 for classification performance—we confirm that our four-layer probe outperforms a single-layer attentive probe across all understanding evaluations (except for Jester), by an average of +1.4 points accuracy for ViT-L and +1.0 points for ViT-g.

Impact of encoder multilayer. We study the impact of feeding tokens from multiple layers from the encoder to the attentive probe during evaluation. Table 17 shows that Diving-48 and Jester strongly benefit from information from deeper layers of the encoder.

表 15 视觉分类 分类评估参数。视觉分类评估所用默认参数，包含
非默认值 各评估（* 表示默认值）。全部注意力探测器包含四个 Transformer 块和 16 个注意力头。

参数	默认（K400）	ImageNet	SSv2	COIN	Jester/Diving-48
帧数	16	16	16	32	32
每视频片段数	8	1	2	8	4
每片段视图数	3	1	*	*	*
帧步长	4	n/a	*	*	2
训练轮次	20	*	*	*	100
全局批量大小	256	1024	*	128	128
分辨率	256×256	*	*	*	*
分类器头数	20 (4x5)	*	*	*	3 (3x1)
分类器学习率	[5e-3 3e-3 1e-3 3e-4 1e-4]	*	*	*	[1e-3 3e-4 1e-4]
分类器权重衰减	[.8 .4 .1 .01]	*	*	*	[.8]

表 16 Jester／Diving-48 的输入层。列出各编码器大小所用四层索引，其词元作为评估线性分类器的输入。

编码器	层数	参与注意力的层
ViT-L	24	17, 19, 21, 23
ViT-H	32	25, 27, 29, 31
ViT-g	40	24, 29, 34, 39

ImageNet 评估。将每张输入图像重复为 16 帧视频；全局批量大小增至 1024，而非 256 或 128。每样本不使用多个片段或视图。

Jester 与 Diving-48 评估。两项动作分类评估与其他理解评估的主要区别，是采用多层策略：不只关注编码器最后层词元，而提取四层（最后一层和三个中间层）词元并共同注意。各编码器所用层见表 16。此外只训练三个分类头，而其他评估为 20 个；但训练 100 轮而非 20 轮，因为这两项任务受益于更长训练。两者全局批量大小均为 128。

优化。每项评估同时训练采用不同学习率和权重衰减的多个分类头，报告最佳分类器准确率。多数评估（Kinetics、SSv2、COIN、ImageNet）训练 20 轮、使用 20 个头，涵盖五种学习率与四种权重衰减，学习率按余弦调度下降。全部超参数汇总见表 15。

C.2 补充结果

探测器大小。由于评估采用四层探测器，我们研究较小探测器是否影响性能。对 ViT-L 与 ViT-g，以仅有一个交叉注意力块、16 个注意力头的探测器，重跑六项理解评估。与第 5 节不同，所有评估（包括 Diving-48 和 Jester）均用 16 帧。结果见表 18：除 Jester 外，四层探测器在所有评估中优于单层，ViT-L 平均准确率高 +1.4 个百分点，ViT-g 高 +1.0 。

编码器多层的影响。考察评估时向注意力探测器提供多层词元。表 17 说明 Diving-48 和 Jester 显著受益于编码器更深层的信息。

Table 17 Encoder Multilayer Ablation. We vary the number of encoder layers fed to the attentive probe. We report the classification performances of attentive probes trained on top of V-JEPA 2 with 16 frames at 256×256 resolution.

Model	Encoder Layers	Diving-48	Jester
ViT-g	1	82.9	96.1
ViT-g	4	86.7	97.6

Table 18 Probe Size Ablation. We vary the number of layers in the attentive probe. We report the classification performances of attentive probes trained on top of V-JEPA 2 with 16 frames at 256×256 resolution.

		Motion Understanding				*Appearance Understanding*		
Model	**Probe Layers**	**Avg.**	**SSv2**	**Diving-48**	**Jester**	**K400**	**COIN**	**IN1K**
ViT-L	1	84.0	72.0	83.2	97.7	83.3	85.9	81.8
ViT-L	4	85.6	73.6	87.1	97.7	85.1	86.8	83.5
ViT-g	1	86.0	74.8	85.3	97.8	85.6	88.9	83.5
ViT-g	4	87.0	75.6	86.7	97.6	86.6	90.7	84.6

D Action Anticipation

We provide additional details, results, and ablations related to the Epic-Kitchen 100 action anticipation evaluation of Section 6.

D.1 Hyperparameters

Probe Architecture. Our probe architecture for action anticipation follows the architecture of our classification probe described in Appendix C.1, consisting of four transformer blocks, including a last cross-attention layer with a set of learnable query tokens, followed by a final linear classifier layer for each query token.

Evaluation setup parameters. We use a focal loss (Lin et al., 2017) with a $\alpha = 0.25$ and $\gamma = 2.0$ when training the probe; this loss is more suited for training with long-tailed imbalanced class distributions. We use a context of 32 frames with a frame-rate of 8 frames per second at resolution 256×256 for V-JEPA 2 ViT-L, ViT-H, and ViT-g; and resolution 384×384 for V-JEPA 2 ViT-g_{384}. During probe training, we randomly sample an anticipation time between 0.25 and 1.75 seconds, and an anticipation point between 0.0 and 0.25. The *anticipation point* identifies a point in the action segment from which to perform anticipation; i.e., an anticipation point of 0 means that we predict the representation of the first frame in the action segment using our V-JEPA 2 predictor before feeding it to the probe, whereas an anticipation point of 1 means that we predict the representation of the last frame in the action segment before feeding it to the probe. The validation anticipation time is set to 1 second and the validation anticipation point is set to 0. We provide a summary of the hyperparameters, including the optimization parameters in Table 19.

D.2 Additional results

Impact of Architecture. Table 20 investigates the impact of providing the output of the V-JEPA 2 encoder, predictor, or both, to the action anticipation probe. Using encoder outputs already leads to competitive performance on the EK100 task. Adding the predictor provides a small but consistent improvement across action, verb, and object categories. In addition, using predictor outputs yields a non-trivial performance, but still at a much lower point compared to using the encoder, showing that the EK100 task mostly requires strong semantic understanding, as opposed to forecasting capabilities.

Impact of Input Resolution. We report in Figure 17, the impact of input resolution and frame sampling parameters. In summary, V-JEPA 2 benefits from a longer context, a higher frame rate, and higher resolution,

表 17 编码器多层分类性能 多层消融。改变输入注意力探测器的编码器层数，报告在 16 帧、分辨率 256×256 的 V-JEPA 2 上训练注意力探测器的性能。

模型	编码器层	Diving-48	Jester
ViT-g	1	82.9	96.1
ViT-g	4	86.7	97.6

表 18 探测器大小消融。改变注意力探测器层数，报告在 16 帧、分辨率 256×256 的 V-JEPA 2 之上训练探测器的分类性能。

		运动理解				外观理解		
模型	探测器层数	Avg.	SSv2	Diving-48	Jester	K400	COIN	IN1K
ViT-L	1	84.0	72.0	83.2	97.7	83.3	85.9	81.8
ViT-L	4	85.6	73.6	87.1	97.7	85.1	86.8	83.5
ViT-g	1	86.0	74.8	85.3	97.8	85.6	88.9	83.5
ViT-g	4	87.0	75.6	86.7	97.6	86.6	90.7	84.6

D 动作预判

补充第 6 节 Epic-Kitchens-100 动作预判评估的细节、结果和消融。

D.1 超参数

探测器架构。动作预判沿用附录 C.1 分类探测器：四个 Transformer 块，最后为带一组可学习查询词元的交叉注意力层，再为各查询词元连接最终线性分类器。

评估设置参数。训练探测器时使用参数为 $\alpha = 0.25$ 和 $\gamma = 2.0$ 的 focal loss（Lin et al., 2017）；该损失更适合类别长尾不平衡的训练。上下文为 32 帧，帧率为每秒 8 帧：V-JEPA 2 ViT-L、ViT-H 和 ViT-g 使用分辨率 256×256，V-JEPA 2 ViT-g_{384} 使用分辨率 384×384。训练探测器时，从 0.25 至 1.75 秒随机采样预判时间，并从 0.0 至 0.25 随机采样预判点。预判点指定动作片段中要预判的位置：0 表示用 V-JEPA 2 预测器预测动作片段第一帧的表征，再送入探测器；1 表示预测最后一帧的表征后送入探测器。验证预判时间设为 1 秒，预判点设为 0。包括优化参数在内的超参数汇总见表 19。

D.2 补充结果

架构影响。表 20 研究向动作预判探测器提供编码器输出、预测器输出或两者的影响。仅编码器已在 EK100 上有竞争力；加入预测器使动作、动词和物体类别小幅而一致改善。仅预测器也取得有意义的性能，但明显低于编码器，说明 EK100 主要需要强语义理解，而非未来预测能力。

输入分辨率影响。图 17 报告分辨率与帧采样参数的影响。总体而言，更长上下文、更高帧率和更高分辨率有利于 V-JEPA 2，

Table 19 Action Anticipation Evaluation params. Default parameters used for the EK100 Action Anticipation evaluation.

Parameter	EK100
Train Anticipation time	0.25s – 1.75s
Train Anticipation point	0.0 – 0.25
Val Anticipation time	1s
Val Anticipation point	0.0
Number of frames	32
Frames per second	8
Epochs	20
Warmup epochs	0
Batch Size (global)	128
Classifier Heads	20 (4x5)
Classifier Learning Rates	[5e-3 3e-3 1e-3 3e-4 1e-4]
Classifier Weight Decay	[1e-4 1e-3 1e-2 1e-1]

Table 20 EK100: Impact of Anticipation Probe Inputs. We investigate the impact of providing the outputs of the V-JEPA 2 encoder, predictor, or both, to the action anticipation probe. Using encoder outputs already leads to competitive performance on the EK100 task. Adding the predictor provides a small but consistent improvement across action, verb and object categories.

		Action Anticipation		
Encoder	**Predictor**	Verb	Noun	Action
✓		61.3	57.0	39.1
	✓	48.7	34.7	20.2
✓	✓	63.6	57.1	39.7

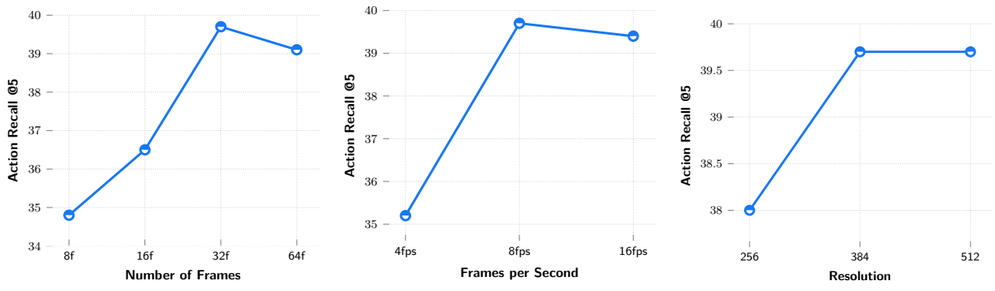


Figure 17 Protocol ablation for action anticipation on EK100. (Left) Performance with respect to the number of context frames used for action anticipation. **(Middle)** Performance with respect to the frame rate (fps) used for inference; number of context frames fixed at 32. **(Right)** Performance with respect to the spatial resolution (height and width) of the context frames used for action anticipation.

up to a point where the performance saturates or slightly decreases. The optimal performance is obtained by training with a 32-frames context length, a frame rate of 8 and a resolution of 384×384.

Longer-term Prediction. We report in Figure 18 (Left), the impact of predicting at a longer horizon, by varying the anticipation time between (1s, 2s, 4s, 10s). For each anticipation time, we report recall at several values (1, 5, 10, 20). The results show that the performance sharply decreases as the anticipation time increases, which is expected since forecasting the future in EK100 is a non-deterministic task.

Analysis of Failure Cases. We report in Figure 18 (Right), the distribution of failure and success prediction, on the EK100 validation set, between each configuration of success/failure for verb, noun, and action. The model performs very well, and the most represented configuration is, therefore, a full success

表 19 动作预判 预判评估参数。EK100 动作预判所用默认参数。
评估。

参数	EK100
训练预判时间	0.25s – 1.75s
训练预判点	0.0 – 0.25
验证预判时间	1s
验证预判点	0.0
帧数	32
每秒帧数	8
训练轮次	20
预热轮次	0
全局批量大小	128
分类器头数	20 (4x5)
分类器学习率	[5e-3 3e-3 1e-3 3e-4 1e-4]
分类器权重衰减	[1e-4 1e-3 1e-2 1e-1]

表 20 EK100：预判探测器输入的影响。比较提供编码器、预测器或两者输出。仅编码器已具有竞争力，加入预测器使动作、动词和物体类别小幅而一致改善。

		Action Anticipation		
Encoder	**Predictor**	Verb	Noun	Action
✓		61.3	57.0	39.1
	✓	48.7	34.7	20.2
✓	✓	63.6	57.1	39.7

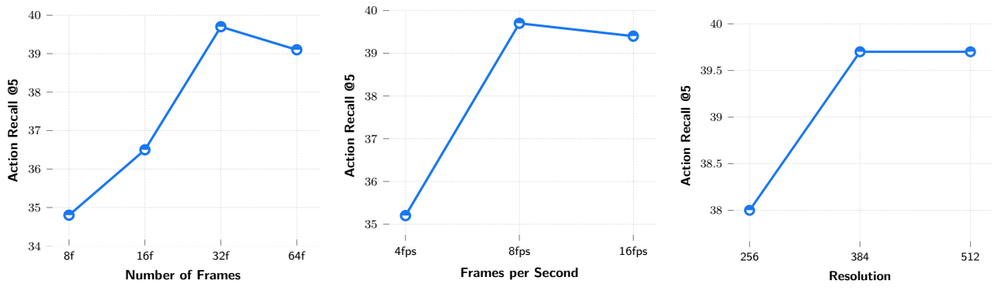


图 17 EK100 动作预判方案消融。（左）性能随上下文帧数变化。（中）随推理帧率变化，上下文固定 32 帧。（右）随上下文帧空间分辨率，即高度和宽度变化。

直至性能饱和或略降。最佳性能在上下文 32 帧、帧率 8、分辨率 384×384 的训练设置中获得。

更长期预测。图 18 左图通过将预判时间设为 1、2、4、10 秒，展示更长时域预测的影响。每个时间报告 recall@1、@5、@10、@20。性能随预判时间增加急剧下降，符合预期，因为 EK100 的未来预测不是确定性任务。

失败案例分析。图 18 右图展示 EK100 验证集中，动词、名词、动作各种成功／失败组合的分布。模型表现良好，因此出现最多的是全部预测成功，

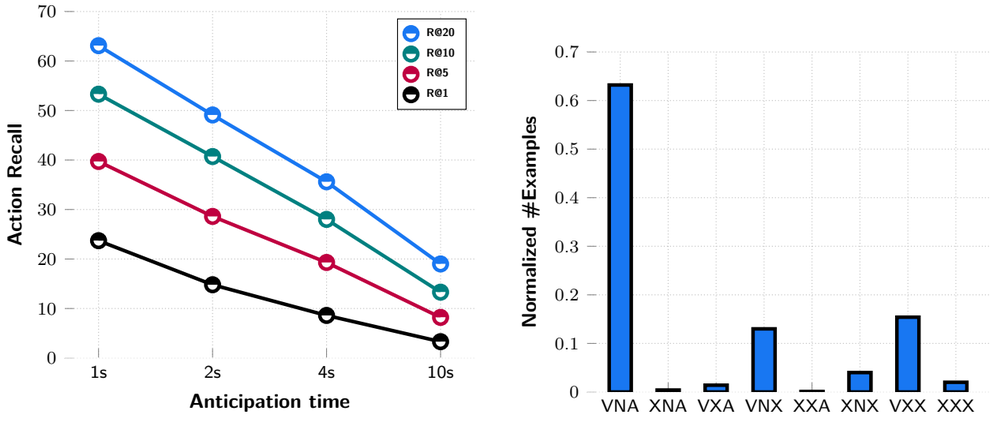


Figure 18 (Left): Impact of longer-term anticipation times. Performance on EK100 action anticipation, at several recall values and anticipation times. **(Right): Distribution of success and failure cases of V-JEPA 2.** Calculated on the validation set of EK100. VNA means that verb, noun and action are all correctly classified by the model. An X symbol means that the corresponding attribute is not correctly classified by the model. Note: the probe is composed of 3 independent classifiers for verb, noun and action, hence why the model can have a different prediction for the action and for the verb/noun pair.

across verb noun and action. The most represented failure configurations all include a failure to find the action.

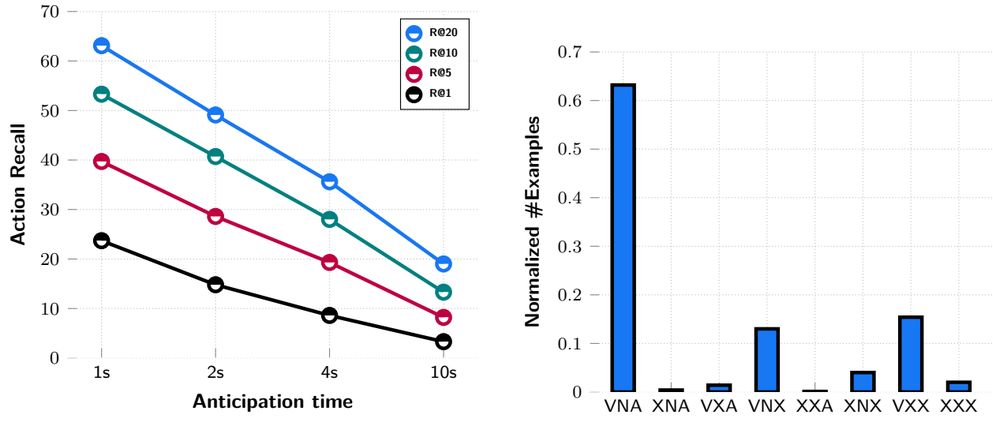


图 18（左）：更长预判时间的影响，展示 EK100 不同 recall 指标和预判时间下的性能。（右）：EK100 验证集上 V-JEPA 2 成功与失败案例的分布。VNA 表示动词、名词和动作均分类正确，X 表示对应属性错误。注意：探测器包含动词、名词和动作三个独立分类器，因此动作预测可与动词／名词组合预测不同。

在动词、名词及 动作。出现最多的失败配置均包含未能找到该动作。

E Video Question Answering

In this section, we provide details on training a V-JEPA 2 Multi-modal Large Language Model (MLLM). We follow the LLaVA framework (Liu et al., 2024a) to train the MLLM, where the vision backbone is using V-JEPA 2, and the LLM backbone can be any off-the-shelf pretrained LLM, akin to the *non-tokenized early fusion* (Wadekar et al., 2024) setup. The MLLM ingests the output embeddings of the vision encoder, which are projected to the hidden dimension of the LLM backbone using a *projector* module. The *projector* is typically a 2-layer MLP. The MLLM is trained using a mix of image-text and video-text paired data, in a series of progressive training steps.

To understand the impact of data scale, we use a dataset of 88.5 million image-text and video-text pairs, similar to what was used for training PerceptionLM (Cho et al., 2025). As mentioned in Section 7, we investigate two setups: (a) *controlled*, where we train with 18M image and video-text pairs, and we evaluate V-JEPA 2 and other encoders on the exact same MLLM training setup, and (b) *scaling*, where we take V-JEPA 2 VITg$_{384}$ and use the full aligment dataset. To further test the versatility of V-JEPA 2, we use Qwen2-7B-Instruct (Yang et al., 2024a) as the language backbone for the controlled experiments, and Llama 3.1 8B Instruct (Grattafiori et al., 2024) for the scaling experiments. We describe the training details in the following sections.

E.1 Processing Images and Videos as Input

Since video question answering uses video instead of image inputs, the number of output visual tokens increases significantly compared to image question answering. If required, we can use pooling methods to reduce the number of visual tokens. Popular pooling methods involve adaptive 2x2 pooling (Cho et al., 2025), Perceiver Sampler (Jaegle et al., 2021), Attentive Pooling (Bardes et al., 2024), etc.

Additionally, we observed that learning from image-text pairs is crucial for high performance in downstream

E 视频问答

本节给出 V-JEPA 2 多模态大语言模型的训练细节。遵循 LLaVA 框架（Liu et al., 2024a），视觉骨干采用 V-JEPA 2，语言骨干可选任意现成预训练 LLM，类似非词元化早期融合设置（ Wadekar et al., 2024 ）。视觉编码器输出嵌入通过投影器映射至语言骨干隐藏维度，再由 MLLM 接收；投影器通常为两层 MLP。采用图像–文本与视频–文本混合配对数据，分多个渐进步骤训练。

为了解数据规模的影响，我们使用 8850 万对图像–文本与视频–文本数据，与 PerceptionLM 的训练数据类似（Cho et al., 2025）。如第 7 节所述，研究两种设置：（a）受控设置，使用 1800 万对数据，在完全相同的 MLLM 训练设置中评估 V-JEPA 2 和其他编码器；（b）扩展设置，使用 V-JEPA 2 ViT-g$_{384}$ 及完整对齐数据集。为进一步检验通用性，受控实验使用 Qwen2-7B-Instruct（Yang et al., 2024a）作为语言骨干，扩展实验使用 Llama 3.1 8B Instruct（Grattafiori et al., 2024）。训练细节见下文。

E.1 图像与视频输入处理

视频问答输入视频而非图像，因此输出视觉词元显著增加。必要时可用池化减少词元，常见方法包括自适应 2×2 池化（ Cho et al., 2025 ）、 Perceiver Sampler （ Jaegle et al., 2021 ）、注意力池化（ Bardes et al., 2024）等。

此外，观察到图像–文本对学习对较高下游性能至关重要，

benchmarks. In order to train with images, a simple approach is to repeat the given image for k frames, where k is the maximum amount of frames supported by V-JEPA 2. However, during our initial experiments we find this strategy is ineffective at improving downstream performance, as it does not allow the model to extract fine-grained information. Therefore, we employ a modified *Dynamic* S^2 strategy introduced by Liu et al. (2024d) to provide V-JEPA 2 higher resolution granularity during training. This method adaptively processes an image at native resolution with different aspect ratios to preserve their original resolution, by creating a sequence of tiles of maximum size supported by V-JEPA 2. In case of videos, we choose to train with a fixed number of frames f_n, by balancing the number of visual tokens with compute budget.

E.2 Controlled Setup

Training details For the controlled setup, we follow the LLaVA-NEXT framework (Liu et al., 2024a; Zhang et al., 2024b), where we use Qwen2-7B-Instruct (Yang et al., 2024a) as the base LLM for all encoders. To reduce the number of visual tokens, we employ an attentive pooler with a factor of 4-16, depending on the compute budget and the number of visual patches. See Table 21 for more details.

Our training setup follows the LLaVA-NeXT pipeline (Li et al., 2024b), which consists of multiple staged training phases. Concretely, the stages consist of: a) aligning the attentive pooler with image captioning data (Stage 1), b) training the full model on high quality image captioning (Stage 1.5), and c) training the full model on large scale image question answering (Stage 2). We add an extra stage to train on large scale video captioning and question answering (Stage 3). We use 18 million image and video-text aligned data. The LLM progressively improves its understanding of the visual tokens after multiple staged training, with the biggest improvement in video question answering tasks after Stage 3.

We explore *frozen* and *finetuned* encoder alignment setups. In both setups, full parameters of the LLM and projector are trained, and in the latter, the V-JEPA 2 parameters are additionally unfrozen. To reduce the number of visual tokens and keep the MLLM context length fixed, we employ an attentive pooler as the projector to reduce the number of visual tokens by a factor of 4, unless otherwise denoted. The implementation used for this controlled study is based on the Llava-NEXT codebase,[4] and uses Pytorch 2.5.1, Transformers 4.46.0, Flash attention 2 and DeepSpeed 0.14.4 for model implementation, faster training and multi-gpu model sharding respectively. We train all models using 128 H100 GPUs with an effective batch size of 256 across all stages. We perform all optimizations using AdamW with 0 weight decay. For Stages 1 and 1.5, we use learning rate of 1e-5 with cosine decay, and for Stages 2 and 3 we use constant learning rate of 5e-6. In all stages, we use linear warmup for the first 3% of training steps. Training hyperparameters are listed in Table 21.

Baselines. To assess the ability of V-JEPA 2 to capture spatiotemporal details for VidQA, we compare to leading off-shelf image encoders. Specifically, we compare to DINOv2 (Oquab et al., 2023), SigLIP2 (Tschannen et al., 2025), and Perception Encoder (Bolya et al., 2025). DINOv2 is a self-supervised image model, while SigLIP2 and Perception Encoder are both trained with language supervision using noisy image-text captions. We apply all image encoders at their "native" pretrained resolution, which is 518px, 384px, and 448px, respectively, on each video frame independently.

We keep all training details the same, except that we increase the attentive pooling ratio to 16 to keep the number of image tokens relatively similar among models. See Table 21 for details.

Evaluation. To evaluate the capability of V-JEPA 2 to understand the world through video and language, we select popular evaluation datasets built to test spatio-temporal reasoning abilities. To ensure reproducible evaluation, we utilize the `lmms-eval` library (Li et al., 2024a; Zhang et al., 2024a) to conduct our experiments, which is a vision model enabled fork of `llm-eval-harness` (Gao et al., 2024), which is a popular evaluation library for evaluating LLMs on text-based tasks. In the controlled setup, for each model and dataset, we evaluate by using uniform frame sampling mechanism, and choosing 128 frames during inference. For PerceptionTest, we further train the model for 5 epochs on the training set.

[4] https://github.com/LLaVA-VL/LLaVA-NeXT

基准。对图像训练，一种简单方法是将图像重复 k 帧，其中 k 为 V-JEPA 2 支持的最大帧数。但初步实验发现，该策略无法提取细粒度信息，因而不能有效改善下游性能。因此采用修改版 Dynamic S^2 策略（Liu et al., 2024d），在训练中提高分辨率粒度。该方法按原始分辨率和不同宽高比自适应处理图像，生成由模型所支持最大尺寸的图块组成的序列，以保持原始分辨率。视频则采用固定帧数 f_n，平衡视觉词元数与计算预算。

E.2 受控设置

训练细节。受控设置遵循 LLaVA-NEXT（Liu et al., 2024a; Zhang et al., 2024b），所有编码器均使用 Qwen2-7B-Instruct（Yang et al., 2024a）作为基础 LLM。按计算预算和视觉图块数，以注意力池化器将视觉词元缩减 4 至 16 倍，详见表 21。

训练遵循 LLaVA-NeXT 流程（Li et al., 2024b），包含多个阶段：（a）用图像描述数据对齐注意力池化器（阶段 1）；（b）用高质量图像描述训练完整模型（阶段 1.5）；（c）用大规模图像问答训练完整模型（阶段 2）。另加入大规模视频描述与问答训练（阶段 3）。采用 1800 万对图像／视频–文本对齐数据。LLM 在多阶段训练后逐步增进视觉词元理解，视频问答的最大改善发生于阶段 3 后。

我们研究冻结和微调编码器两种对齐设置。两者均训练 LLM 与投影器的全部参数；后一种还解除 V-JEPA 2 参数冻结。除另有说明外，使用注意力池化器作为投影器，将视觉词元数缩减为原来的四分之一，以减少词元并保持 MLLM 上下文长度固定。该受控研究基于 LLaVA-NEXT 代码库[4]，分别使用 PyTorch 2.5.1、Transformers 4.46.0、Flash Attention 2 和 DeepSpeed 0.14.4 进行模型实现、加速训练和多 GPU 模型分片。所有阶段均使用 128 张 H100 GPU，有效批量大小为 256。使用 AdamW，权重衰减为 0。阶段 1 和 1.5 使用 1e-5 学习率与余弦衰减，阶段 2 和 3 使用恒定学习率 5e-6。各阶段前 3% 训练步骤采用线性预热。训练超参数见表 21。

基线。为评估 V-JEPA 2 捕获 VidQA 时空细节的能力，与先进现成图像编码器比较：DINOv2（Oquab et al., 2023）、SigLIP2（Tschannen et al., 2025）、Perception Encoder（Bolya et al., 2025）。DINOv2 为图像自监督模型，后两者通过含噪图像–文本描述进行语言监督训练。逐帧独立应用各编码器，采用各自原生预训练分辨率，分别为 518px、384px、448px。

训练细节保持一致，仅将注意力池化比增至 16，使各模型图像词元数相对接近。详见表 21。

评估。为评估 V-JEPA 2 通过视频和语言理解世界的能力，选择测试时空推理的常用数据集。为确保可复现，使用 lmms-eval（Li et al., 2024a; Zhang et al., 2024a），它是常用文本 LLM 评估库 llm-eval-harness（Gao et al., 2024）支持视觉模型的分支。受控设置中，对每个模型与数据集均匀采样帧，推理选择 128 帧。PerceptionTest 另在训练集上训练 5 个轮次。

[4] https://github.com/LLaVA-VL/LLaVA-NeXT

Table 21 **Hyperparameters for controlled comparison of vision encoders. We use each vision encoder with its native pretrained input resolution.**

Model	Pooling Ratio	Vision Tokens	BS	Stage 1/1.5 LR	Stage 2/3 LR	WD	Frames
V-JEPA 2 ViT-L_{256}	4	4096	256	1e-5	5e-6	0.0	128
V-JEPA 2 ViT-H_{256}	4	4096	256				
V-JEPA 2 ViT-g_{256}	4	4096	256				
V-JEPA 2 ViT-g_{384}	4	9216	256				
V-JEPA 2 ViT-g_{512}	8	8192	256				
$DINOv2_{518}$	16	10952	256	1e-5	5e-6	0.0	128
$SigLIP2_{384}$	16	5832	256				
PE_{448}	16	8192	256				

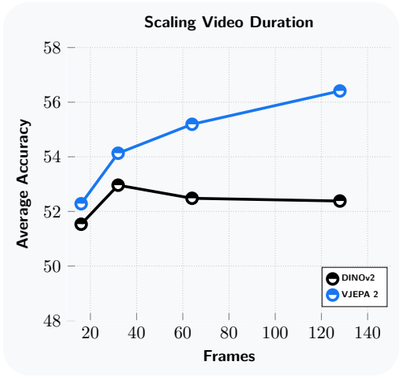


Figure 19 **Impact of video duration during visual instruction tuning.** We investigate the effect of increasing the number of frames during visual instruction tuning, where the encoder is frozen. We observe that with more frames, V-JEPA 2 performance linearly increases compared to DINOv2, an SSL-based image encoder, showing the potential of V-JEPA 2 to scale with more frames.

Impact of Video Duration In the controlled setup, we perform an analysis to understand V-JEPA 2's capability in long-form video understanding. We train MLLMs on V-JEPA 2 and DINOv2, keeping the encoders frozen, and by increasing the number of frames we use in training and testing. We observe as the number of frames increases, performance on downstream tasks *linearly* improves for V-JEPA 2, but decreases and remains flat in case of DINOv2 (Figure 19). This highlights the potential of video encoders such as V-JEPA 2 to understand long-form videos with natural language queries, via adapting an LLM using V-JEPA 2 as the visual encoder.

E.3 Data scaling setup

Training details In the scaling setup, we follow the framework used by Cho et al. (2025) to train Perception LM 8B. Specifically, we utilize the released codebase, which is based on Lingua (Videau et al., 2024). We modify the code to use V-JEPA 2 encoder, and we use the Llama 3.1 8B Instruct (Grattafiori et al., 2024) as the backbone LLM. Unlike Cho et al. (2025), we do not use pooling, instead we train V-JEPA 2 VIT-g_{384} using MLP projector, leading to 288 tokens per frame. The training setup also consists of three progressive stages: Stage 1: aligning the MLP pooler with image captioning data; Stage 2: training on a mix of image-text captioning and QA data; and Stage 3) training on video-text captioning and QA data. We scale up the data size to 88.5 million samples. Our setup uses Pytorch 2.5.1 and Perception LM training code,[5] modified with the V-JEPA 2 encoder. We train on 512 H100 GPUs for Stage 2 and Stage 3 with a global batch size of 2048 and 1024 respectively. Details of the training hyperparams are provided in Table 22.

[5] https://github.com/facebookresearch/perception_models

表 21 超参数以其原生预训练 **视觉编码器受控比较的参数。各视觉编码器使用预训练输入分辨率。**

模型	池化比例	视觉词元	BS	阶段 1／1.5 学习率	阶段 2／3 学习率	WD	帧数
V-JEPA 2 ViT-L_{256}	4	4096	256	1e-5	5e-6	0.0	128
V-JEPA 2 ViT-H_{256}	4	4096	256				
V-JEPA 2 ViT-g_{256}	4	4096	256				
V-JEPA 2 ViT-g_{384}	4	9216	256				
V-JEPA 2 ViT-g_{512}	8	8192	256				
$DINOv2_{518}$	16	10952	256	1e-5	5e-6	0.0	128
$SigLIP2_{384}$	16	5832	256				
PE_{448}	16	8192	256				

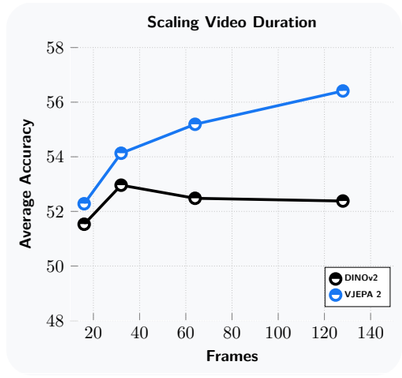


图 19 视觉指令微调中的视频长度影响。冻结编码器，增加训练帧数。相比自监督图像编码器 DINOv2，V-JEPA 2 随更多帧呈线性性能提升，体现其扩展帧数的潜力。

视频长度的影响。受控设置中，分析 V-JEPA 2 的长视频理解能力。在 V-JEPA 2 与 DINOv2 上训练 MLLM，保持编码器冻结，并增加训练和测试帧数。随帧数增加，V-JEPA 2 下游性能线性改善，而 DINOv2 性能下降后持平（图 19）。这体现视频编码器的潜力：以 V-JEPA 2 为视觉编码器适配 LLM，可根据自然语言查询理解长视频。

E.3 数据扩展设置

训练细节。在规模扩展设置中，遵循 Cho et al.（2025）训练 Perception LM 8B 的框架。具体而言，采用其发布的、基于 Lingua（Videau et al., 2024）的代码库，修改代码以使用 V-JEPA 2 编码器，并以 Llama 3.1 8B Instruct（Grattafiori et al., 2024）作为骨干 LLM。与 Cho et al.（2025）不同，我们不采用池化，而使用 MLP 投影器训练 V-JEPA 2 ViT-g_{384}，每帧产生 288 个词元。训练同样分为三个渐进阶段：阶段 1，使用图像描述数据对齐 MLP 池化器；阶段 2，使用图像–文本描述与问答混合数据训练；阶段 3，使用视频–文本描述与问答数据训练。数据规模扩展至 8850 万个样本。使用 PyTorch 2.5.1 与 Perception LM 训练代码[5]，将视觉编码器修改为 V-JEPA 2。阶段 2 和阶段 3 均使用 512 张 H100 GPU，全局批量大小分别为 2048 和 1024。训练超参数详见表 22。

[5] https://github.com/facebookresearch/perception_models

Table 22 **Data scaling training parameters.**

Parameter	Values
Common parameters	
Crop Size	384
Video Frames per Second	1
Sampling method	Uniform
Seed	777
Stage 1	
Steps	16000
Warmup Steps	96
Batch Size (global)	128
Learning Rate	1e-4
Final Learning Rate	1e-6
Weight Decay	0.05
Max sequence length	1920
Stage 2	
Steps	35000
Warmup Steps	200
Batch Size (global)	2048
Learning Rate	4e-5
Final Learning Rate	4e-7
Weight Decay	0.05
Max sequence length	6400
Image tiles	16
Video frames	16
Stage 3	
Steps	28000
Early stopping step	22000
Warmup Steps	168
Batch Size (global)	2048
Learning Rate	1e-5
Final Learning Rate	1e-7
Weight Decay	0.05
Max sequence length	12800
Image tiles	32
Video frames	32

Baselines. We compare our scaling runs with Qwen2VL (Wang et al., 2024a), Qwen2.5VL (Qwen Team et al., 2025), InternVL-2.5 (Chen et al., 2024), and PerceptionLM 8B (Cho et al., 2025). Baseline numbers are sourced directly from the papers, except for MVP which we run ourselves.

Evaluation. We follow similar evaluation pipeline as reported in the controlled setup, using `lmms-eval` library. We report our model evaluations on 32 frames.

表 22 数据规模扩展的训练参数。

参数	数值
通用参数	
裁剪尺寸	384
视频每秒帧数	1
采样方法	均匀
Seed	777
阶段 *1*	
步数	16000
预热步数	96
全局批量大小	128
学习率	1e-4
最终学习率	1e-6
权重衰减	0.05
最大序列长度	1920
阶段 *2*	
步数	35000
预热步数	200
全局批量大小	2048
学习率	4e-5
最终学习率	4e-7
权重衰减	0.05
最大序列长度	6400
图像分块	16
视频帧	16
阶段 *3*	
步数	28000
早停步数	22000
预热步数	168
全局批量大小	2048
学习率	1e-5
最终学习率	1e-7
权重衰减	0.05
最大序列长度	12800
图像分块	32
视频帧	32

基线。将扩展训练结果与 Qwen2VL（Wang et al., 2024a）、Qwen2.5VL（Qwen Team et al., 2025）、InternVL-2.5（Chen et al., 2024）、PerceptionLM 8B（Cho et al., 2025）比较。除 MVP 自行运行外，基线数值直接取自论文。

评估。采用与受控设置类似的 lmms-eval 评估流程，报告 32 帧模型结果。